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Fluids in Porous Media

Transport and phase changes

Fluids in Porous Media

Transport and phase changes

Henk Huinink
Eindhoven University of Technology, The Netherlands

With the help of Philip Ruijten and Thomas Arends

Morgan & Claypool Publishers

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Dedicated to
Trudy and Lieke and Ji

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Preface

During my postdoc I entered the field of porous media. The first thing I did was read a few good books on porous media. In particular, the books of Professor Dullien and Professor Sahimi helped me a lot to create an overview in this field. However, I always kept the feeling that books on porous media could be written more efficiently. In many books the level of detail is so high, that it is pretty difficult to keep the overview as a newcomer in this field. When I became Assistant Professor, I started to give courses to Master students on microfluidics and porous media. While doing this I again felt the need for a more concise book helping students to catch up without being distracted by numerous details.

When I was approached to contribute to the IOP series 'Concise Physics', it was immediately clear to me that this was the opportunity for writing a book that already existed in my mind: a brief but fundamental introduction to the field of porous media with a main text of around a hundred pages. And I am proud that I have achieved that limit.

The main challenge while writing was to eliminate stuff I loved but could distract the reader from the main line. Sad, but necessary to realize my own ideal book. So I had to abandon the idea of being complete. I regret that I could not cover percolation theory, which has given me so much fun during my postdoc period. Further, I could not include an in-depth discussion of gas transport in porous media. However, I am convinced that a newbie in the field will have enough tools to become self-propelling in further literature study.

I wish the reader a pleasant journey

Acknowledgements

First of all, I would like to thank Morgan & Claypool for the opportunity to write this book. It was a challenge and a pleasure to write a lean text, that facilitates students at MSc level to enter the field of porous media. Nicki Dennis: thanks for your support and the trust that I would finalize this task. Further, I have to thank the Eindhoven University of Technology for allowing me to spend time on writing this book.

Without my PhD and MSc students Philip Ruijten, Thomas Arends and Karel van Laarhoven this book would not have come into existence at all. Thank you for criticizing the content, structure and text of this book. Philip and Thomas: all figures except one are from your hands and they are marvelous. In the past months you have endured several sessions in which I criticised the figures.

Further, I would like to thank Chris Benson for editing the manuscript carefully and Jacky Mucklow for proofreading it. I am also happy that Melanie Rijkers allowed me to use her beautiful pictures for the cover. They all contributed to the quality of this book.

The nice viscous fingering pictures could be made because Hans Dalderop has constructing the Hele-Shaw cell for doing the experiments. Further, I am indebted to all my other PhD students: Kees Kuijpers, Kateryna Filippovych, Özlem Gezici, Benjamin Voogt and Leyla Sögütoglu. This book has consumed a significant amount of my time that I could not invest in your supervision.

Finally, this book would not have been there without the ongoing support during the writing process. Trudy, Lieke and Ji: you are a wonderful family to me. Many hours that could have been family time, were invested in this book. It is time to invert this in favor of you all.

About the Author

Hendrik Pieter (Henk) Huinink



Hendrik Pieter (Henk) Huinink is a physical chemist who graduated at the Wageningen University, The Netherlands (1998), with a PhD in soft matter physics. From 1998–1999 he was research associate at the Shell Research and Technology Center Amsterdam (SRTCA), where he investigated the influence of confinement on the morphology of block copolymers. He joined the Applied Physics Department of the Eindhoven University of Technology (The Netherlands) in 1999, where he started to work on modeling of transport in porous media and thin films. In those years he became more and more involved in NMR imaging studies on transport phenonema. In 2005 he became assistant professor at the Eindhoven University of Technology in the group Transport in Permeable Media (TPM). He guides a group of about 5–7 PhD students doing experimental work on transport processes in porous materials and polymer films. He teaches several courses at BSc and MSc level: statistical physics, microfluidics and porous media. Recently, he has started to redirect his research activities towards the kinetics of charging and discharging heat storage materials. For that purpose he spent a sabbatical in 2015 at the Friederich-Alexander University (Erlangen, Germany), where he studied hydration transitions in inorganic crystals with molecular simulations.

Symbols

Below a list of symbols and related units are given. As some symbols are used in different disciplines for other physical properties, sometimes a symbol is used for more than one variable. Further, in the text sometimes subscripts and superscripts are used for specification, i.e. γ refers to the interfacial tension in general and γ_{sl} to the tension of a solid–liquid interface. Here only the more general form of the variable is given.

A	Area [m^2]
a	Particle diameter [m]
Bo	Bond number [–]
$C(\vec{r})$	Correlation function [–]
Ca	Capillary number [–]
$\vec{D}$	Electric displacement [C m^{-2}]
d	Diameter [m]
Di	Diffusion number [–]
$\vec{e}$	Unit vector [m]: subscripts are used to indicate the direction
E	Energy [J]
$\vec{E}$	Electric field [V m^{-1}]
F	Helmholtz free energy [J]
f	Force [N]
G	Gibbs free energy [J]
H	Enthalpy [J] or Front position [m]
h	Height [m] or Average front position [m]
J	Molar flux [mol m^{-2} s^{-1}] along the direction of transport
$\vec{J}$	Molar flux [mol m^{-2} s^{-1}]
$\mathscr{L}$	Contour length [m]: for example to measure the path length of tortuous path
K	Conduction: its units depend on the transport process of interest
k	Permeability [m^2]
L	Length [m]: often used for the system dimension or a length scale of interest
$\mathscr{L}$	Contour length [m]: length measured along a certain path
$\mathcal{L}$	Latent heat [J mol^{-1}]
l	Length [m]: mostly used for the length of pore segment
ℓ_p	Typical pore size [m]
Ma	Mach number [–]
N	Number of molecules [mol]
$\mathcal{N}$	Deposition rate [m s^{-1}]
n	Molar density [mol m^{-3}] or number density [m^{-3}]
p	Pressure [Pa]
p_c	Capillary pressure [Pa]
$P(r)$	Pore size distribution [m^{-1}]
Q	Volumetric discharge [m^3 s^{-1}] or Heat [J]
q	Volumetric flux along the flow direction [m s^{-1}]
$\vec{q}$	Volumetric flux [m s^{-1}]
R	Radius [m] or gas constant [J K mol^{-1}]
RH	Relative humidity [%]
r	Radius [m]: mostly used for the pore radius, but sometimes for the radius of a droplet.

$\vec{r}$	Distance vector [m]
Re	Reynolds number [–]
S	Entropy [J K^{-1}] or saturation [–]
Sr	Strouhal number [–]
T	Temperature [K]
U	Characteristic velocity [m s^{-1}]
$u(\vec{r})$	Interaction energy of particles [J]
$\vec{u}$	Velocity field [m s^{-1}]
V	Volume [m^3] or characteristic velocity [m s^{-1}]
W	Work [J]
x	Cartesian coordinate [m]
Y	Generalized thermodynamical potential [J]
y	Cartesian coordinate [m]
z	Cartesian coordinate [m]: always chosen as the vertical direction
$\vec{x}$	Position vector [m]
α	Wave number [m^{-1}]
γ	Interfacial tension [N m^{-1}]
ϵ	Energy density [J m^{-3}] or dielectric permittivity [F m^{-1}]
$\eta(\vec{x})$	Porosity function [–]
θ	Contact angle [°] or liquid content [–]
ϑ	Angle [°]
$\imath$	The number of ions needed for a neutral unit [–]
κ	Mean curvature [m^{-1}]
λ	Pore size distribution index [–], mean free path [m] and wave length [m]
μ	Chemical potential [J mol^{-1}] or dynamic viscosity [Pa s]
ν	Rate [s^{-1}]
ξ	Characteristic length scale [m]
ρ	Mass density [kg m^{-3}]
ϱ	Space charge density [C m^{-3}]
σ	Conductivity [S m^{-1}]
τ	Tortuosity [–]
ϕ	Porosity [–]
φ	Angular coordinate in cylindrical and spherical coordinate systems [–] or volume fraction [–]
ψ	Potential: its units depend on the transport process of interest
Ω	Grand potential [J]

IOP Concise Physics

Fluids in Porous Media

Transport and phase changes

Henk Huinink

Chapter 1

Introduction

1.1 Porous materials

Rocks, concrete, bricks, zeolites, sandbeds, sediments, wood: a list that can easily be made longer. These materials have in common that they exist of a solid matrix and contain pores. Depending on the geometry of the pore space, liquids and gases can migrate through these materials via pressure driven flows or diffusion under influence of concentration gradients.

The list also demonstrates the economic and societal importance of porous materials in general and transport processes in particular. Despite the fact that there is an urgent need for renewable energy, the global economy still depends on oil. Efficient oil recovery is therefore a billion-dollar-scale industry hungry for new insights in the flow behavior of water/oil mixtures in rocks. Without difficulty another issue of global dimensions can be picked to illustrate the importance of flow in porous media: water management. The whole field of flow in porous media started with the pivotal works of the French engineers Darcy and Dupuit. Actually, the major drive for these nineteenth century engineers was to give the common people in Europe access to clean drinking water. Global access to and availability of clean water is still an important issue, making knowledge on the migration of water in the subsurface extremely valuable.

The aforementioned list also demonstrates that the origin of pore space in materials can have many causes. In clay bricks or other porous ceramics, porosity is a consequence of the fact that the materials are made by sintering particles: packs of particles always contain a certain amount of void space. In wood, the pores are a consequence of a biological growth process leading to a highly structured porous material. In some rocks, pore space is a consequence of air entrapment during solidification of the original magma. Other rocks have been formed by a process of sedimentation of particles, and void space is again a consequence of the packing. Concrete is formed by cement hydration leading to crystals glueing together but incorporating pore space. Although a discussion of the genesis of porous materials is

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beyond the scope of this book, one should always be aware that there is a direct connection between the geometry of the pore space in a material and the original processes creating it.

Porous materials are generally divided into two families: the unconsolidated and consolidated porous media. Unconsolidated media consist of particles that are not bonded together. Typical examples are sand, peat and gravel. Actually all soils are unconsolidated porous materials. In consolidated porous materials the solid matrix is a sample spanning phase. In nature many consolidated porous media have started their lives as unconsolidated media. Due to cementation reactions or sintering processes, the original non-bonded particles have been glued together.

As the title of the book already portrayed, the physics of fluids in porous materials is the key topic. As the book is meant as an introduction to this field, the material has been selected such that it helps the reader to enter the field. This book does not pretend to be complete, but it aims to give its reader the skills and way of thinking needed to attack more advanced topics.

1.2 The geometry of the pore space and porosity

In most cases, the geometry of the pore space of a natural or technological porous material is highly complex, often stochastic in nature and heterogeneous below certain length scales. The structure of some materials even have to be characterized by multiple length scales. In figure 1.1 it is shown how the structure of a cementitious material varies on a wide range of length scales. It is therefore not surprising that detailed quantification of the morphology of the pore space is an extremely difficult task. As a consequence, direct coupling of macroscopic transport properties with the pore space geometry is also extremely difficult. Therefore, the approach of this book is to work with idealized pore space models in order to illustrate the connection between transport laws on macroscopic length scales and the physics governing processes on microscopic length scales.

Despite this bad news with respect to pore structure quantification, there exist a few parameters that can be used and experimentally verified. To that end, a property $\eta(\vec{x})$ [−] is introduced that varies a function of the position $\vec{x}$ [m]: in the pore space $\eta = 1$ and in the solid matrix of the material $\eta = 0$. Integrating η over the volume of

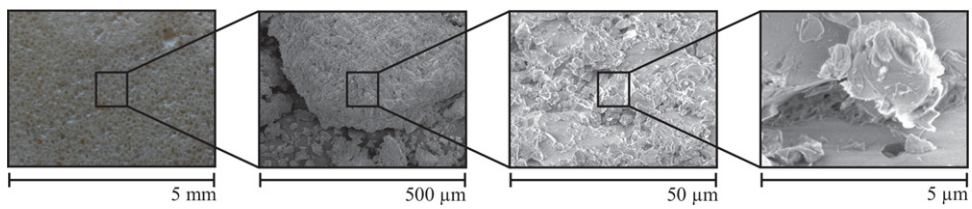


Figure 1.1. Scanning electron microscope (SEM) pictures of a cementititous material on different length scales. Structure variations can be seen on a wide range of length scales. Courtesy of the Eindhoven Multiscale Lab of the TU/e & Ahmet Bakarat.

the material V [m^3] results in the volume of the pore space V_p [m^3]. Now a material can be characterized with a very simple parameter, the porosity ϕ [–]:

$$\phi \equiv V_p/V. \quad [-] \tag{1.1}$$

The porosities of a wide range of materials are shown in table 1.1. The table demonstrates that many materials have porosities between 0.1 and 0.5. This can be understood as follows. Unconsolidated porous materials basically consist of separate particles packed together. The porosity simply increases with a decreasing packing efficiency, which is both related to the particle shape and the speed of the packing process. Here it has to be mentioned that the porosity of perfectly packed identical spheres is about 0.25. As consolidated porous materials result from particle sintering at elevated temperatures or cementation reactions, their porosities stay in the same range as the porosities of unconsolidated materials. In the course of time cementation reactions can lead to a decrease in porosity.

Although we have already connected ϕ with the function $\eta(\vec{x})$, the mathematical connection has not been discussed properly. The porosity equals the spatial average of this function.

$$\phi = V^{-1} \int_V \eta(\vec{x}) \mathrm{d}\vec{x} \equiv \langle \eta \rangle. \quad [-] \tag{1.2}$$

Table 1.1. The porosity ϕ of several porous media. Unconsolidated materials are packs of loose particles. Sources: Freeze R A and Cherry J A 1979 *Groundwater* (Englewood Cliffs, NJ: Prentice-Hall) and Hall C and Hoff W D 2009 *Water Transport in Brick Stone and Concrete* 2nd edn (Boca Raton, FL: CRC).

Material	ϕ
unconsolidated deposits	
gravel	0.25–0.40
sand	0.25–0.50
silt	0.35–0.50
clay	0.40–0.70
rocks	
fractured basalt	0.05–0.50
karst limestone	0.05–0.50
sandstone	0.05–0.30
dolomite	0.00–0.20
shale	0.00–0.10
fractured crystalline rock	0.00–0.10
dense crystalline rock	0.00–0.10
construction materials	
calcium silicate brick	0.36
aerated concrete	0.84
hardened cement paste	0.28
cement-sand mortar	0.17

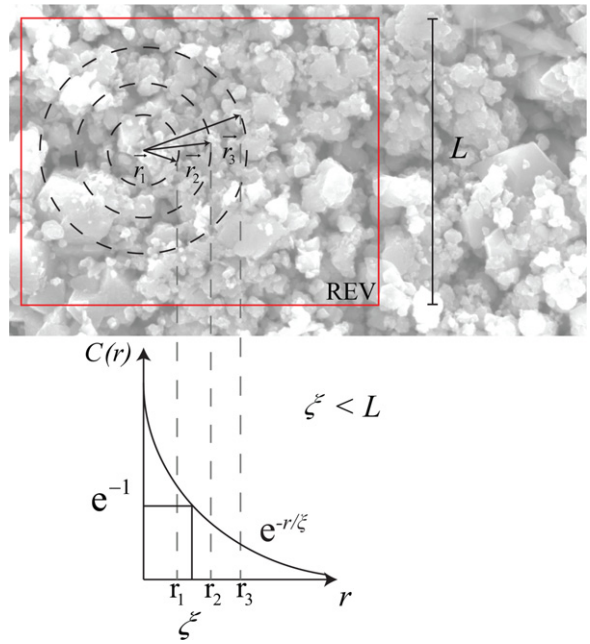


Figure 1.2. Schematic picture of the REV (representative elementary volume) concept and the spatial correlation function. As an example, an exponential decaying correlation function is shown. Further, an SEM image of the porous structure of an Fe_3O_4 particle pack is shown as background image. SEM picture courtesy of the Eindhoven Multiscale Lab of the TU/e & Philip Ruijten.

The brackets $\langle\rangle$ indicate that a volume average is taken. This expression directly points towards a very crucial issue when dealing with porous media: homogeneity. Porosity will only be a material property given that the volume averaging of η is done over a sufficiently large volume. To be more precise: the length scale of heterogeneities in the material should be smaller than the typical dimensions of the material itself, as illustrated in figure 1.2.

To define average quantities at the macroscopic level, a characteristic volume is needed. In porous media theory often this characteristic volume is referred to as the 'representative elementary volume' (REV). It is defined as the minimal volume of a material that can be considered as being homogeneous. To elucidate the REV concept the spatial correlation function of $\delta\eta(\vec{x}) = \eta(\vec{x}) - \langle\eta\rangle$ will be used (note that $\delta\eta$ characterizes the deviation of the local structure from an averaged value):

$$C(\vec{r}) \equiv [\phi(1-\phi)]^{-1}\langle\delta\eta(\vec{x})\delta\eta(\vec{x}+\vec{r})\rangle. \quad [-] \tag{1.3}$$

Here $\vec{r}$ is the vector between the two points that are correlated in space. The brackets $\langle\rangle$ indicate that a volume average of $\delta\eta(\vec{x})\delta\eta(\vec{x}+\vec{r})$ is taken. As a consequence C is a sole function of the $\vec{r}$. As $\langle(\delta\eta(\vec{x}))^2\rangle = \phi(1-\phi)$ this factor is used to normalise the correlation function: $C(0, 0, 0) = 1$. Further, it can be shown that $C(\infty) \to 0$

for $|\vec{r}| = \infty$, because $\delta\eta(\vec{x})$ and $\delta\eta(\vec{x} + \vec{r})$ are no longer correlated and $\langle\delta\eta(\vec{x})\delta\eta(\vec{x} + \vec{r})\rangle = \langle\delta\eta(\vec{x})\rangle\langle\delta\eta(\vec{x} + \vec{r})\rangle$.

With this correlation function in hand, the concept of REV can be discussed in more detail. For simplicity we assume spherical symmetry so that the direction of $\vec{r}$ no longer matters and only the distance between two points $r = |\vec{r}|$ has to be considered. Actually, we assume that our medium is isotropic, meaning that the structure is uniform in all directions. To simplify the problem a bit further we assume that the correlation function decays exponentially: $C(r) = \exp(-r/\xi)$. The interesting parameter in this function is the so-called correlation length ξ [m] that measures the length over which spatial correlations in the structure persist. Given that the length scale of interest is L [m], the material can be considered to be homogeneous given that $\xi \ll L$. Having an explicit equation for the correlation function, a lower bound for the REV can be found by integrating C over the volume:

$$V_c = \int_0^\infty C(r)4\pi r^2 \mathrm{d}r = 8\pi\xi^3. \quad [\mathrm{m}^3] \tag{1.4}$$

As an REV is generally defined as the volume at which the material can be considered to be homogeneous, the volume of an REV V_{REV} should be at least a few times (order 10) bigger than V_c.

Having addressed the topics homogeneity and REV, there remain a few open issues around the property porosity. Firstly, the accessible pore volume V_p' might be less than V_p as there could be isolated pockets of void space. As fluids will never be able to reach these isolated pockets, this part of the porosity does not contribute to transport and sorption. Therefore, in the rest of this book the following definition of porosity will be used:

$$\phi \equiv V_p'/V, \quad [-] \tag{1.5}$$

which deviates from the previous definition (1.1). Given this definition, the porosity of a material can easily be determined by measuring the volume of liquid needed to completely saturate the material. Care has to be taken that no air remains entrapped in the material.

The latter brings us to the second important aspect of the property porosity: its link to transport. Parts of the pore space might be accessible only from one side: the so-called dead end pores. Obviously long distance traveling of molecules in a material can only occur in that part of the porous network that can be accessed at least from two sides. Dead-end pores do not contribute to this long distance motion. Nevertheless, they play an important role in the kinetics via their ability to store mass. Trapping in these dead-pores increases the residence time of molecules in the system.

1.3 Pore sizes and the pore size distribution

Any discussion on the structure of porous media should include the notion of pore size and the concept pore. Actually, we already used the word pore without a proper definition. What is a pore? That this is a difficult question is nicely illustrated by

the previously discussed microscopy picture of concrete in figure 1.1. In theory, pores are discrete objects that together build the pore space. However, in practise the pore space is a highly complex 3D network of void space and there are no objects with an own identity. However, despite the fact that pores are ill-defined objects, they are extremely useful in building models, as will become clear in the course of this book. Therefore and only therefore this concept is widely used in the area of porous media theory.

Before introducing the concept of pores and pore sizes, a length scale ℓ_p [m] is introduced that still has a direct connection with the geometry of the pore space:

$$\ell_p \equiv V_p'/A_p', \quad [-] \tag{1.6}$$

where A_p' [m^2] is the accessible pore surface area. This surface area can be estimated from gas adsorption measurements with for example N_2. When the surface is fully covered by a monolayer of molecules, the absorbed amount no longer increases with the applied gas pressure. With the help of the cross-sectional area of a N_2 molecule the property A_p' can be calculated from the absorbed amount.

The connection with the concept of a pore can be made as follows. When the pore space is assumed to consist of equally sized slits, cylinders or spheres with a diameter $d \equiv 2r$ [m] it can be shown that ℓ_p, respectively, equals r, $r/2$ or $r/3$. In the case of cylinders and spheres r [m] is the radius. While switching from the actual observable ℓ_p to the parameter r (pore radius), the concept of a pore is born. To account for the complexity of the pore space the pore size distribution $P(r)$ [m^{-1}] is introduced that obeys $\int_0^\infty P(r)\mathrm{d}r = 1$ as it is a probability density function.

To simplify the discussion we choose to model the pore space from now on with a set of cylinders. When these cylinders are placed in one line, the length of this line will be called the total accessible pore length $\mathscr{L}_p'$ [m]. In this particular case $P(r)\mathrm{d}r$ represents the fraction of the total pore length having a pore radius r. The pore size distribution can be coupled with the pore volume

$$V_p' = \mathscr{L}_p' \int_0^\infty P(r)\pi r^2 \mathrm{d}r \equiv \pi \mathscr{L}_p' \overline{r^2} \quad [m^3] \tag{1.7}$$

and

$$A_p' = \mathscr{L}_p' \int_0^\infty P(r)2\pi r \mathrm{d}r \equiv 2\pi \mathscr{L}_p' \bar{r}. \quad [m^2] \tag{1.8}$$

The horizontal bars represent the averages over the pore size distribution. From these expressions it follows that

$$\ell_p = \overline{r^2}/2\bar{r}. \tag{1.9}$$

Although we do not have direct experimental access to $P(r)$, mercury intrusion porosimetry (MIP) can give information on the distribution. MIP works as follows. A sample is placed in a mercury bath and the mercury is forced into the sample by gradually increasing the pressure. Assuming cylindrical pores the pressure to enter a

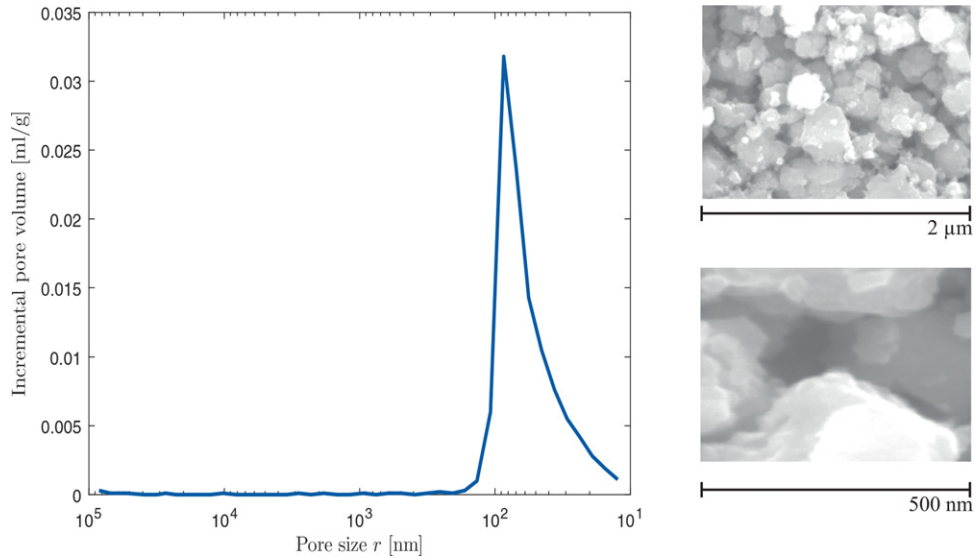


Figure 1.3. MIP curve of a pressed Fe_3O_4 pill. The peak of the curve is located around 200 nm, which is in agreement with the SEM images. SEM pictures courtesy of the Eindhoven Multiscale Lab of the TU/e & Philip Ruijten.

pore equals $2\gamma/r$ [Pa], where γ [N m^{-1}] is the surface tension of mercury. Due to the curvature of mercury interface in a pore, an excess pressure, called capillary pressure p_c, exists in the mercury (for cylindrical pores $p_c = 2\gamma/r$). In section 3.3 the backgrounds of capillary pressure will be discussed in detail. It follows from the expression for the capillary pressure, that the biggest pores are invaded first. The smaller the pores, the bigger the pressure needed for enforcing mercury intrusion. With MIP, the intruded volume is measured as a function of the applied pressure that can be transferred in an estimate of the pore radius r. A typical result is shown in figure 1.3. When cylindrical pores are assumed, the intruded volume $V(r)$ at a given pressure (corresponding to a specific pore radius) equals $\mathscr{L}_p' \int_r^\infty P(r)\pi r^2 \mathrm{d}r$. By using this expression the following equation can be used to extract the pore size distribution from MIP data

$$P(r) = \frac{1}{\mathscr{L}_p' \pi r^2} \frac{\mathrm{d}V}{\mathrm{d}r}. \tag{1.10}$$

The only unknown is $\mathscr{L}_p'$ can be obtained by applying the normalization condition: $\int_0^\infty P(r)\mathrm{d}r = 1$. In the case where the structure of the pore space is such that significant fractions of big pores can only be reached via small pores, expression (1.10) will overestimate the fraction of small pores. This is called the ink-bottle-effect.

In table 1.2 an impression is given of typical pore radii of some common porous materials. Whereas most porous materials have more or less similar porosities, see table 1.1, their pore radii can differ a lot. Often there is a correlation between the

Table 1.2. Typical pore radii of various materials.

Material	r m
zeolite	10^{-10}–10^{-9}
porous silica	10^{-9}–10^{-7}
controlled pore glass	10^{-8}–10^{-6}
sintered Al_2O_3	10^{-8}–10^{-6}
fired-clay brick	10^{-7}–10^{-5}
sand	10^{-4}–10^{-2}

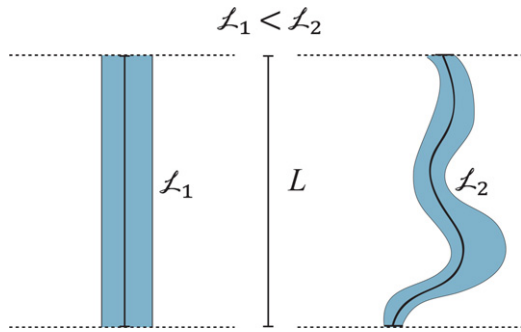


Figure 1.4. The concept of tortuosity. Tortuosity is the ratio between the contour length $\mathscr{L}$ of a pore and the displaced distance L. The right pore is more tortuous than the left pore.

typical radius of material and its genesis. For example, the pore radius of sintered Al_2O_3 is determined by the particle size of the Al_2O_3 powder, used as starting material.

1.4 Tortuosity

In the previous sections the most import concepts regarding the geometry of porous media have been discussed. However, the property 'tortuosity' has not been discussed and will hardly be used in this book. As it sometimes pops up in porous media theory, this topic will be addressed briefly in this section.

The concept of tortuosity is explained in figure 1.4. Two types of porous media are shown that both have a typical dimension L [m].Virtual tubes are drawn to visualize the length of the transport paths $\mathscr{L}$ [m] in both media. Tortuosity τ [–] is now defined as

$$\tau \equiv \mathscr{L}/L, \tag{1.11}$$

and in all cases $\tau > 1$. It can be used as follows. Assume that our transport problem can be mapped on virtual tubes with a radius r [m] and typical contour length $\mathscr{L}$ [m]. A natural choice for the tube radius would be the typical pore radius. The discharge

through an individual tube is Q [m^3 s^{-1}]. As we do not want to specify the nature of the transport process, we will adopt a simple linear law for the discharge

$$Q = -K\frac{\partial \psi}{\partial x}, \quad [\mathrm{m^3\,s^{-1}}] \tag{1.12}$$

where ψ is the potential that drives the discharge. Further, K is the intrinsic conduction of the tube and the dimension of $K\psi$ equals [m^4 s^{-1}]. When a potential drop $\Delta\psi$ is placed over a distance L, corresponding with a contour length $\mathscr{L}$, the discharge obeys

$$Q = -\frac{K}{\tau}\frac{\Delta \psi}{L}. \quad [\mathrm{m^3\,s^{-1}}] \tag{1.13}$$

In the case where the potential drop $\Delta\psi$ is kept constant, the discharge decreases with increasing value of the tortuosity τ as the effective pathway for transport increases with $\mathscr{L}$.

So far the concept of tortuosity looks rather simple. However, its application is complicated by the fact that the contour length $\mathscr{L}$ of the transport pathways can be process dependent. For example, the typical pathways for diffusion or volumetric flow might differ. Therefore, tortuosity cannot be regarded as a material parameter as it is intrinsically coupled with the transport process and fluid of interest. This is precisely the reason that the concept is not used in the rest of this book.

1.5 Guide to the reader

Until now not many words have been spent on the logic behind this book. Here the outline and the ratio behind the chosen topics will be discussed. As thermodynamics plays a role in various topics covered by this book, a recap of thermodynamics is given in chapter 2. The emphasis of this recap is on the tools needed for describing phase transitions and as such it gives the building blocks for understanding chapter 4.

Chapter 3 describes a key topic regarding forces in porous media: capillarity. It is shown how the concept of capillary pressure emerges from the wetting properties of the internal pore surface. Without the concept neither phase transitions, chapter 4, nor multiphase transport, chapters 8 and 9, can be understood.

Chapter 4 shows how the confinement of the porous matrix influences phase transitions such as melting, condensation and crystallization from solution. Before transport in porous media can be discussed, first attention has to be paid to transport theory in general. It will be shown that the capillary pressure is the key factor in all these transitions.

In chapter 5 it is shown how pipe flow can be derived from the Navier–Stokes equation. First, it is demonstrated that in porous media flow is nearly always in the laminar regime and can be described with the Stokes equation. Pipe flow is derived from the Stokes equation. As a pipe is an idealized pore shape, knowledge of pipe flow helps to understand transport in porous media.

Single phase flow in porous media is discussed in chapter 6. The most famous equation in the field of porous media is introduced: Darcy's law. Via this law the

permeability of a medium is discussed and interpreted with simple models based on pipe flow. The importance of the pore size for the flow is underlined.

Chapter 7 is a practical application of Darcy's law to the problem of flow of unconfined aquifers. The Dupuit approach is covered and it is shown how a 3D flow problem can be reduced to a 2D flow problem, given that the vertical dimension is small compared to the horizontal dimensions. Finally, the application of the Dupuit equation to flow around a well is analysed.

After the introduction of Darcy's law in chapter 6, more complicated flow problems can be addressed. First, simultaneous flow of a liquid and air is introduced in chapter 8. It is shown that the air flow in most cases can be neglected. As a liquid can vaporize, it is discussed how vapor diffusion can contribute to the flow problem.

Finally, the flow of two immiscible liquids is explained in chapter 9. It is shown that the ratio of the viscosities of both phases determines the behavior during a displacement process. Attention is paid to the size of the front zone, separating the two liquids. Further, the stability of the front is analysed.

> At various points in the text, examples will be discussed to give the reader insight in the typical values of important parameters. The text of these examples is framed within boxes similar to the box framing the text you are reading right now.

Further reading

Dullien F A L 1991 *Porous Media: Fluid Transport and Pore Structure* 2nd edn (San Diego, CA: Adademic) ch 1: a good introduction to the structure of porous media. It contains an extensive discussion of several types of materials and their characteristics. Anyway, a classical book in the field of porous media.

Hall C and Hoff W D 2009 *Water Transport in Brick Stone and Concrete* 2nd edn (Boca Raton, FL: CRC) ch 1: a clear text with emphasis on construction materials.

Adler P M 1992 *Porous Media: Geometry and Transports* (Boston, MA: Butterworth-Heinemann) ch 2: in depth discussion of geometrical aspects of porous media with a focus on theory.

Wong P Z (ed) 1999 *Methods of the Physics of Porous Media* (San Diego, CA: Academic Press): overview of advanced experimental methods for studying both structural and transport properties of porous materials.

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Fluids in Porous Media

Transport and phase changes

Henk Huinink

Chapter 2

Recap of thermodynamics

2.1 Fundamental equations

As several aspects of porous media theory are based on thermodynamics, this chapter gives a short summary of the relevant thermodynamic relationships. All starts with the notion of energy conservation: the first law of thermodynamics. The energy of system changes for two reasons: heat Q [J] delivered to the system and work W [J] done on the system.

$$\mathrm{d}E = \mathrm{d}Q + \mathrm{d}W. \quad [\mathrm{J}] \tag{2.1}$$

Work can have different origins, i.e. volume V [m^3] changes given a certain pressure p [Pa] ($\mathrm{d}W = -p\mathrm{d}V$), mass changes N [mol] (actually the number of particles) given a certain chemical potential μ [J mol^{-1}] ($\mathrm{d}W = \mu\mathrm{d}N$) and changes in the interfacial area A [m^2] at a specific interfacial tension γ [N m^{-1}] ($\mathrm{d}W = \gamma\mathrm{d}A$). Note that every type of work consists of an conjugated pair of an extensive and intensive variable: V and p in the case of volumetric work. In the example of volumetric work, the volume V is the extensive variable as it scales with the system size given a fixed macroscopic state. The pressure p is the intensive variable that has a fixed value given a fixed macroscopic state.

Having a system also implies the existence of a boundary between the system and the environment, see figure 2.1. Of particular importance are the boundary conditions that can vary from system to system. With boundary conditions we refer to physical properties that are under control of the experimentalist. A few situations often appear and have therefore obtained clear names. In the case where a system has a controllable volume and does not allow heat and mass transport over its boundary, we speak of an isolated system: E, V and N are constant. A closed system has a controlled volume and does not allow mass exchange, however, it allows heat exchange: T [K], V and N are under the control of the experimentalist. When also mass exchange is allowed, we speak of an open system: T, V and μ are constant. Finally, there is also the isobar–isotherm system: T, p and N are the controllable variables.

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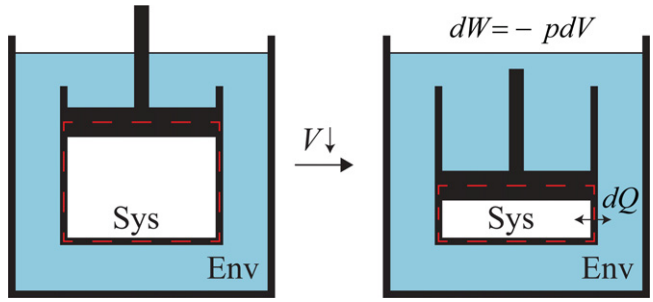


Figure 2.1. System (white) versus environment (blue): the system interacts via the boundary (red dashed line) with its environment. In this particular example heat Q [J] can be transported over the boundary, changing the internal energy E [J] of the system. Further, E can change due to work. In this example work is performed on the system by compressing it.

The next step is to postulate something with respect to the direction of spontaneous processes: the second law of thermodynamics. This law states that there is property called entropy S [J K^{-1}] that always increases in a spontaneous process in an isolated system. It can be shown that there is a direct connection between S and heat given that a process is done reversibly, meaning that the interior of the systems always has sufficient time to equilibrate during the process: $\mathrm{d}S = T^{-1}\mathrm{d}Q$. The differential of the energy can therefore be rewritten as

$$\mathrm{d}E = T\mathrm{d}S - p\mathrm{d}V + \gamma\mathrm{d}A + \sum_i \mu_i \mathrm{d}N_i. \quad [\mathrm{J}] \tag{2.2}$$

Note that this expression can describe mixtures of different molecules/particles as indicated by the subscript 'i'. The terms $-p\mathrm{d}V$, $\gamma\mathrm{d}A$ and $\sum_i \mu_i \mathrm{d}N_i$ represent the different forms of work. As both E and S are uniquely coupled with the state of the system, $\mathrm{d}E$ behaves mathematically as a total differential equation. Although a discussion of the mathematical aspects of thermodynamics is far beyond the scope of this chapter, here it has to be mentioned that it can be proven that

$$E = TS - pV + \gamma A + \sum_i \mu_i N_i. \tag{2.3}$$

In summary, equation (2.2), the differential for the energy, forms the fundamental starting point for all further thermodynamic reasoning. Note that this can be rewritten in a differential for the entropy,

$$\mathrm{d}S = (1/T)\mathrm{d}E + (p/T)\mathrm{d}V - \sum_i (\mu_i/T)\mathrm{d}N_i + \cdots, \quad \left[\mathrm{JK}^{-1}\right] \tag{2.4}$$

which nicely illustrates the importance of entropy for the isolated system. The entropy is the unique function that determines the state of the isolated system as S varies with the boundary conditions E, V, etc. Therefore, S is called the thermodynamic potential of an isolated system. According to the second law, entropy should be maximized to find the equilibrium state of this system. Or differently stated: $-S$ has to be minimized.

2.2 Thermodynamic potentials and equilibrium

As mentioned, every system can be characterized by a set of boundary conditions. When a system is not isolated, the second law teaches that the equilibrium state of this system is governed by a thermodynamical potential other than S. In this section a quick summary without too much proof is given. In Appendix A it is shown how thermodynamical potentials can be derived. In a closed system (N, V and T are fixed), the thermodynamic potential is the Helmholtz free energy F [J] that is defined as

$$F \equiv E - TS. \quad [\mathrm{J}] \tag{2.5}$$

The differential can be obtained by combining $\mathrm{d}F = \mathrm{d}E - T\mathrm{d}S - S\mathrm{d}T$ with equation (2.2).

$$\mathrm{d}F = -S\mathrm{d}T - p\mathrm{d}V + \gamma\mathrm{d}A + \sum_i \mu_i \mathrm{d}N_i. \quad [\mathrm{J}] \tag{2.6}$$

To obtain the equilibrium state of the system F has to be minimized. Note that F varies linearly with $-S$ but incorporates an extra degree of freedom: E. Where in isolated system $-S$ has to be minimized, here F has to be minimized. Further, it can be shown that

$$F = -pV + \gamma A + \sum_i \mu_i N_i \approx \gamma A + F_b. \quad [\mathrm{J}] \tag{2.7}$$

The last expression on the right hand side illustrates that the F can be split in an interfacial and a bulk contribution F_b, assuming that the thickness (volume) of the interfacial zone can be neglected compared to the bulk.

A similar operation can be done in the case of the open system: μ, V and T constant. In this particular system the grand potential Ω [J], defined as

$$\Omega \equiv E - TS - \sum_i N_i \mu_i, \quad [\mathrm{J}] \tag{2.8}$$

is the thermodynamic potential that has to be minimized to find the equilibrium state of the system. The corresponding differential equals

$$\mathrm{d}\Omega = -S\mathrm{d}T - p\mathrm{d}V + \gamma\mathrm{d}A - \sum_i N_i \mathrm{d}\mu_i. \quad [\mathrm{J}] \tag{2.9}$$

Further it can be shown by a combination of equations (2.3) and (2.8) that

$$\Omega = -pV + \gamma A. \quad [\mathrm{J}] \tag{2.10}$$

The last system we want to address is the isobar–isotherm system as it is from an experimental point of view a frequently occuring system. First we introduce the notion of enthalpy H [J]:

$$H \equiv E + pV. \quad [\mathrm{J}] \tag{2.11}$$

The thermodynamic potential of this system is the Gibbs free energy G [J] that is defined as

$$G \equiv H - TS \quad [\mathrm{J}] \tag{2.12}$$

and obeys the following differential

$$\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p + \gamma\mathrm{d}A + \sum_i \mu_i \mathrm{d}N_i. \quad [\mathrm{J}] \tag{2.13}$$

As G is the thermodynamic potential of the isobar–isotherm system, it has to be minimized to find equilibrium. Further, it can be shown that

$$G = \gamma A + \sum_i \mu_i N_i \approx \gamma A + G_b, \quad [\mathrm{J}] \tag{2.14}$$

where G_b is the Gibbs free energy of the bulk.

Finally, this section is concluded with a brief discussion on equilibrium. As mentioned various times, the equilibrium state of a system can be found by minimizing the thermodynamic potential. Equilibrium implies that within the system no net currents of heat, mass, volume, charge, etc exist between different areas. As a consequence, there are no gradients in temperature, chemical potential, pressure, electric potential, etc, which drive such currents.

2.3 Chemical potentials

As both transport processes and many phase changes involve mass transfer, the chemical potential μ of a substance is often an essential ingredient in theories describing these processes. As mentioned before, the work for adding mass is $\mathrm{d}W = N\mathrm{d}\mu$. As μ is not easily accessible by experiments, it would be convenient when this property could be related to the composition of the system. Under certain conditions this can be done quite easily. In this section the chemical potentials of an ideal gas and that of an ideal solute particle will be discussed. Despite the stringent assumption of ideality in both cases, the obtained expression for μ will be of great help explaining the physics of various processes.

A very helpful expression for this purpose is the Gibbs–Duhem relation that can be derived by combining the equations (2.2) and (2.3).

$$-S\mathrm{d}T + V\mathrm{d}p - \sum_i N_i \mathrm{d}\mu_i = 0. \quad [\mathrm{J}] \tag{2.15}$$

Here only the expression a for single phase bulk system is given. When the temperature of a system is kept constant the pressure can be related directly to the chemical potentials. And when both T and p are kept constant, the chemical potentials of different species can be related to each other. This is precisely what will be done for the ideal gas and solution in the remainder of this section.

An ideal gas has a density so low that particle–particle interactions can be neglected. Note that applicability of the ideal gas approximation is wide: i.e. at room temperature and ambient pressure, $T = 298$ K and $p = 10^5$ Pa, air behaves as an ideal gas mixture. Experiments have proven and statistical theory has shown that there is straightforward connection between pressure and particle density:

$$p = RT\sum_i n_i. \quad [\text{Pa}] \tag{2.16}$$

In this expression n_i [mol m^{-3}] is the molar density of component 'i' and R [J Kmol^{-1}] is the gas constant. By keeping the temperature constant and combining the ideal gas law with the Gibbs–Duhem relation, it can be shown that

$$\mathrm{d}p = RT\sum_i \mathrm{d}n_i = \sum_i n_i \mathrm{d}\mu_i. \quad [\text{Pa}] \tag{2.17}$$

As we are dealing with an ideal system there will be no cross-relation between the chemical potentials of the different species. As a consequence, a unique relation between the chemical potential and the density of given component 'i' can be obtained:

$$\mathrm{d}\mu_i = RT\,\mathrm{d}\ln n_i. \quad \left[\text{J mol}^{-1}\right] \tag{2.18}$$

A similar approach can be used for chemical potential of solutes in ideal solutions. The solutes are of course the minority species in the solution. When the solute is diluted such that solute–solute interactions can be ignored, the solution is ideal. To proceed, the variable osmotic pressure Π [Pa] has to be introduced and its meaning is illustrated with figure 2.2. A U-shaped tube is placed such that both its legs point in the direction in which gravity acts. A semi-permeable membrane allowing water to pass, but blocking solute particles, is placed between the two legs. Both legs are filled with an equal amount of liquid. However one leg (I) is filled with a pure liquid and the other contains the same liquid but contaminated with a solute (II). After a certain amount of time the amount of liquid in leg II has increased due to liquid transport through the semi-permeable membrane. The resulting height difference h [m] between legs I and II indicates that there is an extra force acting that can be

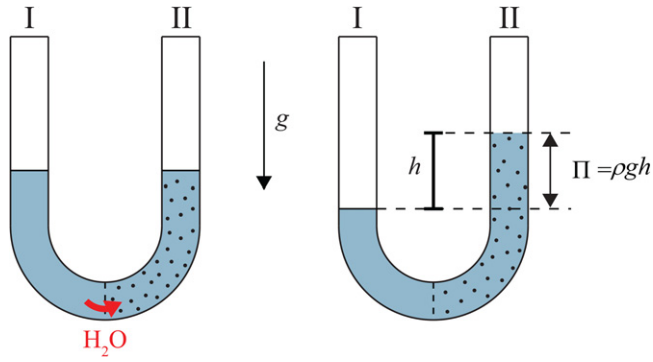


Figure 2.2. An osmotic pressure difference Π driving flow over a semi-permeably membrane and finally balanced by gravity $\Pi = \rho g h$. Initially leg I is filled with pure liquid and leg II is filled with the same liquid but contaminated with a solute (left). The two legs are separated by a semi-permeable membrane allowing only the solvent and not the solute to migrate from I to II. At the start the two legs are not in equilibrium and liquid starts to flow from I to II (left figure). After equilibration a height difference h between the two legs has developed (right).

quantified as a pressure, balancing the gravitational contribution $\rho g h$ [Pa] (ρ [kg m^{-3}] is the mass density of the liquid). As the only difference between the two legs is the presence of solutes, Π has to be a function of the concentration of solute particles. As the liquid can pass the semi-permeable membrane, its chemical potentials are equal in equilibrium: $\mu^{\mathrm{I}}(p) = \mu^{\mathrm{II}}(p + \Pi)$. It follows from the Gibbs–Duhem equation (2.15) that this condition can only be met, when for the solution in leg II it holds that

$$\mathrm{d}\Pi = \sum_j n_j^{\mathrm{II}} \mathrm{d}\mu_j^{\mathrm{II}}, \quad [\mathrm{Pa}] \tag{2.19}$$

where the superscript that refers to the leg has been dropped. The summation 'j' runs over all solute species. This equation illustrates that the only way to keep the chemical potential of the liquid equal in both legs is by balancing the pressure increase $\mathrm{d}\Pi$ with an increase in the chemical potentials of the solute species. Using the experimental fact that for dilute solutions

$$\Pi = RT \sum_j n_j, \quad [\mathrm{Pa}] \tag{2.20}$$

equation (2.19) can be rewritten and an expression for the chemical potential of a solute 'j' is found

$$\mathrm{d}\mu_j = RT \mathrm{d} \ln n_j. \quad [\mathrm{J}] \tag{2.21}$$

In this section equations have been obtained for the chemical potential of a species in an ideal gas mixture, equation (2.18), and a solute in an ideal solution, equation (2.21). Despite the fact that these relations have their application limits due to the assumption of ideality, they are useful building blocks for theories that primarily aim for understanding.

The strength of seawater. Consider the experiment shown in figure 2.2. Leg I is now filled with pure water and leg II is filled with seawater. The question is now: what is the initial osmotic pressure difference driving flow through the semi-permeable membrane at room temperature ($T = 298$ K)? The main ingredient of seawater is NaCl. A typical value for the NaCl concentration in seawater is $n = 0.60$ mol l^{-1} = 600 mol m^{-3}. As NaCl is fully dissociated in solution the total ion concentration is about 1200 mol m^{-3}. With equation (2.20) the osmotic pressure can be calculated:

$$\Pi = (n_{\mathrm{Na}} + n_{\mathrm{Cl}})RT = (600 + 600) \times 8.31 \times 298 = 2.9 \cdot 10^6 \text{ Pa} = 29 \text{ atm}.$$

Note that $R \approx 8.31$ J mol K^{-1}. To get a feeling for this value, we will calculate the height of water column h able to produce the same pressure. The density of water is $\rho = 1$ kg l^{-1} = 1000 kg m^{-3}.

$$h = \frac{\Pi}{\rho g} = \frac{2.9 \cdot 10^6}{1000 \times 9.81} = 296 \text{ m}.$$

Here we used $g = 9.81$ m s^{-2}.

Further reading

Carter A H 2000 *Classical and Statistical Thermodynamics* (Upper Saddle River, NJ: Prentice-Hall) Ch 1–10: a systematic introduction to the framework of thermodynamics without distracting stuff.

Callen H B 1985 *Thermodynamics and an Introduction to Thermostatistics* 2nd edn (New York: Wiley) ch 5 and 6: an insightful discussion of thermodynamic potentials and equilibrium states.

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Fluids in Porous Media

Transport and phase changes

Henk Huinink

Chapter 3

Wetting of solids and capillarity

3.1 Interfaces and interfacial tension

An important factor in porous media is the large internal surface area. As a consequence, the role of the interfaces on thermodynamic properties needs to be considered. In the case of a reversible process at constant temperature, volume and number of particles in both phases, the change in free energy $\mathrm{d}F$ [J] for enlarging the interfacial area with $\mathrm{d}A$ [m^2] is equal to the interfacial work.

$$\mathrm{d}F = \gamma \mathrm{d}A \quad [\mathrm{J}] \tag{3.1}$$

where γ [$\mathrm{N\ m}^{-1}$] is the interfacial tension. For example, when a spherical droplet of radius r [m] is split in two identical spherical droplets at constant temperature, the free energy rises with a factor $\sqrt[3]{2}$ according to this formula given that γ has a positive value and is constant. Both assumptions are valid, which will be demonstrated at the end of this section. In table 3.1 the interfacial tensions of various liquids are listed.

Before the discussion on the background of the variable γ, first some terminology has to be addressed. When an interface between two phases is described, the term interfacial tension can always be used for γ. When the interface actually separates a liquid from a gas, the term surface tension can be used as we are dealing with surface. Actually, this is only a matter of language and does not have any influence on the physics. Therefore, from now on only the term interfacial tension will be used.

To understand the origin of interfacial tension, we consider a gas–liquid interface and compare particles located in the vicinity of an interface and particles located in the bulk. In figure 3.1 a schematic picture is given that illustrates how particles interact in the bulk of the liquid and close to a liquid–gas interface. Particles in the bulk are from all sides attracted by other particles. As is visible in figure 3.1 particles in the neighbourhood of the interface miss the attractive interactions of particles above the interface. As a consequence, the net force

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Table 3.1. Mass densities, interfacial tensions, vaporization enthalpies and dynamic viscosities of various liquids. Most data refer to a temperature of 25°C. In the case of other temperatures this is indicated by a superscript between brackets: i.e. (20) refers to 20°C. Source: CRC 2016 *Handbook of Chemistry and Physics*, 96th edn (Boca Raton, FL: CRC).

Liquid	ρ kg m^{-3}	γ mN m^{-1}	ΔH_{vap} GJ m^{-3}	μ mPa s
mercury	13534	485.48	3.99	1.53
water	997	71.99	2.43	0.89
glycerol	$1261^{(20)}$	62.5		934
ethanol	$789^{(20)}$	19.89	0.72	1.07
n-hexane	661	15.33	0.24	0.30

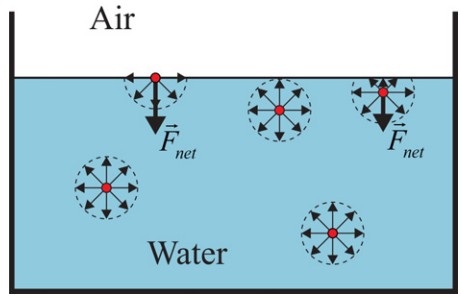


Figure 3.1. The molecular origin of the interfacial tension: Some molecules/atoms are highlighted as red dots. Particles are attracted by fellow particles as represented by the arrows. The net force experienced by particles in the bulk of the liquid equals zero. Particles close to the interface feel a net attraction in the direction to the bulk of the liquid, leading to a tendency of the liquid to minimize this unfavorable interface.

generated by the inter-particle interactions pulls a particle at the surface into the bulk of the liquid phase. As a consequence, a tension lateral to the interface develops promoting minimization of the interfacial area. This effect will become more pronounced with increasing strength of the particle–particle interactions. This is nicely visible in table 3.1. Liquids with multiple options for hydrogen bonding (glycerol and water) have a higher interfacial tension than liquids without this ability. As metallic bonds are much stronger than hydrogen bonds and Van der Waals forces, the interfacial tension of mercury is an order of magnitude bigger than all the others.

Finally, it will be illustrated how the qualitative discussion on the origin of interfacial tension can be made more quantitative. This will be done by evaluating the energy of a liquid layer (film) of thickness L [m] and with area A, bounded by air on two sides (two liquid–air interfaces), figure 3.2. As the gas density in air is low compared to the liquid, the contributions of air to the energy of the liquid film will be neglected. Further, it is assumed that the density changes stepwise to 0 at the liquid–air interfaces. Note that in reality the transition from high to low densities at

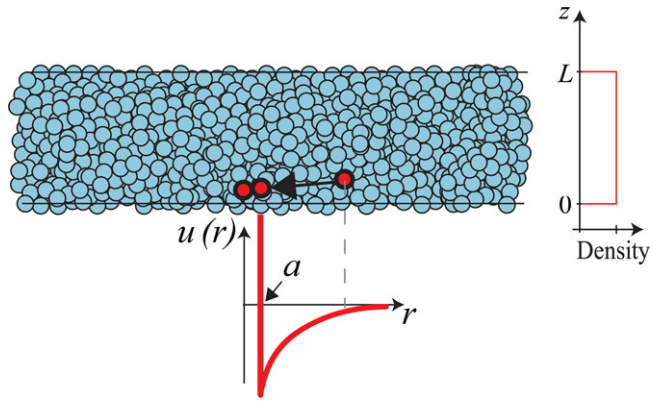


Figure 3.2. A liquid film: particles interact with each other via an interaction potential u [J], shown in the graph below the picture of the film. On the right side of the figure the density profile of the film is plotted.

the liquid–air interface occurs within 1 nm. Two particles separated by a vector $\vec{r}$ [m] interact with each other with an energy $u(\vec{r})$ [J]. The energy of the film can now be calculated by integrating over the film volume V:

$$E = \frac{1}{2}\int_V \int_V n(\vec{r}_1)n(\vec{r}_2)u(\vec{r})\mathrm{d}\vec{r}_2\mathrm{d}\vec{r}_1, \quad [\mathrm{J}] \tag{3.2}$$

where $\vec{r} \equiv \vec{r}_2 - \vec{r}_1$ is the distance between two particles. The vectors $\vec{r}_1$ and $\vec{r}_2$ are the position vectors of particle 1 and 2, respectively. Further, $n(\vec{r})$ [m^{-3}] is the number density at position $\vec{r}$. Note that in this section, we deviate from our standard definition of n as the molar density. The factor 1/2 accounts for double counting of the interactions.

By assuming a homogenous density distribution and a stepwise density profile at the interface, equation (3.2) can be rewritten as

$$E = \frac{1}{2}n^2 A \int_0^L \left[\int_{-z_1}^{L-z_1}\int_0^\infty u(z_1, R_1)2\pi R_2 \mathrm{d}R_2\mathrm{d}z_2\right]\mathrm{d}z_1. \quad [\mathrm{J}] \tag{3.3}$$

To derive equation (3.3) from (3.2) the following things have to be taken into account. First, a switch was made towards cylindrical coordinates. Second, coordinates of particle 2 are defined relative to the coordinates of particle 1, which is reflected by the integration limits including the z-coordinate of particle 1. Third, as visible in the equation, cylindrical symmetry has been assumed. As we will continue with a spherical symmetric particle–particle potential u, cylindrical coordinates suffice. The integration between the brackets over the coordinates z_2 [m] and R_2 [m] gives the potential energy of particle 1 due its interaction with the whole film.

In the previous paragraph a general expression of a liquid film has been derived, which still does not give a detailed view on the γ. To that end an interaction potential

is introduced that consists of a hard-sphere repulsion u_{HS} [J] and a van de Waals attraction $-C/r^6$ [J]:

$$u(r) = u_{\mathrm{HS}}(r) - \frac{C}{r^6}. \quad [\mathrm{J}] \tag{3.4}$$

The expression assumes spherical symmetry. Further C [J m^6] is constant and $C > 0$ to ensure attractive interactions. Further, $u_{\mathrm{HS}} = \infty$ for $r < a$ and 0 for $r \geqslant a$, with a [m] as the particle diameter. Based on this interaction potential the energy per surface area of the film can be computed from (3.3). Details are given in Appendix B.

$$E/A = \frac{\pi C n^2}{2a^2} - \frac{2\pi C n^2 L}{3a^3} - \frac{\pi C n^2}{12L^2}. \quad [\mathrm{J\ m^{-2}}] \tag{3.5}$$

Note that this energy can be split into three contributions. The term proportional to L is the actual energy of the bulk fluid; it scales with the dimension of the film. The property $\epsilon \equiv -2\pi C n^2/3a^3$ [J m^{-3}] is the energy density of the bulk fluid. The last term of the expression disappears in the case where the film thickness goes to infinity. Actually, this is called the disjoining pressure, which exists in thin films. The term $\pi C n^2/2a^2$ [N m^{-1}] accounts for the presence of the two liquid–air interfaces and represents the energetic contribution to the interfacial tension: $\gamma \sim E/A$. Therefore, a first estimate of γ based on the particle–particle potential equals

$$\gamma = \frac{\pi C n^2}{4a^2}. \quad [\mathrm{N\ m^{-1}}] \tag{3.6}$$

The expression nicely illustrates that when the attraction between molecules increases, reflected by C, the interfacial tension increases too. At the beginning of this section, it was assumed that γ has a positive value and behaves as a constant at a given temperature. This equation explains why. A liquid consisting of particles with attractive interactions (could it be different?), always disfavors interface formation as it reduces the number of favorable interactions.

The meaning of the intefacial tension becomes even more clear when the property γa^2 [J] is evaluated, which is the energy of a square shaped element of the interfacial area of the size of the particle.

$$\gamma a^2 = -\frac{3}{8}\epsilon a^3. \quad [\mathrm{N\ m^{-1}}] \tag{3.7}$$

Note that γa^2 equals a fraction of the energy of volume a^3. It represents the energy increase of a particle due to its presence at the interface. This connection between the energy density and the interfacial tension is visible in table 3.1. For the given selection of liquids the interfacial tension indeed increased with the vaporization enthalpy per volume, ΔH_{vap} [GJ m^{-3}].

The interfacial tension of argon. At 1 atm the boiling point of argon is −186°C (87 K). Below this temperature liquid argon can coexist with a vapor. Here the interfacial tension of argon will be estimated with the help of equation (3.7).

As input, the following parameters for argon are chosen. For the particle diameter a the Van der Waals radius (the minimal distance that two nuclei can approach each other) of argon is taken: $a = 0.188$ nm. The energy density ϵ is calculated from the heat of vaporization, −6.53 kJ mol^{-1}, and the density, 1.40 gr cm^{-3}: $\epsilon = 228$ MJ m^{-3}. With equation (3.7) and these parameters one finds:

$$\gamma = -\frac{3}{8}\epsilon a = -\frac{3}{8}228 \cdot 10^{6} \times 0.188 \cdot 10^{-9} = 0.016 \text{ N m}^{-1}.$$

A typical value of the interfacial tension at 87 K is 0.012–0.013 N m $^{-1}$. Given the fact that only the energetic part of the interfacial tension is captured by equation (3.7), the predicted value is in good agreement with the measured value. It has to be mentioned that argon was chosen on purpose as it is a monoatomic fluid, which can be described rather well with the interaction potential (3.4).

3.2 Wetting and contact angle

A porous medium can be filled with two immiscible liquids or a coexisting liquid and gas. In these cases there are at least three types of interfaces with different interfacial tensions. As this is of great importance for multiphase flow in porous media, in this section detailed attention is given to the interaction of two phases with a solid surface. Although a gas–liquid system is chosen as a vehicle for the discussion, the outcomes can be directly applied to liquid–liquid systems.

A perfect spherical cap of liquid is placed on a smooth surface, see figure 3.3. Due to the presence of a solid (s), gas (g) and liquid (l),there are three different interfaces. Each of these interfaces experiences a certain interfacial tension and has an interfacial area. For describing the equilibrium shape of the droplet the contact angle θ is the key parameter, as it reflects the balance between the different forces acting on the droplet. The most favorable configuration is related to the minimum in the Helmholtz free energy F, see section 2.2. In the case where solid–liquid interactions are very

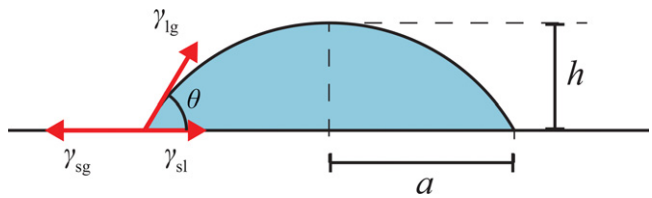


Figure 3.3. A spherical capped droplet on a surface: the actions of the interfacial tensions on the contact line of the droplet is visualized with the arrows. The contact angle θ defines the wettability of the solid surface by the liquid.

Table 3.2. Contact angles of a few liquids on various solid surfaces: PTFE = polytetrafluorethylene (Teflon) and PMMA = polymethylmethacrylate (acrylic glass). All data refer to temperatures between 20 and 25°C. Source: Adamson A W and Gast A P 1997 *Physical Chemistry of Surfaces* 6th edn (New York: Wiley).

Liquid	Solid	θ
mercury	PTFE	150
	glass	128–148
water	PTFE	108, 112
	paraffin	110
	polyethylene	88–103
	stearic acid	80
	human skin	75, 90
	PMMA	60
	platinum	40
	clean glass	small
n-propanol	PTFE	43
	paraffin	22
	polyethylene	7

favorable compared to solid–gas interactions, the droplet will spread and maximize its contact with the solid surface, and the contact angle will be low. The opposite is the case when the solid–liquid interactions are unfavorable. A liquid is called a wetting fluid for a certain solid when $\theta < 90°$ and non-wetting when $\theta \geqslant 90°$.

In table 3.2 the contact angles of mercury, water and n-propanol on various solid substrates are listed. First of all, it is interesting to compare the contact angles of all three liquids on Teflon (PTFE). The contact angle is decreasing with decreasing polarity of the liquid. Teflon is a rather apolar surface that matches better with apolar liquids. Secondly, the table shows that the contact angle of a polar liquid as water decreases with increasing polarity of the substrate. Polar liquids match better with polar substrates. Third, mercury neither wets a polar (glass) nor an apolar (Teflon) surface. The interactions in the mercury are so strong, that even the creation of an interface between mercury and a polar surface is unfavorable.

The relation between the contact angle and the properties of the different interfaces of the droplet can be found by minimizing the free energy of the droplet by adapting its shape ($dF = 0$ and $d^2F > 0$). As the volume and mass of the droplet are conserved and temperature is kept constant, the only change in free energy will be due to changes in the interfacial areas:

$$dF = dF_{sg} + dF_{sl} + dF_{lg} = \gamma_{sg} dA_{sg} + \gamma_{sl} dA_{sl} + \gamma_{lg} dA_{lg}. \quad [\mathrm{J}] \tag{3.8}$$

For simplicity the influence of gravity on the droplet shape will be neglected. In this case the droplet will form a perfect spherical cap with height h [m] and a circular

contact area with the solid having a radius a [m], see figure 3.3. Therefore, the following relationships hold for the droplet volume

$$V = \pi h(3a^2 + h^2), \quad [\mathrm{m}^3] \tag{3.9}$$

and the contact area between the droplet and the solid surface

$$A_{lg} = \pi(a^2 + h^2). \quad [\mathrm{m}^2] \tag{3.10}$$

Further, the radius of curvature of the droplet obeys

$$r = \frac{a^2 + h^2}{2h}, \quad [\mathrm{m}] \tag{3.11}$$

and the contact angle θ is given by

$$\cos\theta = 1 - h/r. \tag{3.12}$$

It holds that $\mathrm{d}A_{sg} = -\mathrm{d}A_{sl}$. With the expressions for the geometry of a spherical cap, equations (3.9)–(3.12), it can be proven that $\mathrm{d}A_{lg} = \cos\theta \mathrm{d}A_{sl}$, see Appendix C. By using these relations in combination with the minimization condition $\mathrm{d}F = 0$, the famous Young–Dupré relation for the contact angle can be derived:

$$\gamma_{sg} = \gamma_{sl} + \gamma_{lg}\cos\theta. \quad \left[\mathrm{N\ m^{-1}}\right] \tag{3.13}$$

The equation can also be interpreted as a force balance equation for the contact line, see figure 3.3, where $\gamma_{lg}\cos\theta$ is the contribution of the liquid–gas interfacial tension to the lateral force. Although, here Young's equation has been derived for a specific situation, a gravity free droplet, it can be applied rather generally. However, this proof is beyond the scope of this book.

Finally, a few words have to be said about θ. Since it has been proven that $\cos\theta \propto \gamma_{sg} - \gamma_{sl}$, the previous definitions of wetting and non-wetting liquids are also clarified. In the case of wetting fluid $\gamma_{sg} > \gamma_{sl}$, which results in $\cos\theta > 0$ and $\theta < 90°$. In the case of wetting fluid $\gamma_{sg} < \gamma_{sl}$, which results in $\cos\theta < 0$ and $\theta > 90°$.

The contact angle of water on glass. Here we analyse the contact angle of water on super clean glass at high and low relative humidity (RH). Where $\mathrm{RH} \equiv 100\rho/\rho^*$ [%] is the ratio between the actual and saturation water vapor densities. For this purpose equation (3.13) will be used.

First, the case of high relative humidity is discussed. At RH > 50% a clean glass surface in contact with air is covered with a liquid-like water layer with a thickness exceeding 1 nm. As the liquid film has two interfaces, the interfacial tension of the solid–gas interface can be approximated as follows: $\gamma_{sg} \approx \gamma_{sl} + \gamma_{lg}$. Combining this with Young's law gives:

$$\cos\theta = \frac{\gamma_{sg} - \gamma_{sl}}{\gamma_{lg}} \approx \frac{\gamma_{sl} + \gamma_{lg} - \gamma_{sl}}{\gamma_{sl}} = 1.$$

The resulting contact angle will be 0° and thus water is perfectly wetting the surface.

In the case of low relative humidity, RH = 0%, the situation is different. There will be no water molecules present at the glass–air interface. As glass and water are both polar, one might expect that γ_{lg}, $\gamma_{sg} > \gamma_{sl}$. For simplicity we assume that $\gamma_{sg} \sim \gamma_{lg}$

$$\cos\theta = \frac{\gamma_{sg} - \gamma_{sl}}{\gamma_{lg}} \sim 1 - \frac{\gamma_{sl}}{\gamma_{lg}} < 1.$$

In this case the contact angle has a finite value: $\theta > 0°$. The wettability of glass by water is reduced.

3.3 Capillary pressure

When an interface is curved, the pressures in the phases separated by the interface differ, see figure 3.4. This pressure difference is called the Laplace pressure p_c [Pa] and obeys

$$p_c = 2\gamma\kappa, \tag{3.14}$$

which is called the Young–Laplace law. In this equation κ [m^{-1}] is the mean curvature of the interface

$$2\kappa \equiv R_1^{-1} + R_2^{-1}, \quad [\text{Pa}] \tag{3.15}$$

where R_1 and R_2 [m] are the radii of curvature of the interface, figure 3.4. These circles defining the curvature radii are oriented perpendicularly to each other.

The Young–Laplace law can be derived by analyzing the forces active on an interfacial element with an area $dA = dl_1 dl_2$ [m^2]. Due to the interfacial tension

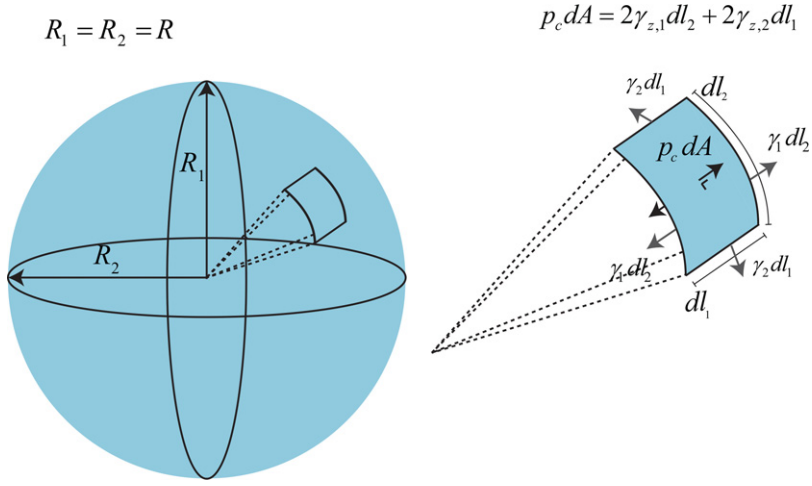


Figure 3.4. Mechanical equilibrium of a curved interface: a spherical droplet is shown that is characterized by two indentical radii of curvature, $R_1 = R_2 = R$. On the right side of the picture the balance between the interfacial tension and the excess pressure in the droplet is visualized.

acting on this element there is small force acting in the direction perpendicular to the interface. This force has to be balanced by a force $p_c \mathrm{d}A$ [N] also acting perpendicularly to this interfacial element, where p_c [N] is the Laplace pressure: the pressure difference between the two phases separated by the interface. This mechanical equilibrium can be described as follows

$$p_c \mathrm{d}A = 2\gamma_{z,1}\mathrm{d}l_2 + 2\gamma_{z,2}\mathrm{d}l_1, \quad [\mathrm{N}] \tag{3.16}$$

where $\gamma_{z,1} = \gamma \sin \vartheta_1$ and $\gamma_{z,2} = \gamma \sin \vartheta_2$ are the interfacial tensions acting in the z-direction on the interfacial element. As a very small interfacial element is considered, the following relations hold $\sin \vartheta_1 \sim \vartheta_1$, $\sin \vartheta_2 \sim \vartheta_2$, $\mathrm{d}l_1 = 2\vartheta_1 R_1$ and $\mathrm{d}l_2 = 2\vartheta_2 R_2$. When these relations are inserted in equation (3.16), the Young–Laplace law is recovered.

To illustrate how the Young–Laplace law can be used, first a spherical droplet with radius R is considered. In such a spherical droplet both radii of curvature are equal: $R_1 = R_2 = R$. When the liquid has an interfacial tension γ, it follows from equation (3.14) that

$$p_c = \frac{2\gamma}{R}. \quad [\mathrm{Pa}] \tag{3.17}$$

As a second example a liquid in a cylindrical pore with radius r is considered. In figure 3.5 two situations are shown: a wetting liquid and a non-wetting liquid in a pore. It follows from mechanical equilibrium that in a wetting liquid the pressure is lowered with an amount $-p_c$ due to the curvature of the interface. In a non-wetting liquid the pressure is increased with an amount $+p_c$. Note that the we have defined the capillary pressure in this way as the pressure in a non-wetting phase p_n minus the

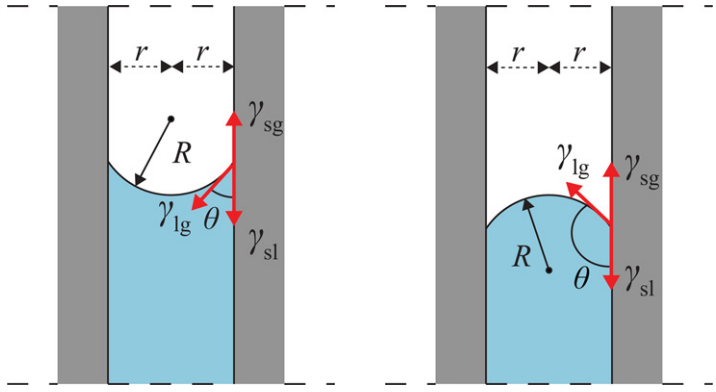


Figure 3.5. A gas–liquid interface in a cylindrical pore: when the liquid is wetting (left), the contact angle is small and the curvature promotes an under pressure $-p_c$ in the liquid. In the case where the liquid is non-wetting (right), the contact angle exceeds 90° and the pressure in the liquid phase is higher than in the vapor phase with an amount $+p_c$.

pressure in a wetting phase p_w: $p_c \equiv p_n - p_w$. The Laplace pressure p_c in a cylindrical pore is now given by

$$p_c = \frac{2\gamma\cos\theta}{r}, \quad [\text{Pa}] \tag{3.18}$$

where γ is the interfacial tension of the interface separating the wetting and non-wetting fluid. Note that in a cylindrical pore the radii of curvature are equal to $R_1 = R_2 = r/\cos\theta$. In the case where interfaces are confined in a pore, Laplace pressure is often called capillary pressure. In the rest of this book we will use the term 'capillary pressure'.

The capillary pressure and pore size. The capillary pressure for water in a cylindrical pore is calculated. Two pore radii are considered: 0.1 mm and 100 nm.

In our calculation we assume that water perfectly wets the pore wall: $\theta = 0$. The interfacial tension of water equals 0.072 N m^{-1}. With equation (3.18) the capillary pressures can be calculated. In the case where $r = 0.1$ mm, $p_c = 2 \times 0.072 \cdot 10^{-4} = 1.44$ k Pa. When the pore radius equals 100 nm, the pressure will be 1.44 M Pa = 14.4 atm. Without proof we state here that this pressure is sufficient to decrease the equilibrium vapor pressure of water with 1%. Although this is still a small number, this effect will become more pronounced when the pore radius is further reduced. This effect will be discussed in section 4.2.

Now the background of the MIP (mercury intrusion porosimetry) measurement for determination of the pore size distribution, discussed in section 1.3, can be understood much better. Due to its strong interactions between the atoms, mercury has a high interfacial tension and contact angle close to 180° for most materials: $p_c = 2\gamma/r$. Therefore, mercury will only enter a pore with a radius r given that the pressure is raised with this amount.

3.4 Stresses in materials

Although mechanics of materials is not the topic of this book, it is interesting to discuss briefly the consequences of capillarity on the structural integrity of materials. A nice example are dykes made of sand and/or earth. It is known that both dry and extremely wet dykes easily collapse. Basically, these dykes consist of particles packed together and disintegration can only be prevented when the particles glue together. The glue is water and the glueing power is related to the strength of the capillary forces. To understand this, we consider two flat parallel plates with area A [m^2] separated by a distance L [m] and the space between the plates is filled with a certain liquid, figure 3.6. The ambient pressure is equal to p_0 [Pa]. Due to the curvature of the meniscus, the pressure in the liquid equals $p_0 - 2\gamma\cos\theta/H$ [Pa]. As the plates are flat, there is no pressure jump over

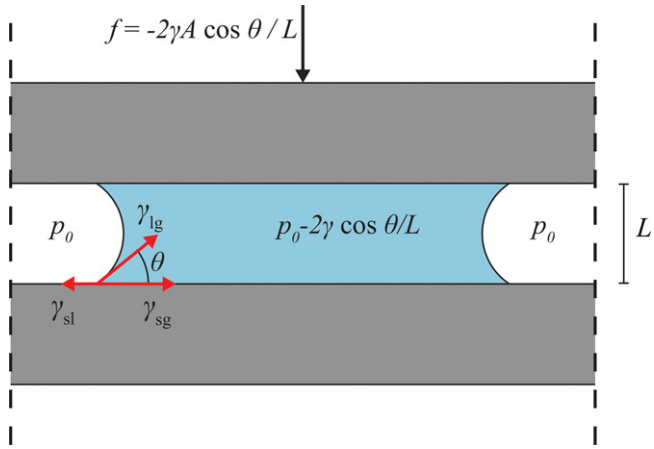


Figure 3.6. Two plates separated by a wetting liquid: Capillarity forces the plates together. Due to capillarity the pressure in the liquid is lower than in the environment. As a consequence, a force f has to be applied to keep the plates in place.

the solid–liquid interface and in equilibrium the pressure just in the plate should equal $p_{\text{plate}} = p_0 - 2\gamma \cos\theta/L$ [Pa]. Obviously, such equilibrium will not exist without intervention as the pressure in the plates no longer matches with the pressure of the ambient air. The only way to keep the plates in position is by applying a force f [N] on the plates:

$$f = A(p_{\text{plate}} - p_0) = -A\frac{2\gamma \cos\theta}{L}. \quad [\text{N}] \tag{3.19}$$

A positive sign is obtained in the case of a non-wetting liquid ($\theta > 90°$), and a compressive force has to be applied, as the liquid tends to push the plates away from each other. A wetting liquid has a small contact angle, $\theta < 90°$, resulting in a negative sign, meaning that a tensile force has to applied, because the liquid pulls the plates together. The latter is what happens in a dyke. Due to the capillary forces generated by the presence of water–air menisci, the sand particles are pulled together.

Generally, the presence of interfaces between two phases in porous materials leads to capillary induced stresses on the solid matrix. When a wetting liquid and a gas are present in a porous system, capillary forces generate a compressive stress on the material. In a similar fashion salt crystallization induces stress on the porous matrix. When ions crystallize, the formed crystal is generally the non-wetting phase. As a consequence there will be over pressure in the crystal, which forces the porous matrix to expand. When these stresses exceed the yield strength of a material, damage can be expected. Failure mechanics is outside the scope this book. However, the next chapter will be used to explain the influence of pore size on phase transitions and the presence of multiple phases.

Lifting a Ferrari F458 with water. Here we explain what you have to do with your Ferrari F458 to lift it with the help of capillary forces: use a 8 micron thick layer of water.

A flat aluminium plate with a roughness less than 1 micron can be connected to a crane. To lift our Ferrari with this device we have to do the following. First remove 1 m^2 coating of the roof of your car in order to uncover the aluminium of the frame. This is to make sure that water wets the surface. The second preparation step consists of polishing the aluminium to obtain a roughness of submicron size. Finally the aluminium should be cleaned with chemicals to remove the dirt that ruins the wetting properties. Let us assume that after this treatment the contact angle of water on the aluminium is 20°. With equation (3.19) we can calculate the mass m [kg] that can be lifted. Here we use $\gamma_{lg} = 0.072$ N m^{-1} and $g = 9.81$ m s^{-2}. A water layer with thickness $L = 8$ micron will be used.

$$m = \frac{2A\gamma \cos\theta}{gL} = \frac{2 \times 1 \times 0.072 \cos(20)}{9.81 \times 8 \cdot 10^{-6}} = 1.7 \cdot 10^3 \text{ kg}.$$

As our Ferrari F458 has a mass of 1565 kg, we will succeed in lifting it.

Further reading

de Gennes P G, Brochard-Wyart F and Quere D 2004 *Capillarity and Wetting Phenomena* (Berlin: Springer) ch 1,2: well written introduction into the field wetting phenomena and capillarity.

Rowlinson J S and Widom B 1982 *Molecular Theory of Capillarity* (New York: Dover): detailed discussion of the microscopic backgrounds of interfacial tension and capillary pressure.

Adamson A W and Gast A P 1997 *Physical Chemistry of Surfaces* 6th edn (New York: Wiley) ch 1–3,10: more than complete, covering the concepts interfacial tensions, contact angles, etc. Besides theory, it gives much information on experimental methods and actual data.

IOP Concise Physics

Fluids in Porous Media
Transport and phase changes
Henk Huinink

Chapter 4

Phase transitions and confinement

4.1 Melting

Matter can exist in different aggregation states (phases): solid, liquid or gas. The transfer of matter from one phase to another is referred to as a phase transition. At specific combinations of temperature and pressure two different phases might coexist separated by an interface. As discussed in the previous chapter, a pressure difference might develop between the two phases due to curvature in the interface. This pressure difference, called capillary pressure, might influence the phase transition.

In this section the melting of a solid confined in a porous material will be discussed. Melting is the transfer of a solid into a liquid; given a certain pressure this happens at a melting temperature T_m [K]. It has to be stressed that in this section the word 'solid' is reserved for solid matter confined inside the pores, and it is not used for the material constituting the porous matrix. As the capillary pressure changes the internal pressures in the two phases, one might expect that the melting temperature is influenced by confining the phases in a pore, see figure 4.1. When the pore is in open contact with an environment with a fixed pressure p_0 [Pa], matter can adapt its volume during the phase transition. Further, it is assumed that the system is in thermal equilibrium with the environment. As the sum of particles in the liquid and solid phase is conserved, $N = \text{const}$, the thermodynamic potential for finding equilibrium is the Gibbs free energy G. At the melting point both the solid and the liquid phases can coexist, and the system is in equilibrium no matter what the distribution of mass over the two phases is. As G is minimal in equilibrium according to the second law, G is constant while transferring the solid into a liquid at the melting point: $\Delta G \equiv G_l - G_s = 0$.

With equation (2.14) it can be shown that for complete melting of a single component solid it holds that

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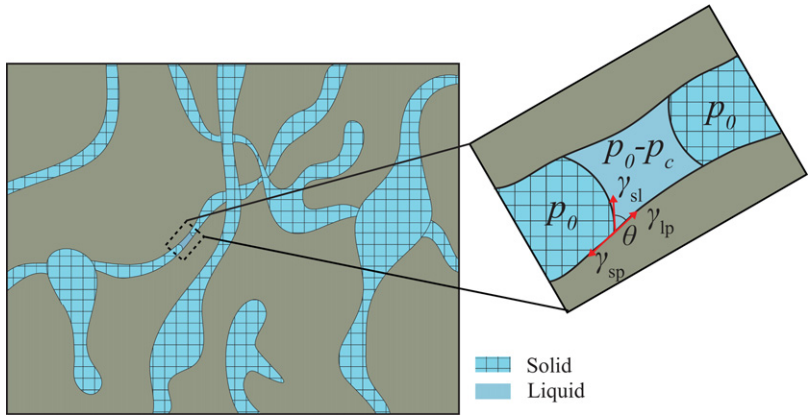


Figure 4.1. Coexisting solid and liquid phases in a porous medium. Enlarged, a pocket of liquid is shown. The liquid preferentially wets the pore surface ($\theta < 90°$), which results in an under pressure in the liquid phase relative to the solid phase. The pressure drop $-p_c$ over the solid–liquid interface increases with decreasing pore size.

$$\Delta G = \Delta\gamma A + \Delta G_b = 0, \quad [\text{J}] \tag{4.1}$$

where $\Delta\gamma \equiv \gamma_{pl} - \gamma_{ps}$ and $\Delta G_b = G_{b,l} - G_{b,s}$. The subscripts p, s and l refer to the material constituting the porous matrix, the solid phase and the liquid phase, respectively.

As the goal is to relate the actual melting temperature in the pore T_m with its value in a bulk system $T_{m,b}$ [K], first a bulk system is analyzed. In a bulk system $\Delta G_b = 0$ at the melting point and therefore

$$\Delta H_b = T_{m,b}\Delta S_b, \quad [\text{J}] \tag{4.2}$$

where the definition (2.12) has been used. Note that $\Delta H_b \equiv N\mathcal{L}$, where $\mathcal{L}$ [J mol^{-1}] is the latent heat and N [mol] is the number of molecules involved. At temperature $T \neq T_{m,b}$, $\Delta G_b \neq 0$ and is equal to

$$\Delta G_b = N\mathcal{L}(1 - T/T_{m,b}). \quad [\text{J}] \tag{4.3}$$

With this general expression for the ΔG_b we can return to the process of melting in a pore.

By combining the expression (4.1) and (4.3), the melting temperature in the pore T_m can be found:

$$T_m/T_{m,b} = 1 + \Delta\gamma A/N\mathcal{L}. \quad [-] \tag{4.4}$$

This expression is the Gibbs–Thomson relation in its purest form. At this stage a few remarks have to be made about assumptions in the derivation of this expression. First, by splitting G in bulk and interface contribution it is assumed that the volume of the interfacial zone is small compared to the total volume (this was already discussed in section 2.2). Second, it is assumed that both the enthalpy and entropy of the bulk phases are constant over the temperature window of interest, which is a

reasonable assumption in the case where the densities of both phases are constant within this temperature window.

To make the connection to the notion of capillary pressure, we introduce the molar density of the liquid phase: $n_l = N/V_l$ [mol m^{-3}]. Therefore,

$$\frac{T_m}{T_{m,b}} \approx 1 + \frac{\Delta\gamma}{\mathcal{L} n_l} \frac{A}{V_l}. \quad [-] \tag{4.5}$$

When Young's equation (3.13) is used to relate $\Delta\gamma$ to the interfacial tension of the solid–liquid interface γ_{sl} and assuming a cylindrical geometry for the pore ($A/V_l = 2/r$), it can be shown that

$$\frac{T_m}{T_{m,b}} \approx 1 - \frac{2\gamma_{sl}\cos\theta}{\mathcal{L} n_l r}, \quad [-] \tag{4.6}$$

where r [m] is the radius of the pore. Interestingly, the pressure jump over the solid–liquid interface (the capillary pressure) emerges naturally in this derivation $-2\gamma_{sl}\cos\theta/r$. In the case where the liquid phase has more affinity with the pore surface than the solid, $\gamma_{lp} < \gamma_{sp}$ and $\theta < 90°$, the melting point is depressed in the pore and melting at lower temperatures is observed. On the other hand, when an increase of the melting temperature is observed, the solid seems to have higher affinity for the pore surface, as compared to the liquid phase.

Equation (4.6) also teaches how bulk properties influence the shift in melting temperature. The melting point of a material is more sensitive to confinement when its latent heat $\mathcal{L}$ is low. The lower the latent heat, the more easily the surface energy can influence the phase transition. To give an impression of the typical values of the latent heat, a list of various materials is shown in table 4.1.

Quantitative comparison of the change in melting temperature predicted by expressions (4.5) or (4.6) with experimental data is difficult, as the quantities $\Delta\gamma$, γ_{sl} and $\cos\theta$ are hardly accessible by experiments. However, the linear relation between the change in melting temperature and inverse pore radius, $\Delta T_m \propto r^{-1}$, has been observed for many liquids, proving the validity of the Gibbs–Thomson relation. In general significant effects are only observed in small pores ($r < 10$ nm).

Table 4.1. Melting enthalpies $\mathcal{L}$ (latent heats), and melting temperatures $T_{m,b}$ of some liquids. Source: CRC 2016 *Handbook of Chemistry and Physics* 96th edn (Boca Raton, FL: CR).

Material	$T_{m,b}$ K	$\mathcal{L}$ kJ mol^{-1}
argon	84	1.18
mercury	234	2.30
water	273	6.01
cyclohexane	280	2.68
glycerol	291	18.3
stearic acid	342	61.2
$CaCl_2$	1048	28.05

Melting of ice in porous glass. With equation (4.6) the change in melting temperature due to confinement can be predicted. Here we want to estimate ice melting in porous glass with a pore radius of 3 nm.

The following numbers will be used as input: $\mathcal{L} = 6.01$ kJ mol^{-1} (the melting enthalpy), $\gamma_{sl} = 0.033$ N m^{-1} (the interfacial tension of the ice–water interface) and $n_l^{-1} = 1.8 \cdot 10^{-5}$ m^3 mol^{-1} (the molar volume of water). Further, we assume that $\theta = 0$. The melting point depression ΔT [K] follows from equation (4.6):

$$\Delta T = -\frac{2\gamma_{sl}}{n_l \mathcal{L} r} = -\frac{2 \times 0.033 \times 1.8 \cdot 10^{-5}}{6010 \times 3 \cdot 10^{-9}} = -18 \text{ K}.$$

This value is in good agreement with the experimental value of −22 K. However, a warning should be given here. Generally, the uncertainty in the input parameters γ_{sl} and r is higher than the difference between the predicted and observed melting point depression. Nevertheless, this example nicely demonstrates that nanopores can depress the melting point of ice significantly.

4.2 Condensation

Another phase transition of interest is the gas–liquid transition. To describe this transition, a single pore is considered that is in open connection with a huge reservoir of a single component gas. As a consequence, the chemical potential μ and molar density ρ_g of the gas in this reservoir are constant, whatever happens in the pore. The pore has a volume V and a surface area A. At a certain combination of μ and temperature T a liquid phase is nucleated in the pore as visualized in figure 4.2. The pore is actually an open system and its thermodynamic state is therefore characterized by the grand potential Ω, which is minimal in equilibrium according to the second Law. Applying this to the situation where the gas and the liquid coexist, the grand potential should not change when the gas is transferred into liquid: $\Delta\Omega \equiv \Omega_l - \Omega_g = 0$. When the pore is completely filled by liquid, one can show that

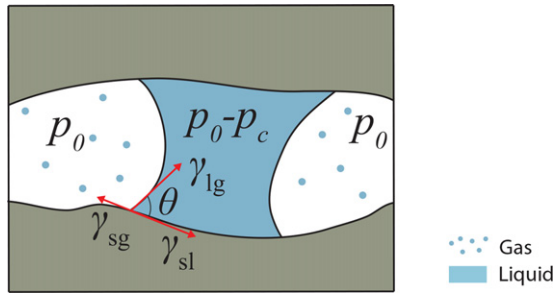


Figure 4.2. Liquid condensation in a porous matrix. The liquid preferentially wets the pore surface, leading to a decrease in the pressure, $-p_c$, in the liquid phase relative to the pressure in the vapor phase.

$$\Delta\Omega = V\Delta p + A\Delta\gamma = 0, \quad [\mathrm{J}] \tag{4.7}$$

where $\Delta p \equiv p_l - p_g$ [Pa] and $\Delta\gamma \equiv \gamma_{sl} - \gamma_{sg}$. Here the subscript s is used for the solid matrix that constituents the porous material of interest. The pressure difference can now be related to the difference in interfacial tensions:

$$\Delta p = \frac{A}{V}\Delta\gamma. \quad [\mathrm{Pa}] \tag{4.8}$$

According to the Gibbs–Duhem expression (2.15) $\mathrm{d}p = n\mathrm{d}\mu$ given that the temperature is constant, where n is the molar density. As a consequence $\mathrm{d}\Delta p = \Delta n\mathrm{d}\mu$, where $\Delta n \equiv n_l - n_g$ is the density difference between a coexisting liquid and gas. Note that these two phases are in equilibrium and therefore have the same chemical potential: $\mu_l = \mu_g = \mu$. Outside the porous matrix the gas condensates at a chemical potential μ^* in a macroscopic sized droplet, where the capillary pressure can be neglected, see equation (3.17). Therefore, μ^* corresponds to $\Delta p = 0$. Using this $\mathrm{d}\Delta p$ can be integrated:

$$\Delta p = \int_{\mu^*}^{\mu} \Delta n\mathrm{d}\mu' \approx n_l(\mu - \mu^*). \quad [\mathrm{Pa}] \tag{4.9}$$

The last term in the expression is obtained by realizing that $n_l \gg n_g$ and using the fact that density of a liquid hardly varies with the pressure.

When the gas behaves ideally, the chemical potential of the ideal gas (2.18) can be introduced in equation (4.9). In this way the gas density at which condensation occurs can be obtained

$$\frac{n_g}{n_g^*} = \exp\left(\frac{A}{V}\frac{\Delta\gamma}{n_l RT}\right), \quad [-] \tag{4.10}$$

where n_g^* is the gas density at which condensation occurs outside the pore. Note that V/A is a measure for the typical dimension of the pore. This equation nicely illustrates the influence of confinement on condensation. When the liquid is the wetting phase, $\gamma_{sl} < \gamma_{sg}$ and $\Delta\gamma < 0$, condensation will occur at a lower vapor density as in the bulk. Further, a vapor will condense at lower vapor densities when the pore size decreases ($A/V\uparrow$).

By using Young's equation (3.13), $\Delta\gamma = -\gamma_{lg}\cos\theta$, and adopting a cylindrical geometry, $A/V = 2/r$, the well-known Kelvin equation for capillary condensation is obtained

$$\frac{n_g}{n_g^*} = \exp\left(-\frac{2\gamma_{lg}\cos\theta}{n_l r RT}\right). \quad [-] \tag{4.11}$$

Note that in the case of water $n_g/n_g^* = \mathrm{R}/100$, where RH [%] is the relative humidity. The equation shows that a gas condensates already at lower densities in a pore in the case where the liquid preferentially wets the pore surface.

Condensation of water in pores. The central question of this intermezzo is: what is the typical pore radius at which water condensates at significantly lower values than RH = 100%?

To answer this question, we will calculate at which pore radius r_e water condensates at RH = $100 \times e^{-1} \approx 37\%$. We will do this at room temperature ($T = 298$ K). Applying this condition to equation (4.11) gives:

$$r_e = \frac{2\gamma_{lg}\cos\theta}{n_l RT}.$$

We assume that the pore wall is perfectly wetted by water: $\theta = 0$. For water $\gamma_{lg} = 0.072\ \mathrm{Nm^{-1}}$ (interfacial tension), $n_l^{-1} = 1.8 \cdot 10^{-5}\ \mathrm{m^3\ mol^{-1}}$ (molar volume). With these parameters the above equation can be used to calculate r_e:

$$r_e = \frac{2 \times 0.072 \times 1.8 \cdot 10^{-5}}{8.31 \times 298} = 1 \cdot 10^{-9}\ \mathrm{m} = 1\ \mathrm{nm}.$$

This calculation clearly demonstrates that capillary condensation only occurs in porous materials with nanopores. In nanoporous materials such as vycor glass, zeolites and porous silica, water indeed condensates at RH < 100%.

As in the case of melting the capillary pressure emerges naturally in the equation as $p_c = 2\gamma_{lg}\cos\theta/r$ in a cylindrical pore. Therefore, capillary condensation can also be understood as a consequence of the fact that the chemical potential of water is lowered due to a pressure drop, which in this case is induced by the curvature of the interface leading to an under pressure $-p_c$ in the liquid phase. Therefore, irrespective of the chosen geometry the Kelvin equation for a wetting fluid can be written as

$$\frac{n_g}{n_g^*} = \exp\left(-\frac{p_c}{n_l RT}\right), \quad [-] \tag{4.12}$$

where $p_c = 2\gamma_{lg}\kappa$ and κ is a measure for the curvature of the meniscus, see (3.15). The advantage of equation (4.12) is that it can be easily applied to many different configurations. However, the approach adopted in this section until now, does not prove its validity.

A few words will be spent to demonstrate how the generic equation (4.12) can be derived. However, the price that has to be paid is that the existence of capillary pressure is assumed beforehand. First, equation (2.15) is used to couple the chemical potential of a wetting liquid with a pressure change in the liquid with an amount: $-p_c$.

$$\mu_l - \mu_l^* = \int_{p*}^{p*-p_c} n_l^{-1}\mathrm{d}p \approx -p_c/n_l \quad [\mathrm{J\ mol^{-1}}] \tag{4.13}$$

In this expression p^* is a reference pressure and μ_l^* is the corresponding chemical potential in the liquid phase. In the second step of the expression a constant density

is assumed. By using the equilibrium condition $\mu_l = \mu_g$ in combination with the equations (4.13) and (2.18) equation (4.12) is obtained.

4.3 Crystallization

Another interesting phenomenon is crystallization from solution. Many solutes crystallize when their concentration increases above a solubility limit: n^*_{sol} [mol m^{-3}]. When a crystal has formed, obviously also an interface between a crystalline phase and the solution is present. As in the case of melting and capillary condensation, such interface plays a crucial role when considering the phase behavior. In this section the formation of a crystal in a pore is discussed. In figure 4.3 a schematic picture is shown of a pore filled with a solution and containing a crystal. Note the geometric model used for the confined crystal is the same as the model used for a confined liquid pocket, see figure 4.2. In this way complicated math is avoided, and the basic physical concepts are easily visible in the theory. In reality, a difference between liquid pockets and crystals is that the latter have well defined geometries due to fact that interfaces with different orientations with respect to the crystal lattice might have different interfacial tensions.

Actually, there is not much difference between capillary condensation and crystallization in the chosen approach. The gas phase is replaced by the solution. The solute particle plays the role of a gas particle in the condensation problem. The solvent is treated as a continuum, which makes it not much different from vacuum. The liquid in the pore is replaced by the crystal.

Applying the same formalism as in the previous section, the pressure difference between the solution and the crystal, $\Delta p \equiv p_{\text{crys}} - p_{\text{sol}}$, can be coupled with the chemical potential of the solute μ and the molar density difference between the solution and crystalline phase, $\Delta n \equiv n_{\text{crys}} - n_{\text{sol}}$ [mol m^{-3}].

$$\Delta p = \int_{\mu^*}^{\mu} \Delta n \mathrm{d}\mu' \approx n_{\text{crys}} (\mu - \mu^*). \quad [\text{Pa}] \tag{4.14}$$

In this equation μ^* is the chemical potential at which the solute crystallizes in a bulk solution. Further, it has been assumed that the solute concentration is low compared to the molar density in the crystal $\Delta n \approx n_{\text{crys}}$. At sufficiently low solute

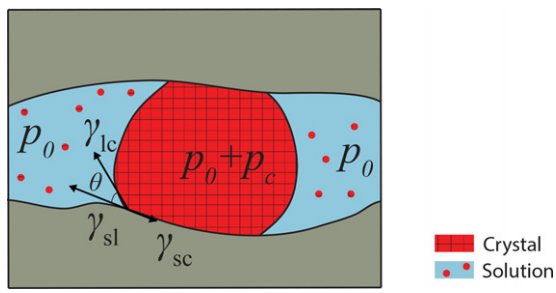


Figure 4.3. Crystallization in a pore from a solution. As the solution preferentially wets the pore surface, the pressure in the crystal is higher than in the solution: $+p_c$. As a consequence the solubility of the crystal is increased. This effect becomes more pronounced when the pore radius decreases.

concentrations the solution behaves ideally and the relation between the chemical potential and the solute concentration is known, see equation (2.21). By combining the chemical potential expression with the equations (4.14) and (4.8), it can be shown how the solubility of the solute changes in a pore:

$$\frac{n_{\text{sol}}}{n^*_{\text{sol}}} = \exp\left(\frac{A}{V}\frac{\Delta\gamma}{n_{\text{crys}}RT}\right). \quad [-] \tag{4.15}$$

The factor V/A is again a measure for the typical dimension of the pore. This expression nicely illustrates that the solubility of a crystal increases when the solution preferentially wets the pore wall: $\Delta\gamma > 0$. The analogy between crystallization and condensation is obvious when equation (4.15) is compared with equation (4.10). As in the case of capillary condensation, crystallization is determined by the wetting properties of the pore wall.

A few experiments have been done to confirm the impact of confinement on the crystal solubility. In these experiments a significant increase in the solubility was found when the typical pore size was in the nanometer range. The observed solubility increase made clear that in these systems indeed the solution preferentially wetted the pore surface.

As in the case of capillary condensation, a generic expression relating the solubility change to the excess pressure in the crystal can be obtained. The derivation is left to reader but should follow the reasoning used to derive (4.12).

$$\frac{n_{\text{sol}}}{n^*_{\text{sol}}} = \exp\left(+\frac{2\kappa\gamma_{sc}}{n_{\text{crys}}RT}\right) \quad [-] \tag{4.16}$$

In this equation γ_{sc} is the interfacial tension of the solution–crystal interface. The term $2\kappa\gamma_{sc}$ is the excess pressure in the crystal p_c, which is sometimes called the crystallization pressure.

The equations derived in this section have to be used with caution. First of all, it has been assumed that the solution phase behaves ideally. Based on this assumption we used equation (2.21) for the chemical potential of the crystallizing species. Nevertheless equations (4.15) and (4.16) are still useful for order of magnitude estimates in case solutions that are no longer ideal. A second point of attention is connected with the considered crystallization process. Until now we have only described the crystallization of a single atom or molecule, see equation (4.14). This approach can be used for describing for example glucose crystallization. However, ionic species cannot be described in this way as they always have to crystallize in neutral pairs; e.g., NaCl crystallization involves two ionic species: Na^+ and Cl^-. We will not redo the derivations presented in this section, but only give the final result for ion crystallization from an ideal solution:

$$\frac{n_{\text{sol}}}{n^*_{\text{sol}}} = \exp\left(+\frac{2\iota\kappa\gamma_{sc}}{n_{\text{crys}}RT}\right). \quad [-] \tag{4.17}$$

In this equation ι is the number of ions making a neutral pair.

Crystallization pressure of sodium carbonate in porous silica. Equation (4.17) can be used to estimate the excess pressure in a crystal confined in pore: $p_c = 2\iota\kappa\gamma_{sc}$. To that end the solubility of the crystal has to be measured. Here we estimate this pressure for Na_2CO_3.

At a temperature of 10°C the solubility increases with a factor 3 when the crystals are confined in porous silica with pore radius of 3.5 nm. The crystal formed is a sodium carbonate hydrate: $Na_2CO_3{\cdot}10H_2O$. The input parameters are: $n_{crys} = 5104$ mol m^{-3} and $\iota = 3$.

$$p_c = \iota n_{crys} RT \ln\left(\frac{n_{sol}}{n^*_{sol}}\right) = 3 \times 5104 \times 8.31 \times 283 \ln 3 = 40.8 \text{ MPa} = 408 \text{ atm}.$$

As discussed in this section, the presented equations have their limitations. Nevertheless, this calculation gives insight into the order of magnitude of the crystallization pressure. We can conclude that this pressure can reach huge values in nano-pores. Although connecting crystallization pressure and material failure is still a formidable task, this calculation shows that salts have the potential to destroy a material.

Further reading

Coussy O 2010 *Mechanics and Physics of Porous Solids* (New York: Wiley) ch 8: thermodynamic derivations of laws for melting point depression, capillary condensation and crystallization in pores.

Evans R, Marconi U M B and Tarazona P 1986 *J. Chem. Phys.* **84** 2376 nonconventional and elegant discussion of capillary condensation.

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Fluids in Porous Media

Transport and phase changes

Henk Huinink

Chapter 5

Pipe flow

5.1 Stokes flow

Before addressing transport in porous media, understanding the much simpler problem of 'pipe flow' is discussed in this chapter. As this problem can be solved analytically it gives much understanding of the influence of confinement on fluid transport, which will be of help when facing the complexity, which is introduced by the geometry of a porous medium. This section is dedicated to an important issue: Stokes flow suffices for most porous media problems.

As a starting point, the motion of a fluid parcel with velocity $\vec{u}$ [m s^{-1}], an infinitely small fluid element, is considered, see figure 5.1. The velocity field can be characterized by a typical value U [m s^{-1}], which could be estimated by a spatial average of $\vec{u}$. It is assumed that the fluid is incompressible and behaves in a Newtonian way. In a Newtonian fluid the local stresses depend linearly on the local velocities, or in other words the dynamic viscosity of the fluid μ [Pa s] is constant. The condition of incompressibility implies that the density of a fluid element is constant. However, density variations on large length- and timescales are allowed! As a rule of thumb, incompressibility holds as long as the Mach number $\mathrm{Ma} \equiv U/c < 0.3$, where c [m s^{-1}] is the speed of sound. In porous media the incompressibility conditions is always met, because the typical fluid velocity is generally many orders of magnitude smaller than c due to the large friction. The motion of the fluid parcel can therefore be described with the Navier–Stokes equation

$$\rho\left(\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u}\right) = -\nabla p + \mu \nabla^2 \vec{u} + \vec{f}_{\mathrm{body}}, \quad [\mathrm{N\ m^{-3}}] \tag{5.1}$$

where p [Pa] is the pressure, ρ [kg m^{-3}] is the mass density and t [s] is time. The term $\vec{f}_{\mathrm{body}}$ [N m^{-3}] is an external force acting on the fluid parcel. If gravity is the only external force present, $\vec{f}_{\mathrm{body}} = -\rho g \vec{e}_z$ with g [m s^{-2}] the acceleration due to gravity and $\vec{e}_z$ [−] the unit vector parallel to the gravity field. The left hand side of the

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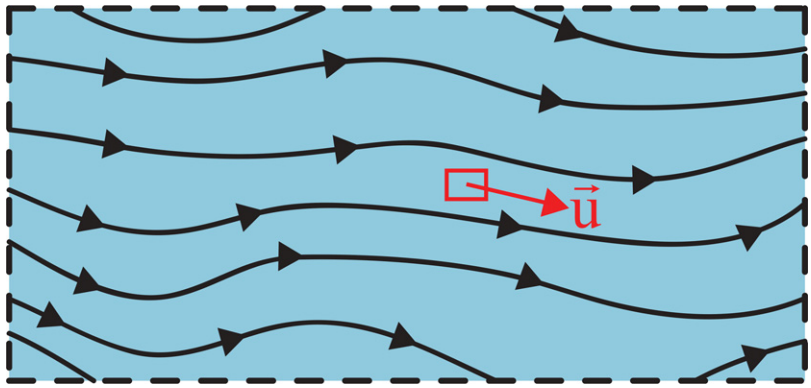


Figure 5.1. A fluid parcel moving through a liquid. The lines with arrow heads represent the streamlines.

equation represents the net force acting on the fluid and leading to acceleration. And as the fluid parcel obeys Newton's laws, the right hand side of the expression is the sum of all forces acting on the fluid parcel. In the rest of the section the body force and the pressure term will be combined $\nabla P \equiv \nabla p - \vec{f}_{\text{body}}$.

Equation (5.1) is still an extremely difficult equation to handle. Therefore, it would be helpful when the left hand side of the equation could be ignored. To that end the Navier–Stokes equation is made dimensionless with $x' \equiv x/L$ [−], $\vec{u}' \equiv \vec{u}/U$ [−], $t' \equiv t/t_s$ [−] and $P' \equiv P/P_s$ [−]. Where L [m], t_s [s] and P_s [Pa] are the scales for, respectively, the dimension, time and pressure characterizing the problem of interest. With these definitions equation (5.1) can be rewritten as

$$Re\left(Sr^{-1}\frac{\partial \vec{u}'}{\partial t'} + \vec{u}' \cdot \nabla' \vec{u}'\right) = -\frac{P_s L}{\mu U}\nabla' P' + \nabla'^2 \vec{u}', \quad [-] \tag{5.2}$$

where $\nabla' \equiv (\partial_{x'}, \partial_{y'}, \partial_{z'})$. It follows from (5.2) that the importance of the acceleration term in the Navier–Stokes equation can be assessed with two characteristic numbers: the Reynolds number

$$Re \equiv \rho U L/\mu \quad [-] \tag{5.3}$$

and Strouhal number

$$Sr \equiv t_s L/U. \quad [-] \tag{5.4}$$

In the case where there is no externally imposed time scale, a natural choice is $t_s = L/U$. In this case $Sr = 1$ and Re is the only dimensionless number that has to be taken into account.

When $Re < 2000$ the acceleration terms of the Navier–Stokes equation can be neglected and one speaks of Stokes' flow or laminar flow. In this regime a relatively simple expression for the motion of a fluid parcel is obtained:

$$-\nabla p + \mu \nabla^2 \vec{u} + \vec{f}_{\text{body}} = 0. \quad [\text{N m}^{-3}] \tag{5.5}$$

The good news is that in porous media we are nearly always in the regime of Stokes' flow. As the typical length scale is of the order of the pore radius, $L \sim r$, and

generally the fluid velocity is also low, $Re \ll 2000$. At the end of the next section a more solid proof for this statement will be given.

Another interesting aspect of the Stokes law is that given a certain pressure distribution, the viscosity rescales the velocity field. A pressure distribution $p(\vec{r})$ is uniquely coupled with $\vec{v}(\vec{r}) = \mu\vec{u}(\vec{r})$. As a consequence, whatever averaged property of the velocity field is calculated, this property is always inversely proportional to the viscosity.

Laminar or not—water in a big pore. Below what velocity U is water still flowing in a laminar mode within in a big pore?

We consider a pore with radius of 0.1 mm, which is a big pore (the typical thickness of a hair). The crossover value for the Reynolds number is 2000. Further, the following numbers are used for the properties of water: $\rho = 1000$ kg m^{-3} and $\mu = 0.001$ Pa s. With the help of equation (5.3) the following calculation can be done:

$$U = \frac{Re\mu}{\rho L} = \frac{2000 \times 0.001}{1000 \times 1 \cdot 10^{-4}} = 20 \text{ m s}^{-1} = 72 \text{ km h}^{-1}.$$

This calculation nicely demonstrates that only at huge velocities might flow become non-laminar in a porous material. In the next section we will estimate what pressures are needed to drive such high speeds.

5.2 Flow through a tube

In this section, the physics of flow in a cylindrical tube with length Δx and radius r will be discussed, as it helps understanding flow in porous media in general. To drive a discharge Q [m^3 s^{-1}] a pressure drop $\Delta p = p_{\Delta x} - p_0$ [N] has to be applied, see figure 5.2. To describe flow in the tube, cylindrical coordinates are handy: x [m] (direction along the tube), R [m] (the radial direction) and φ (the angular coordinate). In this coordinate system $\nabla^2 u_x = R^{-1}\partial_R R\partial_R u_x + R^{-2}\partial_{\varphi\varphi}u_x + \partial_{xx}u_x$.

In the regime of laminar flow $Re \ll 2000$, most of the momentum transfer is along the long axis of the tube ($u_R \approx 0$ and $u_\phi \approx 0$). Making use of cylindrical symmetry, the Stokes equation (5.5) can be rewritten as

$$-\frac{\partial p}{\partial x} + \mu R^{-1}\frac{\partial}{\partial R}R\frac{\partial u_x}{\partial R} + f_x = 0. \quad [\text{N m}^{-3}] \tag{5.6}$$

where f_x [N m^{-3}] is the body force acting along the 'x-direction'. When this expression is solved, a parabolic flow profile is obtained:

$$u_x = -(4\mu)^{-1}(r^2 - R^2)\left(\frac{\partial p}{\partial x} - f_x\right). \quad [\text{m s}^{-1}] \tag{5.7}$$

In the derivation of this equation no-slip boundary conditions have been used: $u_x = 0$ at $R = r$.

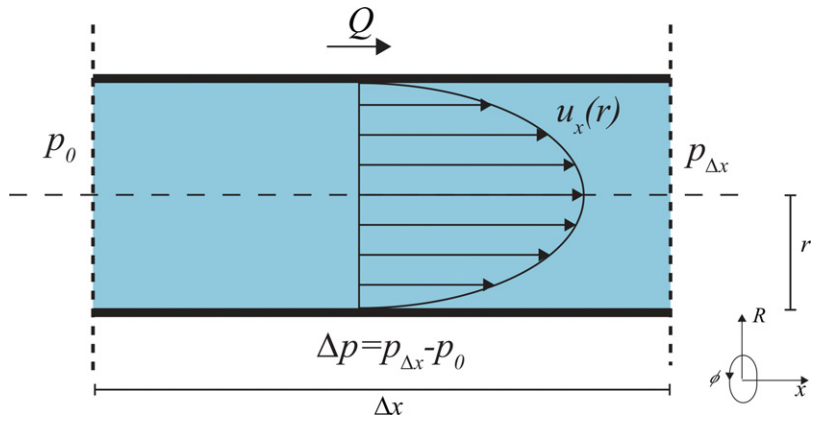


Figure 5.2. The parabolic flow profile in a tube of length Δx. The radius of the tube equals r. A flow Q is driven by a pressure drop Δp. At the surface of the tube no-slip conditions are assumed and the lateral fluid velocity is zero.

By integrating the velocity profile over the cross-sectional area of the tube, the discharge Q can be obtained

$$Q = -\frac{\pi r^4}{8\mu}\left(\frac{\partial p}{\partial x} - f_x\right). \quad [\mathrm{m^3\,s^{-1}}] \tag{5.8}$$

As the fluid is incompressible, the discharge Q is constant and can be related to the pressure drop Δp over the tube

$$Q = -\frac{\pi r^4}{8\mu}\left(\Delta p/\Delta x - f_x\right) \quad [\mathrm{m^3\,s^{-1}}] \tag{5.9}$$

Note that the average fluid velocity equals $U = Q/\pi r^2$ [m $\mathrm{s^{-1}}$]. Further, in the derivation of equation (5.9) the body force is assumed to be constant. Equation (5.9) is the famous Hagen–Pousseuille relation, which is of great value for understanding transport in porous media. It shows that discharge through a pore is extremely sensitive to the pore radius: a factor of 10 in r results in factor 10^4 in Q! As a porous matrix is generally disordered in its structure and pores with different radii exist, this must give rise to large spatial velocity differences. The flow will organise itself along flow paths with the lowest resistance.

With the knowledge of the Hagen–Pousseuille law, typical Reynolds numbers for porous media can now be quantified. In a tube the characteristic length scale is the pore radius r. The scale for the velocity can be based on the average velocity in the tube. By combining (5.9) and (5.3), the following expression can be derived:

$$Re = \frac{\rho r^3}{8\mu^2}\left|\Delta p/\Delta x - f_x\right|. \tag{5.10}$$

To get a feeling for the flow one might expect, the Reynolds number for gravity driven flow is calculated. When the flow direction is chosen parallel to the gravity

field, $|\Delta p/\Delta x - f_x| = \rho g$. For this particular problem the Reynolds'number, equation (5.3), can be calculated with

$$Re = \rho^2 r^3 g/8\mu^2. \tag{5.11}$$

Laminar or not—water in a big pore. In the previous section we have calculated that the typical velocity of water U in a big pore, $r = 0.1$ mm, should be below 72 km h^{-1} (20 m s^{-1}) in order to maintain laminar flow in the pore. Here it will be estimated what pressure drop Δp should be generated to drive such a flow.

We consider water flow in a horizontal tube with length $\Delta x = 1$ m and a radius $r = 0.1$ mm. This represents flow in a porous material with a size of 1 m. Again the following numbers are used for the properties of water: $\rho = 1000$ kg m^{-3} and $\mu = 0.001$ Pa s. With $U = Q/\pi r^2$ equation (5.9) can be rewritten and numbers can be plugged in to obtain an estimate for the pressure drop:

$$|\Delta p| = \frac{8U\Delta x\mu}{r^2} = \frac{8 \times 20 \times 1 \times 0.001}{(10^{-4})^2} = 16 \text{ MPa} = 160 \text{ atm}.$$

This calculation shows that it is extremely difficult to push water flow out of the laminar regime. The needed pressure drop is extremely high. At this point it has to stressed that on field scales (1 km), the pressure drop should be multiplied with a factor 1000!

Further reading

Bruus H 2008 *Theoretical Microfluidics* (Oxford: Oxford University Press) ch 2,3: derivations of the Navier–Stokes equation and Stokes' law for flow at low Reynolds numbers.

Massey B and Ward-Smith J 2006 *Mechanics of Fluids* 8th edn (New York: Taylor & Francis) ch 6,7: extensive discussion of laminar flow problems for various geometries including pipe flow.

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Fluids in Porous Media

Transport and phase changes

Henk Huinink

Chapter 6

Single phase flow

6.1 Darcy's law

This chapter is the first devoted to transport in porous media and focusses on the transport of a single phase. At a first glance, the mission of this chapter is rather simple and a continuation of the previous chapter. Just as in the case of pipe flow: take Stokes' law (5.5), solve it given a certain pore geometry and analyse the outcome. Unfortunately it is not as simple as that, because the complex geometry of the pore space enters via the no-slip conditions the solutions of equation (5.5). Although there have been attempts to approach porous media transport in this way, this book does not follow this approach as the focus is more on elucidating the physical concepts and less on generating detailed solutions. Thereby, these solutions often involve a number assumptions that are difficult to justify and a number of parameters that are difficult to assess by experiments. In this book a phenomenological starting point is chosen: Darcy's law.

The law about flow through porous media, as found by Darcy via experimentation, is the cornerstone in the field of transport in porous media. Darcy's law relates the volume flux $\vec{q}$ [m s^{-1}] with the force driving the flow

$$\vec{q} = -\frac{k}{\mu}(\nabla p + \rho g \vec{e}_z), \quad [\text{m s}^{-1}] \tag{6.1}$$

where k [m^2], μ [Pa s] and ρ [kg m^{-3}] are the permeability of the medium, the fluid viscosity and the fluid mass density, respectively. In the equation the driving force is split into a viscous pressure gradient ∇p and gravity $\rho g \vec{e}_z$. Gravity acts along the minus z-direction. Although this issue will be discussed at the end of the section in more detail, here it is already mentioned that in the case of 1D incompressible flow $\partial_x p = \Delta p / \Delta x$ (Δp [Pa] is the pressure drop over a distance Δx [m]). Below, three aspects of the equation will be discussed: homogeneity, the connection with Stokes' flow and the role of the permeability k.

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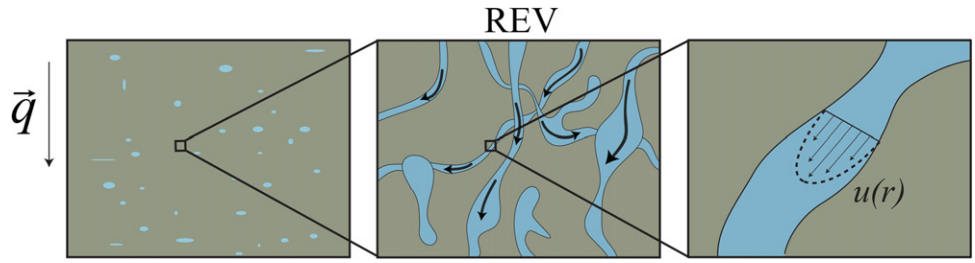


Figure 6.1. A schematic picture of fluid flow at different length scales. The volume flux, as used in Darcy's equation, describes flow on length scales where the material can be considered as being homogeneous (left): length scales exceeding the size of an REV. Fluid velocities fluctuate from pore to pore on length scales below the size of an REV(middle) and even vary within a single pore (right).

Darcy's law describes flow on length scales bigger than the size of an REV, see figure 6.1, where the material can be considered as being homogeneous. This has been discussed in section 1.2. The only parameter connecting the flow problem with material properties is the permeability k. The fact that all information about the complex structure of the pore space can be lumped in a single parameter k, indicates that the law can only be used to describe flow on length scales at which the medium can be considered as homogeneous. Therefore, the volume flux $\vec{q}$ is a quantity that is an average of the velocity field over a volume V bigger than the REV.

$$\vec{q} = V^{-1} \int_V \vec{u}(\vec{x}) \mathrm{d}\vec{x} \quad [\mathrm{m\ s^{-1}}] \tag{6.2}$$

The right hand side of this expression can also be regarded as the liquid content in the considered volume times the average fluid velocity.

Knowing the connection between $\vec{q}$ and the microscopic velocity field, see equation (6.2), the link with Stokes' flow can be made clear. In the case of Stokes' flow the microscopic velocity field at a given pressure distribution can be rescaled by the viscosity, see section 5.2. Darcy observed similar behavior for the volume flux in a porous medium, $\vec{q}\mu \propto \nabla p$, which can only be understood when the microscopic velocity field obeys Stokes' law.

In the previous chapter pipe flow has been considered, which could be described with the Hagen–Pousseuille equation. It is the most simple and idealized porous material and therefore an easy access point to understanding 'permeability'. Consider a rectangular block with area A [m^2] and thickness Δx [m] that is punched with a perfect cylindrical pore with radius r. The volume flux q [m] in the direction parallel to the pore can be derived from equation (5.9):

$$q = Q/A = -\frac{\pi r^4}{8\mu A} \frac{\Delta p}{\Delta x} \tag{6.3}$$

By comparing this expression with Darcy's law, equation (6.1), one can find that

$$k = \pi r^4 / 8A. \quad [\mathrm{m^2}] \tag{6.4}$$

Although, the porous media discussed is not very realistic, this exercise nicely demonstrates that the parameter k, the permeability, is a constant quantifying the

impact of the geometry of the pore space on the flow. The permeability increases both with increasing pore radius r and increasing fraction of the surface area open for transport, $\pi r^2/A$. In the next section the relation between k and the structure of the pore space will be discussed in more detail.

Permeability of polycarbonate membranes. Biochemical laboratories often use polycarbonate filters in their analysis. These filters have cylindrical pores with a well defined pore size. Here the permeability of such a filter is estimated.

For this operation we have to use Darcy's law, equation (6.1). The viscosity of water equals $1.0 \cdot 10^{-3}$ Pa s. The supplier gives in the product brochure the following details. The filter has a thickness of $\Delta x = 11$ micron. Water flows with a volume flux $|q| = 2.17 \cdot 10^{-2}$ m s^{-1} through the membrane at an applied pressure drop $|\Delta p| = 68.9$ kPa. With these numbers the permeability can be estimated:

$$k = \frac{|q|\mu\Delta x}{|\Delta p|} = \frac{2.17 \cdot 10^{-2} \times 1.0 \cdot 10^{-3} \times 11 \cdot 10^{-6}}{6.89 \cdot 10^{4}} = 3.5 \cdot 10^{-15}\ \mathrm{m}^2.$$

After having discussed Darcy's law in detail, attention has to be paid to its application on single phase flow. Darcy's law itself only describes the volume flux, but to obtain a solution it has to be coupled with the local fluid content θ.

This can only be done as the law of mass conservation is understood, which will be introduced in the following intermezzo. Consider a volume V [m^3] containing a certain fluid with a molar density n [mol m^{-3}]. The amount of fluid present in this volume equals $\int_V n\mathrm{d}V$ [mol] and its time variation is connected to the molar fluxes $\vec{J}$ [mol m^{-2} s] through the surface of the volume:

$$\frac{\partial}{\partial t}\int_V n\mathrm{d}V = -\oint_S \vec{J}\cdot\vec{n}\mathrm{d}S. \quad [\mathrm{mol\ s^{-1}}] \tag{6.5}$$

In this equation $\vec{n}$ is the normal of the surface enclosing the volume. Locally the molar flux into the volume equals $-\vec{J}\cdot\vec{n}$. The surface integral on the right hand side of the equation can be written as a volume integral by using Gauss's theorem:

$$\frac{\partial}{\partial t}\int_V n\mathrm{d}V = -\int_V \nabla\cdot\vec{J}\mathrm{d}V. \quad [\mathrm{mol\ s^{-1}}] \tag{6.6}$$

And therefore

$$\frac{\partial n}{\partial t} = -\nabla\cdot\vec{J}. \quad [\mathrm{mol\ m^{-3}\ s}] \tag{6.7}$$

By using the the molar volume υ [m^3 mol m^{-1}] this equation can be rewritten as

$$\frac{\partial\theta}{\partial t} = -\nabla\cdot\vec{q}. \quad [\mathrm{s^{-1}}]. \tag{6.8}$$

Note that $\theta = \upsilon n$ and $\vec{q} = \upsilon\vec{J}$.

Now we have equation (6.8), we can return to the discussion of Darcy's law. In the case of an incompressible fluid the $\partial_t \theta = 0$ and

$$-\nabla \cdot \vec{q} = \frac{k}{\mu} \nabla^2 p = 0. \quad [\mathrm{s}^{-1}] \tag{6.9}$$

The second term in the equation is obtained by incorporation of Darcy's law. It follows from (6.9) that the main task for solving a single phase flow problem is to find solutions for the pressure field: $\nabla^2 p = 0$. Further, it can be concluded that in a 1D situation $\partial_x p = \Delta p / \Delta x$, where Δp is the pressure drop over a distace Δx.

An interesting analog in physics that might help in solving equation (6.9) can be found in the area of electrostatics. The electric displacement field $\vec{D}$ [C m^{-2}] can be written as

$$\vec{D} = \epsilon \vec{E} = -\epsilon \nabla \psi, \quad [\mathrm{C\ m}^{-2}] \tag{6.10}$$

where ϵ [F m^{-1}], $\vec{E}$ [V m^{-1}] and ψ [V] are the dielectric permittivity, electric field and electric potential, respectively. Darcy's relation for the volume flux $\vec{q}$ has a similar mathematical form: ϵ is the analogue of the factor k/μ and ψ has the same function as the pressure p. In a charge free system $\nabla \cdot \vec{D} = 0$ and to find the potential distribution $\nabla^2 \psi = 0$ has to be solved. Note that this is similar to the effect of compressibility resulting in $\nabla^2 p = 0$.

6.2 Permeability and geometry

The permeability k in Darcy's law (6.1) is a material property and reflects the resistance against flow due to the geometry of the pore space. A fluid experiences friction as it is in contact with the internal surface of the porous matrix: the smaller the pore sizes, the larger the internal surface area, the higher the friction and the lower the permeability will be. In table 6.1 the permeabilities of a variety of soils and rocks are given. The values of the different materials vary by orders of magnitude, which is different from the behavior of the porosities, see table 1.1. To understand fluid transport, the connection between flow and the pore geometry have to be understood better. In this section idealized models will be used to investigate how geometry influences the permeability.

The first model consists of identical parallel tubes with a perfect cylindrical shape, see figure 6.2. A porous slab of thickness Δx [m] and area A [m^2] is permeated by N identical cylindrical pores with radii r [m]. Using equation (5.9) for the discharge Q through a single tube it can be found that

$$Aq = NQ = -N \frac{\pi r^4}{8\mu} \frac{\Delta p}{\Delta x}. \quad [\mathrm{m}^3\ \mathrm{s}^{-1}] \tag{6.11}$$

When this equation is rearranged in a Darcy-type of expression, the permeability equals

Table 6.1. Ranges of permeabilities found for different types of soils and rocks. Source: Bear J 1972 *Dynamics of Fluids in Porous Media* (New York: Dover).

Material	k [m^2]
unconsolidated sand and gravel	
well sorted gravel	10^{-7}–10^{-9}
well sorted sand	10^{-9}–10^{-12}
very find sand, silt	10^{-12}–10^{-16}
unconsolidated clay and organic	
peat	10^{-11}–10^{-13}
layer clay	10^{-13}–10^{-16}
unweathered clay	10^{-16}–10^{-20}
consolidated rocks	
highly fractured rocks	10^{-7}–10^{-11}
oil reservoir rocks	10^{-11}–10^{-14}
sandstone	10^{-14}–10^{-16}
limestone	10^{-16}–10^{-18}
granite	10^{-18}–10^{-20}

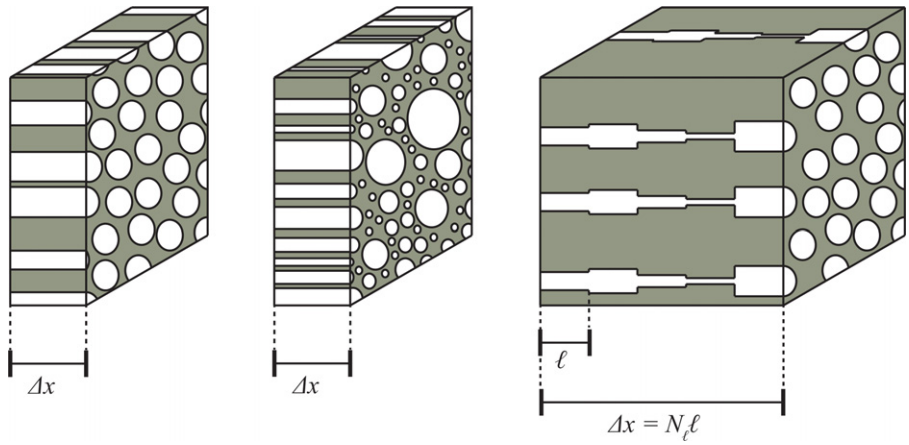


Figure 6.2. Three different models for permeability: identical cylinders (left), cylinders with different pore radii (center) and identical pores consisting of segments with different radii (right).

$$k = \frac{1}{8}\phi r^2, \quad [\mathrm{m}^2] \tag{6.12}$$

where ϕ [–] is the porosity of the matrix: $\phi = V_p/V = N\pi r^2 \Delta x/A\Delta x$. The beauty of equation (6.12) is in its simplicity and its message is very important: materials with the same porosity can still have permeabilities that differ orders of magnitude due to their pore sizes. This is why permeabilities of materials can vary by orders of magnitude as shown in table 6.1.

The number of pores in polycarbonate membranes. In the previous section we have calculated the permeability of a commercially available polycarbonate filter. Here the number of pores per area and the porosity of the same filter will be estimated.

The supplier has one additional property described in his brochure: the pore radius that has a value of $r = 1.0$ micron. Previously, we have calculated the permeability: $k = 3.5 \cdot 10^{-15}$ m^2. With equation (6.12) the porosity can be estimated: $\phi = 8k/r^2 = 8 \times 3.5 \cdot 10^{-15}/(1.0 \cdot 10^{-6})^2 = 0.028$. In the case of parallel identical pores, $\phi = N\pi r^2/A$, where N is the number of pores. With this formula the number of pores per area can be calculated: $N/A = \phi/\pi r^2 = 0.028/(3.14 \times (1.0 \cdot 10^{-6})^2) = 8.8 \cdot 10^9$ m^{-2}.

The second model that is discussed is a variation on the previous model, see figure 6.2. Again a porous slab is permeated by N cylindrically shaped pores, but now each pore 'i' has a different radius r_i [m] and therefore a different discharge Q_i. The total discharge through the porous slab is equal to

$$Aq = \sum_{i=1}^{N} Q_i = -\sum_{i=1}^{N} \frac{\pi r_i^4}{8\mu} \frac{\Delta p}{\Delta x}. \quad [\mathrm{m^3\,s^{-1}}] \tag{6.13}$$

Note that $\sum_{i=1}^{N} r_i^4 = N\overline{r^4}$, where $\overline{r^4}$ is the average value of r_i^4, which in the case of sufficient statistics can be coupled with the pore size distribution 1.10: $\overline{r^4} = \int_0^\infty P(r) r^4 \mathrm{d}r$. Note that this is valid for volumes bigger than an REV. When equation (6.13) is written as the Darcy equation, the permeability turns out to be equal to

$$k = \frac{1}{8}\phi \frac{\overline{r^4}}{\overline{r^2}}, \quad [\mathrm{m^2}] \tag{6.14}$$

where $\phi = N\pi\overline{r^2}/A$. Note that this expression is rather similar to the one derived for identical tubes (6.12), except that the factor r^2 is replaced by $\overline{r^4}/\overline{r^2}$. In statistics, the property $\overline{r^n}$ is called the nth moment of the pore size distribution $P(r)$. In figure 6.3 the location of the $\bar{r}$ (the average pore radius), $\overline{r^2}$ and $\overline{r^4}$ is illustrated. What this figure nicely demonstrates is that the permeability of the material is dominated by the pores with the biggest radii as $\overline{r^4}$ emphasizes this part of the pore size distribution. This is rather obvious. Due to the extreme radius dependency of the discharge Q in a tube, see equation (5.9), the pores with big radii act as short cuts in the system. Although the focus here was on a very specific system, the lesson is more general. In a porous matrix with pore spaces of different dimensions, the fluid will search for flow paths with minimal friction and thus these pathways will be strings of pores with relatively large dimensions compared to the average.

The third model is also a variation of the system of identical pores. A system of N identical tubes is considered, but each individual tube is a string of N_l segments of length l [m]: $\Delta x = N_l l$. Each segment i is a cylinder with an own radius r_i [m] and the discharge of each tube can be described with the equation of Hagen–Pousseuille. As all tubes are equal, the discharge of each tube is Q. As the fluid is incompressible,

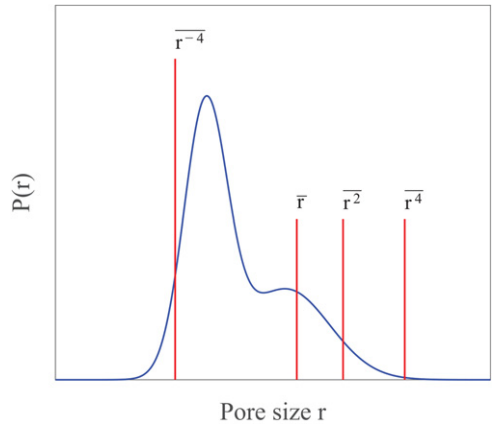


Figure 6.3. An example of a pore size distribution. Various relevant averages of the pore radius are marked with red vertical lines.

the discharge in each individual segment of a tube also equals Q. The pressure drop over a pore segment 'i' follows from equation (5.9) and equals

$$\Delta p_i = -8l\mu Q/\pi r^4. \quad [\text{Pa}] \tag{6.15}$$

The total pressure drop over the porous slab $\Delta p = \sum_{i=1}^{N_l} \Delta p_i$ and therefore

$$\Delta p = -8\Delta x \mu Q \pi^{-1} \overline{r^{-4}}, \quad [\text{Pa}] \tag{6.16}$$

where $\overline{r^{-4}}$ is again an average over the pore size distribution. Using the fact that $Aq = NQ$, this relation can be cast into a Darcy equation, which is left to the reader as an exercise. The resulting permeability obeys the following relationship:

$$k = \frac{1}{8}\phi \frac{1}{\overline{r^{-4}} \cdot \overline{r^2}}. \quad [\text{m}^2] \tag{6.17}$$

In this particular case $\phi = V_p/V = NN_l l\pi\overline{r^2}/A\Delta x$. It follows from the factor $\overline{r^{-4}}$ that all emphasis is here on the pores with small radii, see also figure 6.3. In this configuration small pores dominate the flow behavior as they act as bottlenecks in the flow path. As a consequence of r^4-dependence of Pousseuille flow the major pressure drop in the system will be over a few pore segments. Again there is a more general lesson in this finding. Given a certain flow path in the system, the major contribution to the resistance against flow through this pathway is due to the smallest pores in this flow path.

Having discussed various idealized pore models, a more general but qualitative picture can be synthesized, as illustrated in figure 6.3. In the case of a porous network with disorder in the pore radii, flow paths will avoid the smallest pores and develop through a subset of pores on the higher end of the pore size distribution. Given the existence of these flow paths, the resistance of such an individual flow path will be determined by the smallest pores that are actively engaged in flow.

6.3 Heterogeneity in the permeability

In practice the permeability can vary in space: $k(\vec{x})$. Consider for example the case of concrete, which is a mixture of impermeable inclusions and permeable porous cement, figure 1.1. By using Darcy's law and using the law of mass conservation (6.8), the following expression can be derived

$$\frac{\partial \theta}{\partial t}(\vec{x}) = \nabla \cdot k(\vec{x}) \nabla p(\vec{x}). \quad [\mathrm{s}^{-1}] \tag{6.18}$$

Therefore, to solve the flow profile of an incompressible fluid, the following differential equation has to be solved:

$$\nabla \cdot k(\vec{x}) \nabla p(\vec{x}) = 0. \quad [\mathrm{s}^{-1}] \tag{6.19}$$

As already discussed, similar problems have been faced in other fields: e.g. electrostatics. The dielectric behavior of charge-free heterogeneous media can be described with $\nabla \cdot \epsilon(\vec{x}) \nabla \psi(\vec{x}) = 0$ [C m^{-3}], where ϵ [F m^{-1}] and ψ [V] are the local dielectric permittivity and potential, respectively. The conduction of heterogeneous material can be calculated with $\nabla \cdot \sigma(\vec{x}) \nabla \psi(\vec{x}) = 0$ [S m^{-2}], where σ [S m^{-1}] and ψ [V] are the conductivity and electric potential, respectively.

Here we can directly profit from findings in this field. To calculate the dielectric constant or conductivity of mixtures, effective medium approximations (EMA) have been developed. For spherical inclusions the famous Bruggeman equation can be used, which calculates the average properties of a medium by embedding an object in an average environment. Thanks to the analogy with electrostatics, the Bruggeman equation can also be used to calculate the averaged permeability k_m [m^2] of a mixture of N types of spherical inclusions with different permeabilities:

$$\sum_{i=1}^{N} \varphi_i \frac{k_i - k_m}{k_i + 2k_m} = 0. \quad [-] \tag{6.20}$$

Here φ_i [–] and k_i [m^2] are the volume fraction and permeability of an inclusion of type 'i', respectively ($\sum_{i=1}^{N} \varphi_i = 1$). For a derivation of this equation we refer to Appendix D.

For a binary mixture of impermeable $k = 0$ and permeable $k = k^*$ inclusions, it can be shown that

$$k_m = k^*(3\varphi - 1)/2, \tag{6.21}$$

where φ is the volume fraction of the permeable inclusions. As the permeability cannot become negative, the Bruggeman approach clearly breaks down in the case where $\varphi < 1/3$. However, this is also what makes the outcome interesting as it points to the existence of a percolation transition. Below a certain threshold value of permeable inclusions, the medium becomes impermeable. This can be understood as follows. Permeability demands a system spanning network of permeable inclusions. Below a certain value for the volume fraction the probability to find a system-spanning path of permeable inclusions is close to zero.

Further reading

Quintard M and Whitaker S 2000 Theoretical analysis of porous media *Handbook of Porous Media* ed K Vafai (New York: Marcel Dekker) ch 1: for the tough boys and girls. On volume-averaging techniques to derive macroscopic empirical laws, i.e. Darcy's law, from equations describing transport on the microscopic scale (smaller than the pore size), i.e. the Navier–Stokes equation.

Sahimi M 2011 *Flow and Transport in Porous Media and Fractured Rock: From Classical Methods to Modern Approaches* 2nd edn (New York: Wiley) ch 9,10,12: a detailed overview of various approaches to connect the Darcy equation, basically the permeability, with the microstructure of porous media. The text covers volume-averaging techniques, pore scale modeling and approaches to account for fractures.

Choy T C 2016 *Effective Medium Theory: Principles and Applications* 2nd edn (Oxford: Oxford University Press): fundamental treatment of effective medium theory (EMT) and application to several fields not related to porous media.

IOP Concise Physics

Fluids in Porous Media

Transport and phase changes

Henk Huinink

Chapter 7

Unconfined aquifers

7.1 Dupuit equation

Darcy's law is important for solving problems related to groundwater flow. In this chapter it is shown how Darcy's law can be applied to a specific class of ground water problems: the Dupuit approach for describing the behavior of unconfined aquifers. The aim is twofold: illustrating an application of Darcy's law and demonstrating how a flow problem can be simplified by reducing the number of dimensions. First, it has to be explained what an aquifer is: a water bearing layer in the subsurface. Two types of aquifers are distinguished: confined and unconfined. In figure 7.1 the difference between a confined and unconfined aquifer is illustrated. Confined aquifers are water bearing layers that are sandwiched between impermeable rock layers. In the case of an unconfined aquifer the upper boundary of the aquifer is the so-called water table that can vary in height over time. As the 'layer' on top of the unconfined aquifer is highly permeable the pressure at the water table is equal to the atmospheric pressure. Dupuit developed a theory that greatly simplified the mathematical description of unconfined aquifers.

In this book we discuss the Dupuit approach for the case where the interface between the aquifer and the underlying impermeable rock layer is horizontal and located at $z = 0$. The Dupuit equation describes the spatio-temporal variations of the height of the water table h [m] of an unconfined aquifer:

$$\phi \frac{\partial h}{\partial t} = \frac{k\rho g}{\mu} \nabla \cdot h \nabla h + \mathcal{N}(x, y) \cdot \quad [\mathrm{m\ s^{-1}}] \tag{7.1}$$

In this equation ϕ [–], k [$\mathrm{m^2}$], μ [Pa s], ρ [$\mathrm{kg\ m^{-3}}$] and g [$\mathrm{m\ s^{-2}}$] are the porosity, permeability, viscosity, the fluid mass density and gravity acceleration, respectively. The variable $\mathcal{N}(x, y)$ [$\mathrm{m\ s^{-1}}$] describes the change in the water table due to processes like rainfall, evaporation and transpiration (evaporation of water via vegetation). Further, $\nabla = (\partial_x, \partial_y)$ as flow in the perpendicular direction z is neglected. The term containing the parameters k and μ betrays a connection with Darcy's law. The fact

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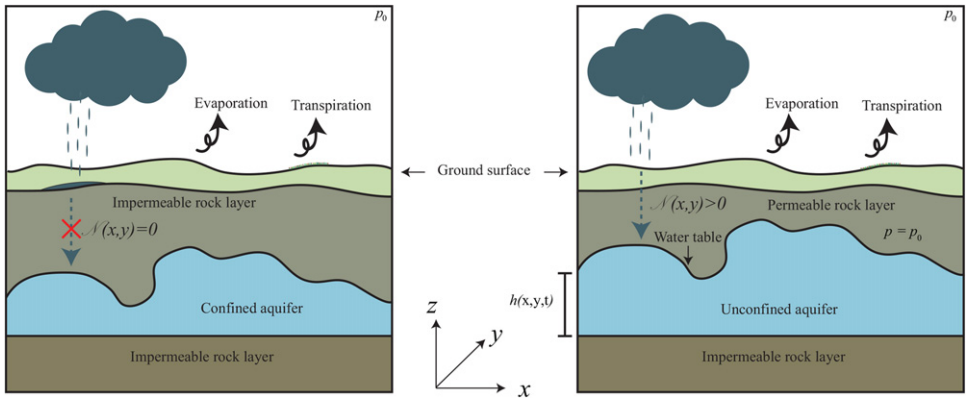


Figure 7.1. Confined (left) and unconfined aquifers (right). Whereas a confined aquifer is covered by an impermeable layer, the unconfined aquifer can communicate via a permeable layer with the atmosphere. Therefore, the height h (the water table) of the unconfined aquifer is influenced by rainfall, evaporation and transpiration via vegetation, mimicked by the variable $\mathcal{N}$. Due to the permeability of the layer on top of the unconfined aquifer, the pressure at the water table equals the atmospheric pressure p_0.

that ∇h drives the flow suggests that gravity is the driving force. An interesting conceptual aspect of the Dupuit approach is that a three-dimensional flow problem has been reduced to a two-dimensional problem only depending on two lateral coordinates: x and y. Below, the derivation of equation (7.1) is discussed in detail.

First of all, it has to be shown that the magnitude of the fluid velocity in the perpendicular (z) direction is small compared to the velocities in the lateral directions. Typical length scales are the lateral dimension of the aquifer L [m] and the typical height of the water table H [m]. The typical velocity scales in the lateral and perpendicular directions are set to U [m s^{-1}] and $V = \varepsilon U$ [m s^{-1}], respectively. The velocities V and U are coupled via $\nabla \cdot \vec{u} = 0, \vec{u} = \vec{q}/\phi$, and the value of ε can be found by making the equation for mass conservation dimensionless:

$$\frac{U}{L}\frac{\partial u_x'}{\partial x'} + \frac{U}{L}\frac{\partial u_y'}{\partial y'} + \frac{V}{H}\frac{\partial u_z'}{\partial z'} = 0, \tag{7.2}$$

where $x' \equiv x/L$ [−], $y' \equiv y/L$ [−], $z' \equiv z/H$ [−], $u_x' \equiv u_x/U$ [−], $u_y' \equiv u_y/U$ [−] and $u_z' \equiv u_z/V$ [−]. It follows from equation (7.2) that

$$V \sim U\frac{H}{L}. \quad [\text{m s}^{-1}] \tag{7.3}$$

As aquifers span large areas, the vertical fluid velocity can be neglected: $H/L \ll 1$. From Darcy's law, equation (6.1) it follows that $V \sim (k/\mu\phi)(|\Delta p_z|/H)$ and $U \sim (k/\mu\phi)(|\Delta p_x|/L)$. Therefore, with equation (7.3) the typical pressure variations in lateral and vertical directions can be compared:

$$|\Delta p_z| \sim \left(\frac{H}{L}\right)^2 |\Delta p_x|. \quad [\text{Pa}] \tag{7.4}$$

This shows that the viscous pressure drop in the z-direction can be neglected compared to the pressure drop in the lateral direction. As a consequence, the

pressures are equilibrated in the z-direction on timescales relevant for flow in the lateral direction. Therefore,

$$p(x, y, z) \approx p_0 + \rho g[h(x, y) - z], \quad [\text{Pa}] \tag{7.5}$$

where p_0 [Pa] is the air pressure at the water table. Note that this expression represents the assumption of local hydrostatic equilibrium.

To derive the Dupuit equation, a column with area $\mathrm{d}x\mathrm{d}y$ [m^2] is analysed, see figure 7.2. The water table in this column is $h(x, y)$. Changes in the water table should be due to flow or processes like rain, etc. The following relation describes the water table:

$$\begin{aligned}\phi \mathrm{d}x\mathrm{d}y\frac{\partial h}{\partial t} = &-\left[Q_x(x + \mathrm{d}x, y) - Q_x(x, y)\right] - \left[Q_y(x, y + \mathrm{d}y)\right. \\ &\left. - Q_y(x, y)\right] + \mathrm{d}x\mathrm{d}y\mathcal{N}(x, y). \quad [\text{m}^3\ \text{s}^{-1}]\end{aligned} \tag{7.6}$$

where for example Q_x [m^3 s^{-1}] is the volume of water passing through the yz-surface of the column per time unit, which can be calculated by integrating the volume flux q_x [m s^{-1}] and applying Darcy's law:

$$Q_x = \mathrm{d}y \int_0^{h(x,y)} q_x(x, y)\mathrm{d}z = -(k/\mu)\mathrm{d}y \int_0^{h(x,y)} \partial_x p \mathrm{d}z. \quad [\text{m}^3\ \text{s}^{-1}] \tag{7.7}$$

With the help of equation (7.4) it can be shown that

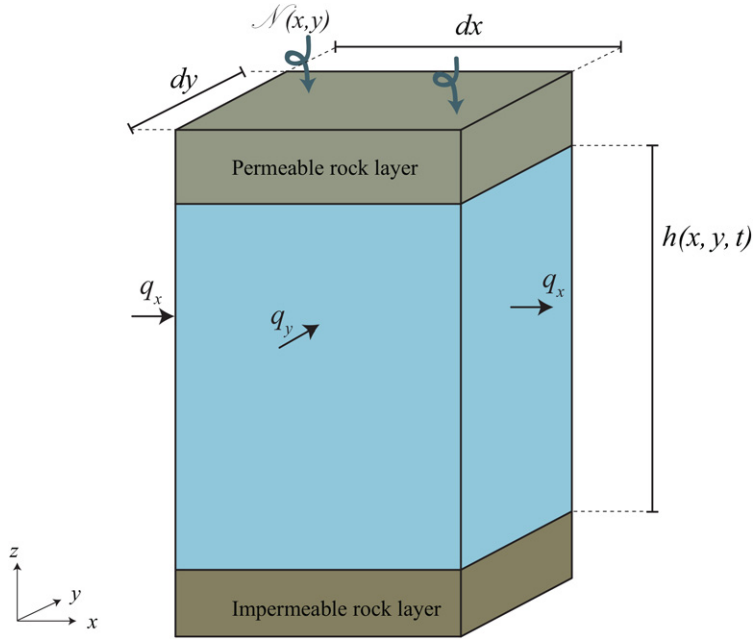


Figure 7.2. A schematic picture of a column in an unconfined aquifer. The height of water table h varies due to volume fluxes $q_{x,y}$ from the sides and influx from the top $\mathcal{N}$ (rainfall, evaporation, transpiration).

$$Q_x = -\frac{k\rho g}{\mu} \mathrm{d}y h \frac{\partial h}{\partial x}. \quad [\mathrm{m^3\ s^{-1}}] \tag{7.8}$$

Finally, this operation has to be done for the volume rates in all directions. When the outcomes are plugged into the mass conservation relation (7.6), the Dupuit equation (7.1) is obtained.

The beauty of this approach is in the reduction of the dimensionality of the problem by analyzing the velocity scales.

7.2 Steady state applications and wells

In some applications the water table remains constant over the time scale of interest: $\partial_t h = 0$. In these cases a steady-state approach can be adopted and equation (7.1) can be rewritten to

$$\frac{k\rho g}{2\mu} \nabla^2 h^2 + \mathcal{N}(x, y) = 0. \quad [\mathrm{m\ s^{-1}}] \tag{7.9}$$

Note that $h\nabla h = \nabla h^2/2$. This equation can be solved given that the function $\mathcal{N}(x, y)$ and the boundary conditions are known.

Here a simple example is discussed. Consider a rectangular piece of land with an area $L_1 \times L_2$ [$\mathrm{m^2}$] that is bounded by two parallel canals, see figure 7.3. Further $L_2 \gg L_1$. The coordinate systems is chosen such that the impermeable layer bounding the aquifer is located at $z = 0$ and that $z = H$ coincides with the surface. The canals are separated by a distance L_1. The canals have a depth of H [m] and the bottom of the canals coincide with $z = 0$. The average rainfall per area is $\mathcal{N}$ [$\mathrm{m\ s^{-1}}$]. Note that the landowner has to pump with a rate of $\mathcal{N} L_1 L_2$ $\mathrm{m^3\ s^{-1}}$ to maintain a stable water table. One could ask under which conditions the channels are sufficient

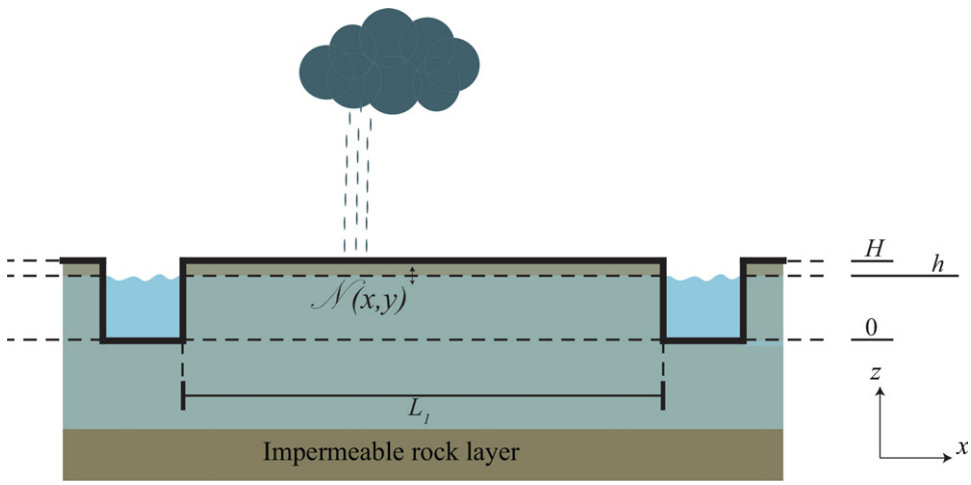


Figure 7.3. Two parallel canals bounding a piece of land.

for preventing flooding: $h \leqslant H$. As $L_2 \gg L_1$ the problem can be reduced to a 1D problem, where the canals are located at $x = \pm L_1/2$. For this particular case the solution of equation (7.9) is

$$h^2(x) = h_c^2 - \left[x^2 - \frac{\mathscr{N}\mu}{4k\rho g} L_1^2\right], \quad [\mathrm{m}^2] \tag{7.10}$$

where h_c [m] is the water level in the canals. The maximum of the water table is located halfway between the two canals at $x = 0$. Therefore, there will be no flooding given that $h(0) \leqslant H$. By combining this condition with equation (7.10) the required water level in the canal can be calculated:

$$h_c \leqslant \sqrt{H^2 - \frac{\mathscr{N}\mu}{4k\rho g} L_1^2}. \quad [\mathrm{m}] \tag{7.11}$$

This equation shows two interesting features. First, the equation shows that above a certain amount of rainfall $\mathscr{N}$ the canals do not function anymore ($h_c = 0$). In this case the landowner is pumping so fast that the canals dry out without preventing flooding in the center of the land. A second feature of equation (7.11) is the role of the permeability k. The more permeable the soil is, the less the variation in the water table. In order words, canals can only have a function in the water management given that the soil is sufficiently permeable.

Finally we want to show how Dupuit's equation (7.1) can be applied to study the flow behavior around wells. Production and injection wells are points in the aquifer where water is extracted and injected, respectively. The problem can be simplified by focussing on steady-state situations in the aquifer, described with equation (7.9).

Water table in between canals. Here a typical Dutch question will be answered. How does the height of the water table in a farmland h (close to Amsterdam) compare with the water level in the canals h_c adjacent to the land?

As in the main text, a piece of land is considered that is bounded by two parallel canals. The canals are separated by a distance $L_1 = 40$ m. The soil has a permeability $k = 1.25 \cdot 10^{-8}$ m^2. For water we take $\mu = 0.001$ Pa s (viscosity) and $\rho = 1000$ kg m^{-3} (density). In January the average rainfall in Amsterdam is $\mathscr{N} = 2.24 \cdot 10^{-8}$ m s^{-1} (6 cm/month). With equation (7.10) we can calculate that

$$\sqrt{h^2 - h_c^2} = \frac{\mathscr{N}\mu}{k\rho g}(L_1/2)^2 = \frac{2.24 \cdot 10^{-8} \times 0.001}{1.25 \cdot 10^{-8} \times 1000 \times 9.81} \times 20^2$$
$$= 0.0085\,\mathrm{m} = 8.5\,\mathrm{mm},$$

where h is taken at $x = 0$. Note that thanks to the high permeability of the soil there is hardly any difference between the height of the water table and the water level in the canal.

Already a few times the analogy with electrostatics has been addressed, and here it is done again as there is nice similarity between the behavior of point charges and wells. The Poisson equation

$$\epsilon \nabla^2 \psi + \varrho = 0 \quad [\mathrm{C\ m^{-3}}] \tag{7.12}$$

relates the electric potential ψ [V] with the space charge density ϱ [C m^{-3}]. In this equation ϵ [F m^{-1}] is the dielectric permittivity. Note that a well is a discrete spot in the aquifer where water is extracted with a volume rate $-Q$ [m^3 s^{-1}] or injected with a rate $+Q$. In electrostatics this corresponds with the presence of discrete charges. In the case of an aquifer with a single well, the potential h^2 driving flow should thus behave as a Coulomb potential in electrostatics. When an aquifer is in contact with a production well with rate $-Q$ and injection well $+Q$, the potential h^2 should mimic the electric potential distribution of a dipole. Further, the space charge density $\varrho(x, y)$ has in electrostatics the same function $\mathcal{N}(x, y)$ in the aquifer.

For a single well with radius a [m] Dupuit derived a relation between the height of the water table and the volumetric production Q [m^3 s^{-1}]:

$$h(r)^2 - h(a)^2 = \frac{Q\mu}{k\rho g} \ln(r/a). \quad [\mathrm{m^2}] \tag{7.13}$$

From this expression it can be seen that with increasing pumping rate Q, a larger zone around the well dries out.

Further reading

Bear J 1972 *Dynamics of Fluids in Porous Media* (New York: Dover) ch 8: an in depth discussion of the Dupuit approximation, covering many examples of unconfined flow.

Freeze R A and Cherry J A 1979 *Groundwater* (Upper Saddle River, NJ: Prentice-Hall) a classical book devoted to the field of groundwater flow and as such giving the context for the topic of this chapter.

IOP Concise Physics

Fluids in Porous Media

Transport and phase changes

Henk Huinink

Chapter 8

Unsaturated flow

8.1 Capillary suction

In practice, flow problems often involve two or more phases. The following two chapters are devoted to the description of multiphase flow. In this chapter, we start this topic by studying the simultaneous flow of a liquid and air, known as unsaturated flow. In this section the most basic form of unsaturated flow is discussed: capillary suction. Capillary suction is the phenomena that a liquid is drawn into a porous medium driven by capillary action and replaces the air present in the material. The primary aim of this section is to elucidate the role of the various forces acting in the problem. For this purpose a simple model is adopted, which assumes that air and liquid are separated by a sharp interface. In the next section attention will be paid to the sharpness of the interface itself.

A dry homogenous porous block with a porosity ϕ [–] is brought in contact with a bath of liquid that is wetting fluid for the material. The liquid spontaneously imbibes the material. The pressure in the air outside the porous block equals p_0 [Pa]. The size of the block in the vertical direction is L [m]. At time t [s] a liquid front has moved into the material and has reached a height $H(t)$ [m], see figure 8.1. The front speed $\mathrm{d}_t H$ is directly connected to the volume flux $q(H)$ [m s^{-1}] at the interface: $\phi \mathrm{d}_t H = q$. Therefore, the front speed can be coupled with the local pressure gradients on both sides of the interface via Darcy's law (6.1):

$$\phi \frac{\partial H}{\partial t} = q(H) = -\frac{k}{\mu_l}\left(\frac{\partial p}{\partial z} + \rho_l g\right)_{z=H^-} = -\frac{k}{\mu_g}\left(\frac{\partial p}{\partial z} + \rho_g g\right)_{z=H^+}, \quad [\mathrm{m\ s^{-1}}] \qquad (8.1)$$

where ρ_l [kg m^{-3}], μ_l [Pas], ρ_g [kg m^{-3}] and μ_g [Pa s] are the mass density of the liquid phase, the liquid viscosity, the mass density of air and the viscosity of air, respectively. As both liquid and air are incompressible at the speeds of interest, the volume flux does not depend on z and the pressure varies linearly in both the liquid and gas phase. As a consequence, $(\partial_z p)_{z=H^-} = \Delta p_l / H$ and $(\partial_z p)_{z=H^+} = \Delta p_g / (L - H)$. At the

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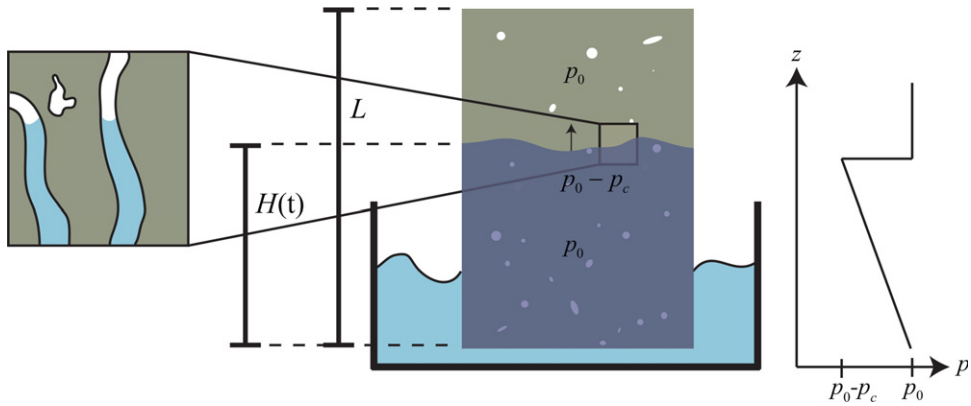


Figure 8.1. Spontaneous capillary suction: the water front is located at $H(t)$. The pressure profile (right) decreases from p_0 in the bath to $p_0 - p_c$ just below the front. Just above the front the pressure jumps back to atmospheric pressure p_0.

liquid front the pressure jumps with an amount p_c and it can be shown that $\Delta p_l = -p_c - \Delta p_g$.

As both the viscosity and density of air are orders of magnitude smaller than their corresponding values for the liquid phase, one might expect that the contribution of air to the motion can be neglected and $\Delta p_l \approx -p_c$ ($|\Delta p_g| \ll |\Delta p_l|$). To justify this, the quantities $\Phi_l \equiv \Delta p_l/H + \rho_l g$ and $\Phi_g \equiv \Delta p_g/H + \rho_g g$ are introduced, which represent the forces needed to drive the flow in the liquid and air phase, respectively. As the ratio of these forces reflects the ratio of the viscosities

$$\Phi_g/\Phi_l = \mu_g/\mu_l \ll 1, \, [-] \tag{8.2}$$

it can be concluded that the pressure drop in the air layer can be neglected: $\Delta p_g \approx 0$, $\Delta p_l \approx -p_c$, $p(H^+) \approx p_0$ and $p(H^-) \approx p_0 - p_c$. Air still plays a role in the value of the capillary pressure via the interfacial tension γ and the contact angle θ.

When the contribution of the air is neglected, equation (8.1) can rewritten as an ordinary differential equation

$$\phi \frac{\mathrm{d}H}{\mathrm{d}t} = \frac{k}{\mu_l}\left(\frac{p_c}{H} - \rho_l g\right). \quad [\mathrm{m\ s^{-1}}] \tag{8.3}$$

This diffential can be solved by using $\int x(1-x)^{-1}\mathrm{d}x = \int -1 + (1-x)^{-1}\mathrm{d}x$:

$$-H/\xi - \ln(1 - H/\xi) = \frac{\rho_l^2 g^2 k}{\phi \mu_l p_c} t. \quad [-] \tag{8.4}$$

In this equation ξ is length scale that measures the distance over which capillary forces can compete with gravity:

$$\xi = p_c/\rho_l g. \, [\mathrm{m}] \tag{8.5}$$

The first interesting limit of equation (8.4) corresponds with $t \to \infty$. In that case $H/\xi \uparrow 1$ as the $-\ln(1 - H/\xi)$ diverges. Assuming cylindrical pores with a radius r,

$p_c = 2\gamma \cos\theta / r$, equation (3.18), the maximum height $H_{\max}$ reached by the front equals

$$H_{\max} = \xi = p_c / \rho_l g \sim 2\gamma \cos\theta / \rho_l g r. \quad [\mathrm{m}] \tag{8.6}$$

The important lesson here is that liquids can rise higher in materials with smaller pores.

Rise height and wetting properties. When water is brought in contact with a cylindrical glass tube, the tube immediately sucks the water. Here the influence of the contact angle of water on glass is estimated.

A cylindrical glass capillary with a radius $r = 0.1$ mm is considered. Water imbibition in this glass capillary is investigated before and after intensive plasma cleaning. Due to the cleaning procedure the contact angle of water decreases from 54 to 15°. The density of water is $\rho = 1000\ \mathrm{kg\ m^{-3}}$. With equation (8.6) we can calculate the rise height before

$$H_{\max} = \frac{2 \times 0.072 \times \cos 54}{1000 \times 9.81 \times 1 \cdot 10^{-4}} = 0.086\,\mathrm{m} = 8.6\,\mathrm{cm}$$

and after cleaning

$$H_{\max} = \frac{2 \times 0.072 \times \cos 15}{1000 \times 9.81 \times 1 \cdot 10^{-4}} = 0.142\,\mathrm{m} = 14.2\,\mathrm{cm}.$$

These calculations demonstrate three interesting features. First, the extreme importance of the wettability of a material for a liquid is visible. Second, contamination can lead to a dramatic decrease of the capability of a material to absorb water. Third, in real porous media the rise height can be of the order of meters depending on the typical pore size.

Another interesting limit of the equation (8.4) is the limit that the front is much lower than its maximal possible value, $H/\xi_g \ll 1$. In this case it can be shown that

$$H^2 = \frac{2kp_c}{\phi\mu_l} t, \quad [\mathrm{m^2}] \tag{8.7}$$

which has been derived using the following expansion $\ln(1 + x) = x - x^2 + O(x^3)$. When the porous materials consists of parallel cylindrical tubes with radii r [m], it can be derived from equation (8.7) that

$$H^2 = \frac{\gamma \cos\theta r}{2\mu_l} t. \quad [\mathrm{m^2}] \tag{8.8}$$

To obtain this equation we have used the equation for permeability (6.12) and capillary pressure (3.18). Equation (8.8) is known as the Lucas–Washburn equation and nicely demonstrates how the initial front motion is influenced by the pore radius. The bigger the pore radii of a material, the faster the motion of the liquid front.

Here we conclude that an important lesson on porous media transport has been learnt by analyzing the front height of a liquid sucked into a material by capillary action. This lesson is summarized in figure 8.2, where the rise height H, equation (8.4), is plotted as a function of time for systems with different pore sizes. As before it is assumed that the pores are cylindrical: $p_c = 2\gamma \cos\theta / r$ and $k = \phi r^2/8$. Small pores favor high rising as the capillary pressure is high in these pores. Big pores favor high initial front speeds as the friction is low due to the low surface-to-volume ratio.

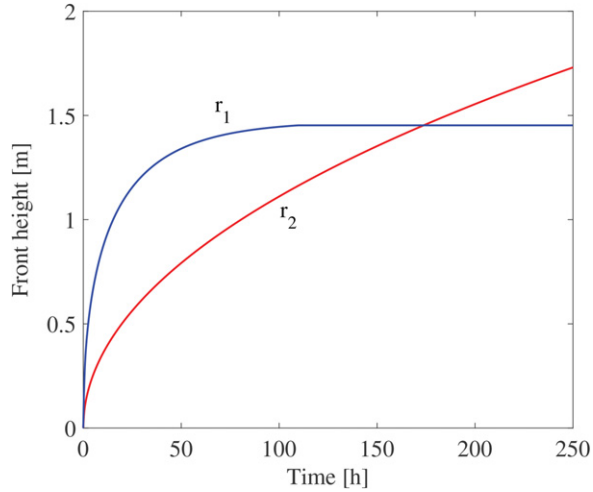


Figure 8.2. The time variation of the front position H. Curves for materials with big pores r_1 and small pores r_2 are shown. Big pores promote fast initial uptake, but quickly lose the competition with gravity.

Rise time of water. Here we calculate how fast a porous medium is saturated, when it is brought into contact with water.

Two materials with typical pore radii of 0.1 and 10 microns are compared. For simplicity we assume that the pores are perfect cylinders. Both materials are cubes with a typical size of $L = 10$ cm. Water perfectly wets the pore surface, $\theta = 0°$. For the properties of water we take $\gamma = 0.072$ N m^{-1} (interfacial tension) and $\mu = 0.001$ Pa s (viscosity). The Washburn equation is used to estimate the rise times of both materials. With equation (8.8) one finds for $r = 0.1$ micron

$$t = \frac{2\mu L^2}{\gamma r} = \frac{2 \times 1.0 \cdot 10^{-3} \times (0.10)^2}{0.072 \times 1.0 \cdot 10^{-7}} = 27.8\,\text{s} = 0.46\,\text{min}$$

and for $r = 10$ micron

$$t = \frac{2 \times 1.0 \cdot 10^{-3} \times (0.10)^2}{0.072 \times 1.0 \cdot 10^{-5}} = 2.78 \cdot 10^3\,\text{s} = 46\,\text{min}.$$

The typical rise time of water in sintered Al_2O_3 ($L = 5$ cm and $r = 0.1$ micron) is about 40 min. So the order of magnitude of our predictions is good. Further, these calculations nicely demonstrate that the rate of liquid absorption strongly increases with the pore radius. Despite the fact that the capillary force drops with increasing pore radius, the friction with the pore surface drops too.

8.2 Beyond the sharp front approach

In the previous section a sharp front between liquid and air has been assumed in the analysis of capillary suction: the liquid content was either equal to the porosity ϕ or to 0. However, it has been found experimentally that the transition between the liquid saturated part of the material and the air saturated part of the material is more gradual, see figure 8.3. In this section we will address the issue of unsaturated flow and our vehicle in the discussion is capillary suction, meaning that we discuss the problem of a wetting liquid intruding in a porous matrix. Despite the specific application to capillary suction, the concepts introduced in this section have a much wider applicability.

At first, the saturation S [−] is introduced to describe the fluid content in porous material and to characterize the state of the porous material:

$$S \equiv \theta/\phi. \quad [-] \tag{8.9}$$

Since S is the fluid content scaled on the porosity, it is a parameter varying between 0 and 1. In an unsaturated medium, $S < 1$, filled and empty pores coexist. In the case of wetting liquid first the small pores are filled, and the saturation S can be coupled with the pore size distribution via

$$S(r) = \frac{\mathscr{L}_p'}{V_p'} \int_0^r P(r)\pi r'^2 \mathrm{d}r', \quad [-] \tag{8.10}$$

where V_p' [m^3] and $\mathscr{L}_p'$ [m] are the total accessible pore volume and pore length. Analog to equation (1.7), $\mathscr{L}_p' \int_0^r P(r)\pi r'^2 r'$ is the pore volume filled with the wetting fluid. As a non-wetting fluid wants to minimize the contact with the pore surface, a non-wetting fluid first fills the big pores and the fluid saturation can be found with

$$S(r) = \frac{\mathscr{L}_p'}{V_p'} \int_r^\infty P(r)\pi r'^2 \mathrm{d}r'. \quad [-] \tag{8.11}$$

Note that a cylindrical pore shape has been assumed. Both equations postulate a unique relation between the saturation S and the pore radius of the maximal or

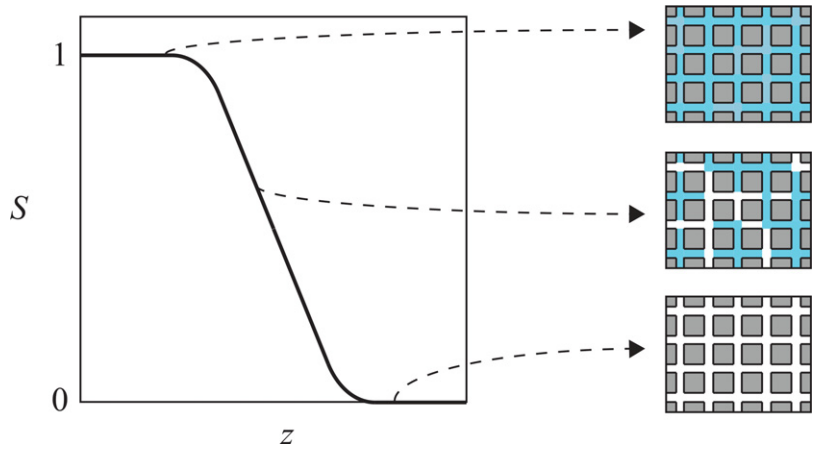


Figure 8.3. The liquid saturation profile $S(z)$ during capillary suction. On the right, schematic pictures of the pore network are shown for different degrees of saturation. It has to be mentioned that the actual pore sizes of the pores are not visualized in these pictures. At low saturation mainly the small pores are filled (the liquid is the wetting phase). With increasing saturation the big pores will also be filled with the wetting liquid.

minimal pore filled by the fluid. Actually, it is assumed that there is local mechanical equilibrium and that the state of the system can be characterized by S.

The equation of mass conservation, equation (6.8), serves as a starting point for the description of the capillary suction process:

$$\phi \frac{\partial S}{\partial t} = -\nabla \cdot \vec{q}. \quad [\mathrm{s}^{-1}] \tag{8.12}$$

The volume flux $\vec{q}$ [m s^{-1}] is described again with Darcy's law, equation (6.1). However, both the permeability and the pressure field will be effected by the saturation S. By combining Darcy's law with the law of mass conservation the Richards equation is obtained

$$\phi \frac{\partial S}{\partial t} = \nabla \cdot \frac{k(S)}{\mu} \left[\nabla p(S) + \rho g \vec{e}_z \right], \quad [\mathrm{s}^{-1}] \tag{8.13}$$

which describes the time evolution of the liquid distribution in a porous material. The subscripts referring to the liquid and air phases have been dropped, because the role of air is completely ignored based on the analysis in the previous section. Therefore, μ and ρ refer to the viscosity and density of the liquid phase. In the Richards equation the pressure in the liquid phase $p(S)$ and the local permeability $k(S)$, are both unique functions of S. This indicates that local equilibrium is assumed and that this equilibrium state of the porous matrix can be characterized by the saturation S. In the subsequent paragraphs the pressure and the permeability will be discussed in more detail.

First, the pressure term in equation (8.13) will be addressed. As said before, the air phase in the porous matrix is assumed to be in equilibrium with the environment of the system and equals p_0. As discussed in the previous section, the pressure drop in

the air phase can be neglected as the viscosity of air is very low. When local mechanical equilibrium is assumed, the pressure in the liquid phase is $p = p_0 - p_c$. In equilibrium there is a unique relation between the fluid saturation S and the capillary pressure $p_c(S)$, $p = p_0 - p_c(S)$. Therefore, the Richards equation can be rewritten as

$$\phi \frac{\partial S}{\partial t} = -\nabla \cdot \frac{k(S)}{\mu}\left[\nabla p_c(S) - \rho g \vec{e}_z\right]. \quad [\mathrm{s}^{-1}] \tag{8.14}$$

The good news is that unique relations between p_c and S have been determined experimentally. The bad news is that often this relationship is not as unique as wanted, meaning that hysteresis effects are observed. However, for building a theoretical framework and illustrating the physical concepts, it is extremely helpful to assume that p_c is a unique function of S. This assumption will be used in the rest of this book. Finally, a few words have to be devoted to the shape of the function $p_c(S)$, which is illustrated in figure 8.4. It follows from equation (8.10) that at a given value of S all pores with a radius smaller than r are filled. Therefore, S is a monotonously increasing function of r. Since the p_c decreases with r, see for example equation (3.18), p_c will be a monotonously decreasing function of S.

The other quantity in the Richards equation that needs discussion is $k(S)$. Obviously, the permeability should increase with the saturation, see figure 8.5. Liquid can only flow through a network of connected liquid-filled pores, filled edges and films. When S increases, the amount of liquid filled pores and the amount of liquid present in edges increases, and the network facilitating flow densifies. As a result the permeability increases. In the Richards equation, S is a state variable that characterizes the local structure and thereby the permeability, and a unique relation between k and S is assumed. Despite the fact that the experimental literature has

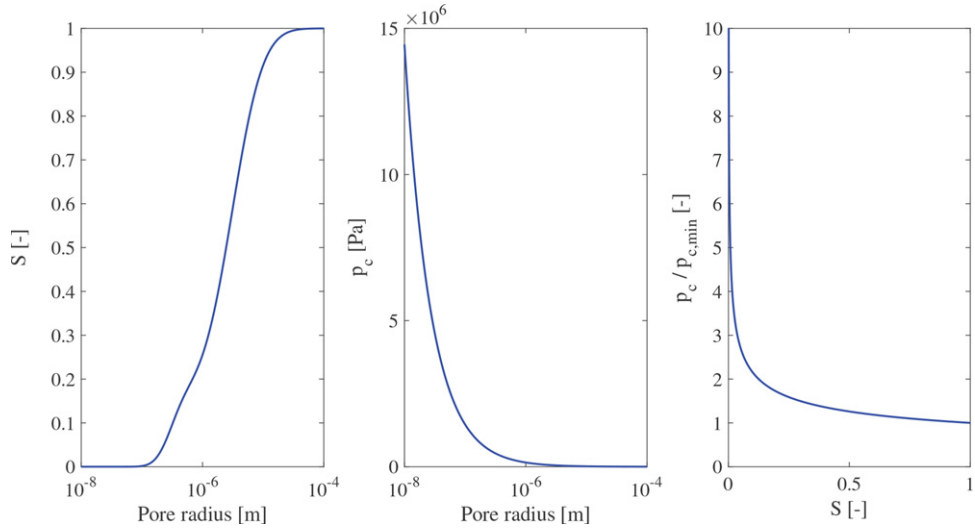


Figure 8.4. The relation between saturation S, capillary pressure p_c and pore radius r. To calculate these curves the pore size distribution, as shown in figure 6.3, is used in combination with $p_c = 2\gamma/r$ ($\theta = 0°$).

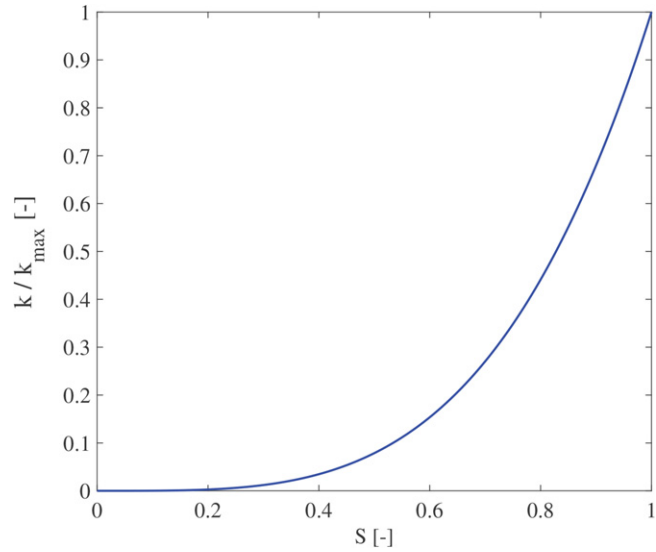


Figure 8.5. The permeability k as function of the saturation S. The curves has been calculated with the Brooks–Corey relation, equation (8.18), with $\lambda = 3$.

regularly reported deviations from this behavior, it is a fruitful approach for modeling. Therefore, this approach is adopted in the rest of this book.

When the gravity can be neglected, $\xi/L > 1$, see equation (8.5), the Richards equation can be written as a non-linear diffusion equation,

$$\phi \frac{\partial S}{\partial t} = \nabla \cdot D(S) \nabla S, \quad [\mathrm{s}^{-1}] \tag{8.15}$$

where the diffusivity $D(S)$ [$\mathrm{m}^2\ \mathrm{s}^{-1}$] equals

$$D(S) = -\frac{k(S)}{\mu\phi} \frac{\partial p_c}{\partial S}. \quad [\mathrm{m}^2\ \mathrm{s}^{-1}] \tag{8.16}$$

It has to be stressed that equation (8.15) does not describe a diffusion process, but only makes clear that the Richards equation mathematically behaves as a diffusion equation.

Many empirical relations for $p_c(S)$ and $k(S)$ have been put forward in the literature. As an illustration, simplified versions of the Brooks–Corey relations are used:

$$p_c(S) = p_{c,\min} S^{-1/\lambda}, \quad [\mathrm{Pa}] \tag{8.17}$$

and

$$k(S) \equiv k_{\max} k_r(S) = k_{\max} S^{(2+3\lambda)/\lambda}, \quad [\mathrm{m}^2] \tag{8.18}$$

where $p_{c,\min}$ [Pa] and $k_{\max}$ [m^2] are the capillary pressure and permeability at $S = 1$, respectively. Note that $k_{\max}$ is the permeability in the presence of single phase.

Further, $k_r(S)$ [–] is the relative permeability and $k_r(1) = 1$. The parameter λ is called the pore size distribution index ($\lambda > 0$), which is a measure for the width of the pore size distribution. In practice λ varies between 1 (a broad pore-size distribution) and 10 (a narrow pore-size distribution). In the case of a broad pore size distribution, p_c varies strongly with S. When the pore size distribution is narrow, the capillary pressure is less sensitive to S. Therefore, it follows from equation (8.17) that λ decreases with increasing width of the pore size distribution. By incorporating the equations (8.17) and (8.18) into the diffusivity, equation (8.16), it can be shown that

$$D(S) = D_{\max} S^{(1+2\lambda)/\lambda}, \quad [\mathrm{m^2\ s^{-1}}] \tag{8.19}$$

where

$$D_{\max} = k_{\max} p_{c,\min} / \mu\phi\lambda \sim \gamma \cos\theta r / 4\mu\lambda. \quad [\mathrm{m^2\ s^{-1}}] \tag{8.20}$$

Note that $p_{c,\max} \propto \gamma \cos\theta$, where γ [N m^{-1}] is the interfacial tension of the liquid and θ the contact angle. The last term of the expression can be obtained by assuming parallel cylindrical pores: $k_{\max} = \phi r^2/8$ and $p_{c,\min} = 2\gamma \cos\theta / r$. This is useful for estimating typical values of $D_{\max}$.

In the previous section it was found that in the case of capillary suction the speed of a liquid front increases with the pore radius of a material as reflected by the equations (8.7) and (8.8). Despite the fact that we did not solve the Richards equation in this section, the diffusivity expressions (8.19) and (8.20) indicate that similar behavior is to be expected. The diffusivity D increases with S as the flow network of the liquid densifies and that $D_{\max} \propto r$, which are both consequences of the fact that the friction decreases with S as the area per volume decreases. Note the similarity between the proportionality between equation (8.8) and $D_{\max}$.

The maximal diffusivity of water in bricks. Here the $D_{\max}$ of water in a fired-clay brick is estimated.

The material has a typical pore radius of 10 micron and pore size distribution index $\lambda = 2$. The contact angle of water is 40°. For the properties of water we take $\gamma = 0.072$ N m^{-1} (interfacial tension) and $\mu = 0.001$ Pa s (viscosity). Equation (8.20) is used:

$$D_{\max} = \frac{0.072 \times \cos 40^\circ \times 1.0 \cdot 10^{-5}}{4 \times 1.0 \cdot 10^{-3} \times 2} = 6.9 \cdot 10^{-5}\ \mathrm{m^2\ s^{-1}}.$$

8.3 Capillary suction revisited

Having introduced the formalism for describing unsaturated flow, the issue of capillary suction can be addressed again. An important aim of this section is to demonstrate with the example of capillary suction, the practical use of the non-linear diffusion equation. Here we discuss liquid uptake by a porous medium of infinite dimensions in 1D, see figure 8.6. In this case equation (8.15) can be rewritten to

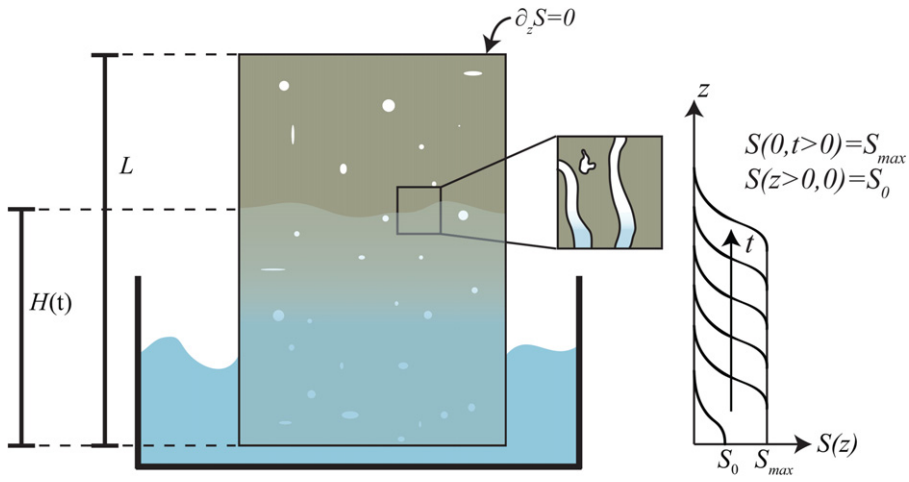


Figure 8.6. Capillary suction and unsaturated flow. Saturation curves are shown on the right.

$$\phi\frac{\partial S}{\partial t} = \frac{\partial}{\partial z}\left(D(S)\frac{\partial S}{\partial z}\right). \quad [\mathrm{s}^{-1}] \tag{8.21}$$

The boundary conditions are $S(0,\, t > 0) = S_{\max}$ and $\partial_z S|_{z=\infty} = 0$ (no flux). The initial condition is $S(z > 0,\, 0) = S_0$. Further, we define: $\Delta S(t) \equiv S(t) - S_0$ and $\Delta S_{\max} \equiv S_{\max} - S_0$. Note that by using this approach the influence of gravity is ignored.

The beauty of the case under discussion is that the saturation profiles behave self similarly, which can be demonstrated by introducing a new variable $\xi \equiv z/\sqrt{t}$ $[(\mathrm{m\ s}^{-1})^{1/2}]$. With this variable the partial differential equation (8.21) can be transferred into a non-linear differential equation:

$$-\frac{1}{2}\xi\frac{\mathrm{d}S}{\mathrm{d}\xi} = \frac{\mathrm{d}}{\mathrm{d}\xi}D(S)\frac{\mathrm{d}S}{\mathrm{d}\xi}. \quad [-] \tag{8.22}$$

This differential equation will give a solution for the saturation profile of the form $S(\xi)$. NMR imaging experiments have shown indeed, that saturation profiles have this self-similar behavior.

Due to the non-linearity of $D(S)$ equation (8.22) has to be solved numerically in most cases. In figure 8.7 a numerical solution for a Brooks–Corey type of D, equation (8.19), is compared with an analytical solution for a constant D. The impact of the strong increase of $D(S)$ is clearly visible in figure 8.7. The shape of the saturation changes from convex at constant diffusivity to concave in the Brooks–Corey case. Due to the low diffusivity at low saturations, the fluid motion increases at higher saturations leading to a more block shaped profile.

Based on this fact, it is easy to estimate an average front position H [m] via the volume V [m^3] intruding into the material:

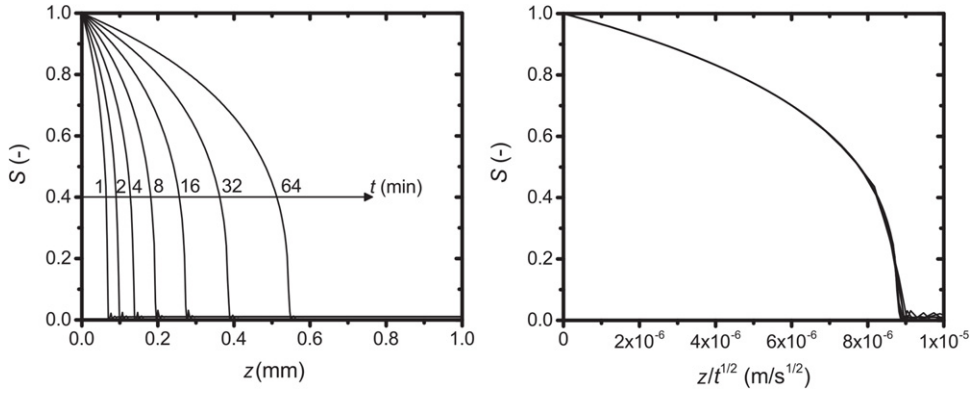


Figure 8.7. The time evolution of the saturation profiles during spontaneous imbibition purely driven by capillary forces (left). The self-similarity of the profiles is demonstrated by plotting the profiles as a function of $z/\sqrt{t}$ (right). The profiles have been calculated with equations (8.21) and (8.19): $D_{\max} = 1.0 \cdot 10^{-10}$ m^2 s^{-1}, $\lambda = 1$, $S(0, t) = 1$ and $S(z, 0) = 0.01$.

$$H = \frac{V(t)}{\phi A \Delta S_{\max}} = \sqrt{t}\,\Delta S_{\max}^{-1} \int_0^\infty \Delta S(\xi) \mathrm{d}\xi, \quad [\mathrm{m}] \tag{8.23}$$

where A [m^2] is the surface area of the material through which the liquid enters. As the saturation profiles are self-similar, the $t^{1/2}$ time dependence of the saturation profiles and the average front position easily emerges. Note that the same time behavior was obtained from a sharp front analysis, equation (8.8). The $t^{1/2}$ shows that the front motion slows down, which is a consequence of the fact that the friction increases while the front progresses ($\propto H$) and pressure difference for driving the front remains constant.

With a slight adaption of the coordinate ξ, an even more meaningful result can be obtained. As visible in the Brooks–Corey-type of relations, for example equation (8.19), $D(S) \equiv D_{\max} F(S)$. With $\xi' \equiv \xi/\sqrt{D_{\max}}$ [−] it can be shown that

$$H = \sqrt{D_{\max} t}\,\Delta S_{\max}^{-1} \int_0^\infty \Delta S(\xi') \mathrm{d}\xi' \sim \sqrt{\frac{\gamma \cos\theta r t}{4\mu\lambda}}\,\Delta S_{\max}^{-1} \int_0^\infty \Delta S(\xi') \mathrm{d}\xi', \quad [\mathrm{m}] \tag{8.24}$$

where the last term has been obtained by using the Brooks expression (8.20) for cylindrical pores. At this point it has to be stressed that generically, whatever approach is used, always $D_{\max} \sim \gamma \cos\theta r/\mu$. Thanks to the operation $\xi' \equiv \xi/\sqrt{D_{\max}}$ the resemblance with equation (8.8) is even clearer. Both the liquid's properties and the geometry of the pore space affect in the same way the predicted front motion.

The last part of this section is used to show how $D(S)$ can be obtained from experiments. With NMR imaging or neutron tomography, liquid distributions can be obtained during the uptake process. It follows from equation (8.22) that

$$D(S) = -\left(\frac{\mathrm{d}S}{\mathrm{d}\xi}\right)^{-1} \int_S^\infty \xi \mathrm{d}S'/2 \quad [\mathrm{m^2\ s^{-1}}]. \tag{8.25}$$

In summary, we have shown in this section how capillary suction can be described when there is no sharp front. It was shown that spontaneous imbibition goes with $t^{1/2}$, and as such it was proven that this is not a peculiarity of the Washburn equation, discussed in section 8.1. Finally, we showed how effective transport parameters can be extracted from experiments.

8.4 The role of vapor transport

In the first section of this chapter we have ruled out the influence of volume flow of air in the zone above the front, as its viscosity is low. From that point on the discussion focussed on liquid flow. Without explicit notice, it has been assumed that molecules in the liquid state cannot evaporate and travel via diffusive transport in air. Note that diffusive transport in air is not the same as the earlier discussed volume flow of air. Therefore, the role of diffusion in the gas phase is discussed at the closure of this chapter on unsaturated flow. The main goal here is to find out when vapor diffusion might play a role and our aim is not to give a full introduction to the topic of vapor diffusion as such. First, the importance of vapor diffusion under isothermal conditions will be analysed. Secondly, the effect of temperature gradients will be addressed briefly.

Diffusion is the random motion of particles. The driver for the motion is their kinetic energy that can be linked with the temperature. The stochastic nature of the motion originates from the collisions between particles. As diffusive motion plays a role, the molar flux can be split in a liquid $\vec{J}_l$ [mol m^{-2} s^{-1}] and a gas contribution $\vec{J}_g$ [mol m^{-2} s^{-1}],

$$\vec{J} = \vec{J}_l + \vec{J}_g, \quad [\mathrm{m\ s^{-1}}] \tag{8.26}$$

where $\vec{J}_l = n_l \vec{q}$ can be described with Darcy's law, equation (6.1). Here n_l [mol m^{-3}] is the molar density of the liquid. The gas flux $\vec{J}_g$ is not due to air flow but due to diffusive motion

$$\vec{J}_g = -D_g \nabla n_g, \quad [\mathrm{mol\ m^{-2}\ s^{-1}}] \tag{8.27}$$

where n_g [mol m^{-3}] is the molar density of the specie of interest in the gas phase. This equation is Fick's law for diffusion, which introduces the assumption that the gas phase behaves as an ideal gas.

To evaluate the importance of vapor diffusion, the magnitudes of the vapor and liquid fluxes have to be compared. As the volume flux of the liquid equals $\vec{q} = (k/\mu)\nabla p_c$, it is convenient to rewrite the molar flux due to vapor diffusion, equation (8.27), also in terms of ∇p_c:

$$\vec{J}_g = -D_g \frac{\partial n_g}{\partial p_c} \nabla p_c = \left(\frac{D_g}{RT}\right)\left(\frac{n_g}{n_l}\right) \nabla p_c. \quad [\mathrm{mol\ m^{-2}\ s^{-1}}] \tag{8.28}$$

The last term is obtained by application of the Kelvin equation (4.12), which relates p_c with the vapor density. Now we can introduce a characteristic number, which we shall abbreviate as Di (Diffusion number):

$$Di = |\vec{J}_g|/|\vec{J}_l| = \frac{\mu D_g}{k n_l R T}\frac{n_g}{n_l}. \quad [-] \tag{8.29}$$

Here we have to stress that we could not find a similar number as Di in the literature. When $Di \ll 1$ when vapor transport can be neglected. Equation (8.29) demonstrates the importance of the ratio between the gas and liquid density: n_g/n_l. Low values of the vapor density disfavor motion via vapor diffusion compared to liquid flow.

Equation (8.29) teaches that the importance of vapor diffusion is connected with the degree of liquid saturation. In section 8.2 it has been explained that the permeability $k(S)$ in an unsaturated medium decreases with decreasing liquid saturation S. As a consequence, Di increases with decreasing saturation. We now define a critical permeability k^*, related to a certain degree of saturation S^*, which corresponds to the case that $Di = 1$. The critical permeability can be derived from equation (8.29) and obeys

$$k^*(S^*) = \frac{\mu D_g}{n_l R T}\frac{n_g}{n_l}. \quad [\mathrm{m}^2] \tag{8.30}$$

To get a feeling for this critical permeability k^*, the identical tube model is again of great use. The tubes now represent the flow paths in the system. The pore radius r in equation (6.12) should be replaced by typical diameter/radius of the flow path ℓ [m]. Further, the porosity ϕ in expression (6.12) has to be transferred in the liquid content ϕS. By doing so the following equation for the critical permeability is obtained:

$$k^*(S^*) = \frac{1}{8}\phi S^* \ell^2. \quad [\mathrm{m}^2] \tag{8.31}$$

It has to be stressed this equation is only useful for order of magnitude calculations.

Vapor transport and permeability. Here we investigate when vapor transport starts to play a role. To that end the critical permeability k^* and the typical radius of the flow paths are estimated.

With equation (8.30) k^* can be calculated. The diffusion constant of water in air D_g at 1 atm and at 20 °C is $2.8 \cdot 10^{-5}$ m^2 s^{-1}. For the vapor density of water we take the saturation vapor density: $n_g = 0.96$ mol m^{-3}. Further, we use $\mu = 0.001$ Pa s and $n_l = 5.56 \cdot 10^4$ mol m^{-3}

$$k^* = \frac{0.001 \times 2.8 \cdot 10^{-5}}{5.56 \cdot 10^4 \times 8.314 \times 293}\left(\frac{0.96}{5.56 \cdot 10^4}\right) = 3.6 \cdot 10^{-21}\,\mathrm{m}^2.$$

Here a material with a porosity $\phi = 0.3$ is considered. Assuming that vapor transport becomes important below a saturation $S^* = 0.05$, it can be shown that

$$\ell = \sqrt{\frac{8k^*}{\phi S^*}} = \sqrt{\frac{8 \times 3.6 \cdot 10^{-21}}{0.3 \times 0.05}} \sim 10^{-9}\,\mathrm{m}.$$

Still a few words have to be spent on the diffusion constant. In the vapor phase the travelled distance between two collisions λ [m] (the mean free path) can be significant. At a pressure of 1 atm $\lambda \approx 68$ nm. In the case where a pore size is smaller than the mean free path $\lambda > r$, the molecules mainly scatter by collisions with the pore wall. As a consequence, the actual mean free path λ' will become of the order of the pore radius. For a medium with cylindrical pores and a porosity ϕ it can be shown that

$$D_g = \phi D_{g,\mathrm{tube}} \approx \phi D_{g,\mathrm{air}}(\lambda'/\lambda), \quad \left[\mathrm{m^2\ s^{-1}}\right] \tag{8.32}$$

where $D_{g,\mathrm{tube}}$ [$\mathrm{m^2\ s^{-1}}$] and $D_{g,\mathrm{air}}$ [$\mathrm{m^2\ s^{-1}}$] are the diffusivities in a tube with radius r and in bulk gas phase, respectively. The parameters λ [m] and λ' [m] are the mean free path in a bulk gas phase and the actual mean free path in the porous medium, respectively.

Until now only the isothermal case has been discussed. At the end of this section the influence of temperature gradients will be discussed briefly. To introduce the topic we consider an initially partially saturated porous material in equilibrium. The liquid content S is therefore constant in space. At a certain moment a temperature gradient is applied and a vapor flux develops. The vapor flux can be connected with the temperature gradient ∇T by rewriting equation (8.27):

$$\vec{J}_g = -D_g\left(\frac{\partial n_g}{\partial T}\right)\nabla T. \quad [\mathrm{mol\ m^{-2}\ s^{-1}}] \tag{8.33}$$

For simplicity we only discuss the case where the capillary pressure is small (big pores). As liquid and vapor are in equilibrium inside the pore matrix, it follows from equation (4.12) that the vapor density equals the saturation vapor density $n_g = n_g^*$. The factor $\partial n_g/\partial T$ in equation (8.33) is the key to understanding temperature driven vapor fluxes. A nice example is water. The saturation vapor density of water varies between 0.005–0.59 kg $\mathrm{m^{-3}}$ for temperatures ranging from 273 to 373 K.

Further reading

Hunt A G 2005 *Percolation Theory for Flow in Porous Media* (Berlin: Springer) ch 3, 4: pore scale approaches for modeling the permeability, both saturated and unsaturated, and capillary pressure curves based on percolation theory. As such, also a nice introduction to percolation theory.

Hall C and Hoff W D *Water Transport in Brick, Stone and Concrete* 2nd edn (Boca Raton, FL: CRC Press) ch 4: well-written text on macroscopic descriptions of unsaturated transport in general and the use of the non-linear diffusion equation in particular. All framed within the context of construction materials.

Cunningham R E and Williams R J J 1980 *Diffusion in Gases and Porous Media* (Berlin: Springer): a detailed introduction to gas transport, which is hardly covered in the book you are reading.

IOP Concise Physics

Fluids in Porous Media

Transport and phase changes

Henk Huinink

Chapter 9

Two phase flow

9.1 Front motion

In the previous chapter unsaturated flow has been discussed, which is actually a specific case of two-phase flow: air–liquid. In this chapter the air is exchanged for a second liquid. In the case of unsaturated flow, the role of the air phase could be ignored as its viscosity was low. This will not work any longer when two liquids are considered. The viscous pressure gradients in both liquids have to be described explicitly, as their viscosities have values that cannot be neglected. In this section the motion of a sharp front between a wetting and a non-wetting liquid is discussed, which has two aims. First of all, the influence of the viscosities of both liquids on the front motion will be investigated. Secondly, the order of the displacement process will be elucidated: what is the difference between 'drainage', the displacement of a wetting by a non-wetting liquid, and 'imbibition', the displacement of a non-wetting by a wetting liquid.

Although we use the words wetting and non-wetting in connection with specific liquids, it has to be stressed that these are not properties of the liquids themselves. In chapter 3 the concept of wetting has emerged from the interactions between three phases: the solid matrix of the porous material, a liquid and a vapor. In the present chapter the vapor is exchanged for a second liquid phase. So we have two liquids in contact with a solid surface. Depending on the nature of the solid and the other liquid, a liquid can be wetting or non-wetting. This is nicely visible in table 9.1. Given that a mixture of benzene and water is considered, water and benzene are the wetting liquids in contact with Al_2O_3 and stibnite, respectively.

We consider a porous medium with a length L [m], porosity ϕ [–] and permeability k [m^2]. The medium is initially filled with a liquid I. From one side a liquid II is pumped into the material, which replaces liquid I, see figure 9.1. The displacement process will be analyzed in 1D, meaning that we assume homogeneity in the directions parallel to the front. Note that it is assumed that the permeability does not depend on the liquid type. Over the material a pressure drop $\Delta p \equiv p_{ex} - p_{in}$ [Pa] is

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Table 9.1. Contact angles of two liquids on a solid substrate: PTFE = polytetrafluorethylene (Teflon) and PE = polyethylene. All data refer to temperatures between 20 and 25 °C.

solid	liquid I	liquid II	γ mN m^{-1}	θ
stibnite	water	benzene	35.0	130
Al_2O_3	water	benzene	35.0	22
PTFE	water	n-decane	51.2	180
	water	benzyl alcohol		150
PE	water	n-decane	51.2	180
	water	paraffin oil		150
glass	mercury	gallium		0

Sources: Girifalco L A and Good R J 1957 *J. Phys. Chem*, 61 904 (interfacial tensions) and Adamson A W and Gast A P 1997 *Physical Chemistry of Surfaces* 6th edn (New York: Wiley) (contact angles).

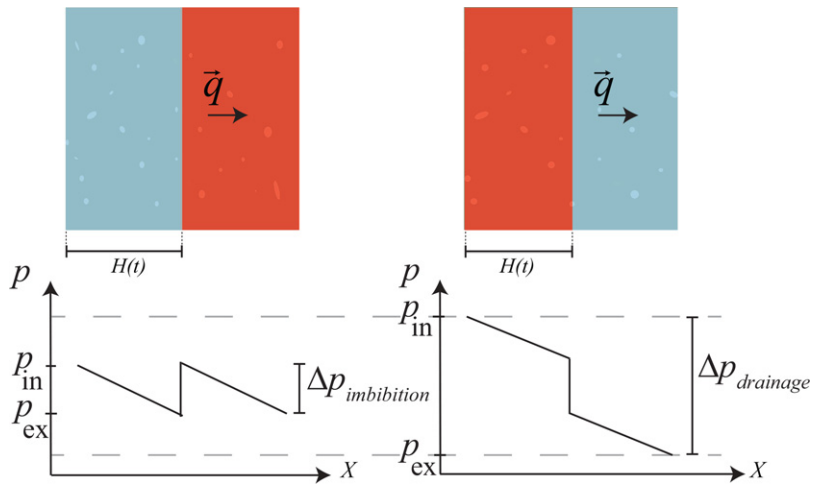


Figure 9.1. Two-phase flow with a constant volume flux: imbibition (left) and drainage (right). The wetting and non-wetting fluids are colored blue and red, respectively. The pressure profiles are shown at the bottom. Due to the capillary pressure, the applied pressure $\Delta p \equiv p_{ex} - p_{in}$ needed to drive the flow is lower in the case of imbibition than in the case of drainage (see pressure profiles below the pictures).

applied, where p_{ex} [Pa] and p_{in} [Pa] are the pressures at the inlet and outlet of the porous material. The orientation of the material is such that the flow with volume flux q [m s^{-1}] develops in the horizontal direction and gravity can be neglected.

Similar to the approach chosen in the description of capillary suction, see equation (8.1), the front motion can be coupled with the volume flux and the pressure gradients at the front:

$$\phi\frac{\partial H}{\partial t} = q(H) = -\frac{k}{\mu_{II}}\left(\frac{\partial p}{\partial x}\right)_{z=H^-} = -\frac{k}{\mu_I}\left(\frac{\partial p}{\partial x}\right)_{z=H^+}. \quad [\mathrm{m\ s^{-1}}] \tag{9.1}$$

Here μ_{I} [Pa s] and μ_{II} [Pa s] are the viscosities of the liquids. As both liquids are incompressible, it follows from equation (6.9) that $\partial_x q = 0$ and $\partial_{xx} p = 0$. Therefore, the volume flux will not depend on x and the pressure varies linearly in both the liquid and gas phase. As a consequence $(\partial_x p)_{z=H^-} = \Delta p_{\mathrm{II}}/H$ and $(\partial_x p)_{z=H^+} = \Delta p_{\mathrm{I}}/(L-H)$, where $\Delta p_{\mathrm{I}} \equiv p_{\mathrm{ex}} - p(H^+)$ [Pa] and $\Delta p_{\mathrm{II}} \equiv p(H^- - p_{\mathrm{in}})$ [Pa]. At the liquid front the pressure jumps with an amount Δp_f [Pa], where $\Delta p_f = -p_c$ for imbibition and $\Delta p_f = +p_c$ for drainage. Therefore, the total pressure drop can be written as $\Delta p = \Delta p_{\mathrm{I}} + \Delta p_{\mathrm{II}} + \Delta p_f$. With equation 9.1 and this knowledge of the pressure, one can derive that

$$\Delta p - \Delta p_f = \Delta p_{\mathrm{I}} + \Delta p_{\mathrm{II}} = -\frac{\phi}{k}\left(\mu_{\mathrm{I}} L + \Delta\mu H\right)\frac{\mathrm{d}H}{\mathrm{d}t}, \quad [\mathrm{Pa}] \tag{9.2}$$

where $\Delta\mu \equiv \mu_{\mathrm{II}} - \mu_{\mathrm{I}}$ [Pa s]. Note that $\mathrm{d}_t H = \phi q$.

With equation 9.2 a closed form differential equation for the front position H is obtained. The solution of this equation is equal to

$$H^2 + \left(\frac{2\mu_{\mathrm{I}} L}{\Delta\mu}\right)H = -\frac{2k}{\phi\Delta\mu}\left(\Delta p + \Delta p_f\right)t. \quad [\mathrm{m}^2] \tag{9.3}$$

First of all, this equation should be able to describe capillary suction as a limiting case. In the case of capillary suction no pressure drop is applied, $\Delta p = 0$, and a wetting phase replaces a non-wetting phase (imbibition), $\Delta p_f = -p_c$. The displaced phase (fluid I) is air, which has a very low viscosity compared to the intruding phase, liquid II: $\mu_{\mathrm{I}} \approx 0$ and $\Delta\mu \approx \mu_{\mathrm{II}}$. Incorporating this into equation 9.3 gives

$$H^2 = \left(\frac{2kp_c}{\phi\mu_{\mathrm{II}}}\right)t, \quad [\mathrm{m}^2] \tag{9.4}$$

which is equal to equation (8.8). So capillary suction is indeed received from the model as a limiting case.

A second interesting aspect of the displacement equation 9.3 is that it illustrates nicely the difference between drainage and imbibition. Here we focus on flows that exclusively go in the positive x direction. According to the right hand side of the equation, flow occurs when $\Delta p + \Delta p_f < 0$. In the case of drainage $\Delta p_f = +p_c$ and flow will only be generated when the pressure drop is sufficiently large: $\Delta p < -p_c$. The reverse happens in the case of imbibition $\Delta p_f = -p_c$. As seen before, spontaneous flow will develop in the absence of an applied pressure drop and flow happens as long as $\Delta p < p_c$.

The third point is the effect of viscosity. Actually, one limiting case has been discussed already: capillary suction, where $\mu_{\mathrm{II}}/\mu_{\mathrm{I}} \gg 1$. In figure 9.2 the front position as a function of time is shown for three limiting cases: $\mu_{\mathrm{II}}/\mu_{\mathrm{I}} \gg 1$, $\mu_{\mathrm{II}}/\mu_{\mathrm{I}} = 1$ and $\mu_{\mathrm{II}}/\mu_{\mathrm{I}} \ll 1$. The situation $\mu_{\mathrm{II}}/\mu_{\mathrm{I}} \gg 1$ has been discussed extensively in section 8.1. As the pressure driving the flow is fixed, $\Delta p + \Delta p_f$, and the friction increases with the

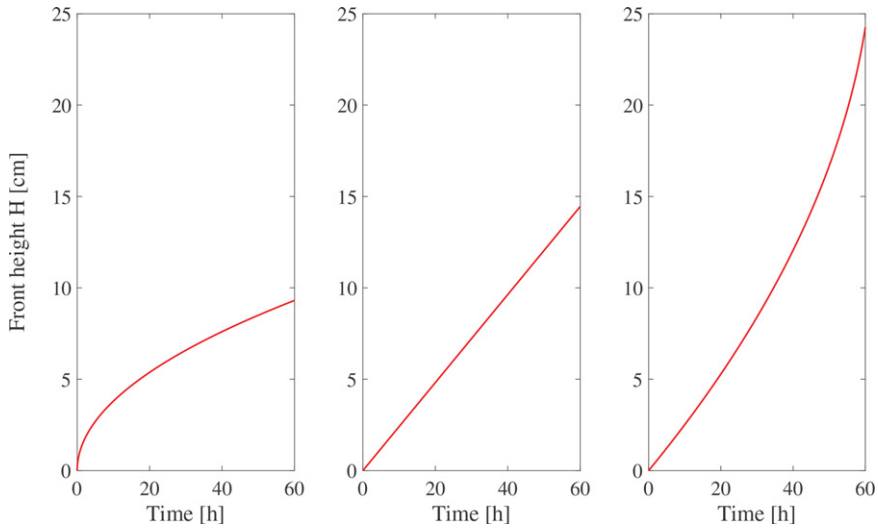


Figure 9.2. Front motion ($H - t$) for three limiting cases: $\mu_{II}/\mu_I \gg 1$ (left), $\mu_{II}/\mu_I = 1$ (middle) and $\mu_{II}/\mu_I \ll 1$ (right).

progress of the front H, the front slows down. Interestingly, in the case where $\mu_{\mathrm{II}}/\mu_{\mathrm{I}} = 1$ the front motion is linear in time and does not depend on the front position itself:

$$H = -\frac{k}{\phi\mu_{\mathrm{I}}L}\left(\Delta p + \Delta p_f\right)t. \quad [\mathrm{m}] \tag{9.5}$$

As both the viscosities of both phases are equal, the friction does not change with position of the front, see figure 9.2, which is represented by the factor L on the right hand side of the equation.

We have discussed two of the viscosity scenarios and there is still one left: $\mu_{\mathrm{II}}/\mu_{\mathrm{I}} \ll 1$. As can be seen from figure 9.2, equation (9.3) predicts that the front will accelerate when time progresses:

$$H^2 - 2HL = \frac{2k}{\phi\mu_{\mathrm{I}}}\left(\Delta p + \Delta p_f\right)t. \quad [\mathrm{m}^2] \tag{9.6}$$

Following our previous line of reasoning, the acceleration is simply a consequence of the fact the liquid experiencing the most friction (liquid I) is pressed out of the system. The amount of liquid I decreases with time and therefore the friction also decreases with time. Whereas equation 9.3 captures the physics of the regimes $\mu_{\mathrm{II}}/\mu_{\mathrm{I}} \gg 1$ and $\mu_{\mathrm{II}}/\mu_{\mathrm{I}} = 1$ correctly, it does not predict front motion very well for $\mu_{\mathrm{II}}/\mu_{\mathrm{I}} \ll 1$. Experiments have shown that fronts are unstable and fingers develop in this particular case, see figure 9.3. The issue of front stability is postponed to the last section of this chapter. Due to its 1D character, the approach chosen in this section cannot describe structure formation in the front.

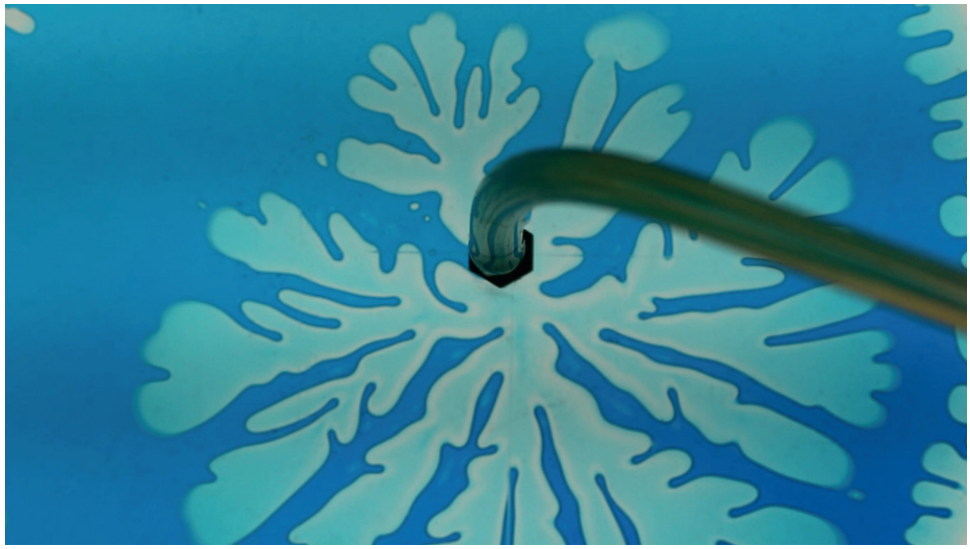


Figure 9.3. Viscous fingering in a Hele–Shaw cell: a fluid II with a low viscosity (air) is injected into a fluid I with a high viscosity (glycerol). As $\mu_{II} < \mu_I$ the front becomes unstable and the typical fingering pattern develops. A Hele–Shaw cell is the most simple porous material: two parallel plates. Picture: courtesy of Philip Ruijten and Thomas Arends.

We already discussed one shortcoming of the chosen approach: it cannot capture processes like finger formation. However, this problem can be overcome easily by adopting a 2D or 3D approach. There are other issues that need attention. First of all, the analysis in this section used a sharp front separating the two liquids. As discussed already in chapter 8, the front may be blurred: the saturations for liquid I and II might increase and decrease gradually passing the front. Section 9.2 is devoted to the analysis of the front width. Secondly, we did not discuss whether or not liquid I is fully displaced by liquid II. In practice, pockets of the displaced liquid always remain. This topic is intrinsically coupled with the notion of relative permeability, which will be discussed in section 9.3.

9.2 The front zone

In practise, the front separating the invading liquid (II) from the displaced liquid (I), see figure 9.1, is never infinitely sharp. Capillary forces that can counteract the viscous forces blur the interface. Although this topic has already been addressed within the framework of unsaturated flow, see section 8.2, here this subject is treated again as we want to introduce simple characteristic numbers for front broadening.

To understand front broadening we start with a static problem: the stratification of two immiscible liquids under the influence of gravity, see figure 9.4. The mass density of liquid I is higher than the mass density of liquid II: $\rho_{\mathrm{I}} > \rho_{\mathrm{II}}$ and $\Delta\rho \equiv \rho_{\mathrm{I}} - \rho_{\mathrm{II}}$ [kg m^{-3}]. Due to this density difference two layers have developed: a top layer enriched with liquid II and a bottom layer enriched with liquid I. The typical size of the transition zone ξ [m] between the two layers can be estimated by

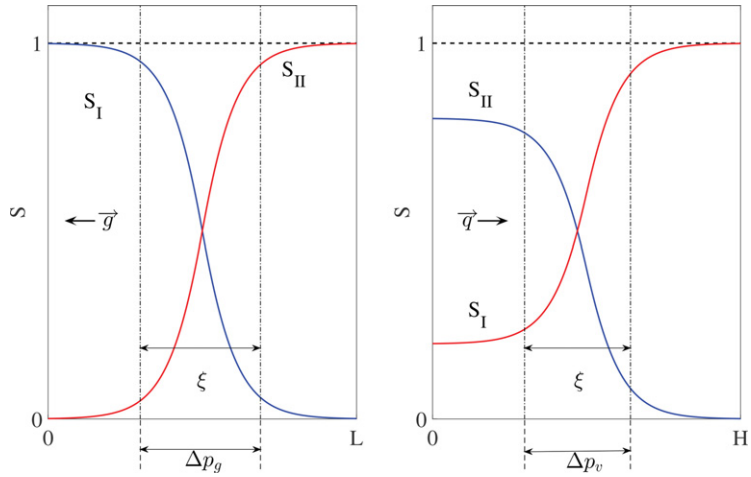


Figure 9.4. Saturation profiles around the front zone: a gravity induced front (left) and a front formed due to viscous forces (right). Capillary forces can blur the front zone up to a width ξ. At larger length scales gravity (left) or viscous forces (right) beat capillarity.

analyzing the capillary forces. For simplicity we assume here that liquid I is the wetting liquid. Given a certain value of the saturations for both liquids, S_{I} [−] and S_{II} [−] and $S_{\mathrm{I}} + S_{\mathrm{II}} = 1$, it can be shown that

$$S_{\mathrm{I}}(r) = \frac{\mathscr{L}_p{}'}{V_p{}'} \int_0^r P(r')\pi r'^2 \mathrm{d}r' \quad [-] \tag{9.7}$$

and

$$S_{\mathrm{II}}(r) = \frac{\mathscr{L}_p{}'}{V_p{}'} \int_r^\infty P(r')\pi r'^2 \mathrm{d}r'. \quad [-] \tag{9.8}$$

In these equations, $V_p{}'$ [m^3] and $\mathscr{L}_p{}'$ [m] are the accessible pore volume and pore length, respectively. Further $P(r)$ [m^{-1}] is the pore size distribution. Note that these equations are similar to the equations (8.10) and (8.11). All pores with radii smaller than r are filled with liquid I, the wetting liquid, and all bigger pores are filled with liquid II, the non-wetting liquid. Note that we have assumed cylindrically shaped pores. It follows from equations (9.7) and (9.8) that the local capillary pressure can be expressed as a function of the local saturation: $p_c(S_{\mathrm{I}})$. Just above the front zone $S_{\mathrm{I}} \sim 0$ and $p_c \sim p_{c,\max}$, where $p_{c,\max}$ is the maximal capillary pressure that can be reached in the system corresponding to the minimal pore radius $r_{\min}$. Below the front zone $S_{\mathrm{I}} \sim 1$ and $p_c \sim p_{c,\min}$, where $p_{c,\min}$ is the minimal capillary pressure that can be reached in the system corresponding to the maximal pore radius $r_{\max}$. Note that the disorder in the porous matrix means that gravity can be counteracted by capillary forces. The pressure drop Δp_c that can be generated by capillary forces equals

$$\Delta p_c \equiv p_{c,\max} - p_{c,\min} = 2\gamma \cos\theta (r_{\min}^{-1} - r_{\max}^{-1}) \sim \gamma \cos\theta / r, \quad [\mathrm{Pa}] \tag{9.9}$$

which balances a hydrostatic pressure drop

$$\Delta p_g = \Delta\rho g\xi \quad [\text{Pa}] \tag{9.10}$$

over the front zone. The last term of expression (9.9) is a crude approximation of the capillary pressure. The property $r_{\min}^{-1} - r_{\max}^{-1} = (r_{\max} - r_{\min})/r_{\min}r_{\max}$ is approximated by the characteristic radius r of the porous material of interest. By combining the equations (9.9) and (9.10) the front width can be approximated:

$$\xi \approx \left|\Delta p_c\right|/\Delta\rho g. \quad [\text{m}] \tag{9.11}$$

With equation (9.11) a characteristic number Bo [−] can be defined that compares the front width ξ with the system size L:

$$Bo \equiv L/\xi = \frac{\Delta\rho g L}{\left|\Delta p_c\right|} \sim \frac{\Delta\rho g L r}{2\gamma\cos\theta}. \quad [-] \tag{9.12}$$

Bo is called the Bond number. The last term in this equation follows from the previously discussed approximation for Δp_c. When $Bo > 1$, stratification occurs. When $Bo \gg 1$ the front width is small compared to the length scale of interest and a sharp front approach is valid. In the case where $Bo \ll 1$, both liquids are homogeneously distributed. There is no mixing on a molecular scale, but on pore scale level. Locally, big pores are filled with the non-wetting fluid and the small pores are filled with the wetting liquid. Note that it does not matter if liquid I is the wetting or non-wetting liquid for the final value of Bo, despite the fact that the discussion was started with assuming that liquid I was the wetting phase.

Do oil and water stratify in a sandstone layer? Oil and water do not mix. As a consequence they stratify in two separate layers. Here both the width of the interfacial zone ξ and the Bond number Bo will be calculated for a crude oil–water mixture in sandstone.

The sandstone layer has a typical thickness of 100 m and its pore size distribution is peaked around 10 micron (radius). The densities of water ρ_w and crude oil ρ_o are 1000 and 873 kg m^{-3}. Here it has to be mentioned that the physical properties of crude oil vary from reservoir to reservoir. The interfacial tension of the oil/water interface equals $3.0 \cdot 10^{-2}$ N m^{-1}. With equation (9.11) and $p_c = 2\gamma/r$ (assumptions: cylindrical pores and water as the perfect wetting fluid) we find

$$\xi = \frac{2 \times 3.0 \cdot 10^{-2}}{1.0 \cdot 10^{-5} \times (1000 - 873) \times 9.81} = 4.8 \text{ m}$$

and

$$Bo = \frac{100}{4.8} = 21.$$

This calculation demonstrates that the interfacial zone has a significant width. In this zone the big pores will be filled with the crude oil and the small pores with water. The numbers also show that on the length scale of interest stratification will occur and the crude oil will occupy the upper part of the sandstone layer.

Interestingly, moving fronts can be analysed in a similar fashion as a static interface between two liquids in the presence gravity. Again, capillary forces are the source for the blurring of the front, see figure 9.4. The role of gravity is taken over by the viscous forces. Therefore, we can quickly move forwards by recalling the essential ingredients used in the discussion on the role of gravity. For simplicity it is assumed that the invading liquid (II) is non-wetting and the displaced liquid (I) is wetting. However, the final result will be again invariant for this choice. In the case of flow the capillary pressure difference over the interface, equation (9.9), balances the viscous pressure drop

$$\Delta p_v = q\mu\xi/k, \quad [\text{Pa}] \tag{9.13}$$

where q and k are typical values for the volume flux and the permeability. As we are only interested in orders of magnitude, we will do the calculations with the viscosity of the most viscous liquid: μ. It follows from the equations (9.9) and (9.13) that the width of the front equals

$$\xi \approx k\left|\Delta p_c\right|/q\mu. \quad [\text{m}] \tag{9.14}$$

With equation (9.14) the so-called capillary number Ca [–] can be defined that compares the front width ξ with the system size L:

$$Ca \equiv L/\xi = \frac{q\mu L}{k\left|\Delta p_c\right|} \sim \frac{q\mu L}{\phi r\gamma\cos\theta}. \quad [-] \tag{9.15}$$

The last term in this equation again follows from the previously discussed approximation for Δp_c and the adoption of the identical tube model for the permeability, see equation (6.12). When $Ca > 1$, a front will be present during the displacement process. When $Ca \gg 1$ the front width will be small compared to the length scale of interest and a sharp front model is valid. When $Ca < 1$, capillary forces always dominate over the viscous forces. In this case both liquids will distribute homogeneously through the material during a major part of the imbibition or drainage process.

The validity of the sharp front approach for oil reservoir calculations. When water is used to replace crude oil, a front zone develops. In this front zone water and oil are simultaneously present, i.e., locally there are water- and oil-filled pores present. Here we estimate the size of this front zone for a certain example. This enables us to judge the validity of a sharp front approach.

Again, a sandstone layer is considered: $\phi = 0.2$ (porosity), $r = 10$ micron (typical pore radius), $k = 3.0 \cdot 10^{-12}$ m^2 (permeability), and $L = 1$ km (length scale of interest). The typical fluid velocity u in the reservoir is $1.0 \cdot 10^{-5}$ m s^{-1} ($q = \phi u = 2.0 \cdot 10^{-6}$ m s^{-1}). We assume that water perfectly wets the sandstone. In the calculations the viscosity of the oil will be used as this is the most viscous fluid: $\mu = 7.5 \cdot 10^{-3}$ Pa s (as mentioned, the viscosities of crude oils vary a lot). For the interfacial tension the value 0.030 N m^{-1} will be used. With equation (9.14) and the assumption of a cylindrical pore shape ($p_c = 2\gamma/r$) we obtain

$$\xi = \frac{3.0 \cdot 10^{-12} \times (2 \times 0.030/1.0 \cdot 10^{-5})}{2.0 \cdot 10^{-6} \times 7.5 \cdot 10^{-3}} = 1.2 \text{ m}.$$

Although the front zone has a significant width, its size is small compared to the length of interest in the case of oil reservoirs (kms). In this type of application, modeling with a sharp front approach seems to be valid.

9.3 Relative permeability and residual saturation

Until now explicit equations for simultaneous flow of two immiscible phases have not been discussed. Obviously such equations are needed to describe the details of the flow in the front zone. More important, they are needed to understand the phenomenon of residual saturation. In section 9.1 front motion was described, while assuming that the invading liquid II was completely displacing liquid I. In reality a significant amount of the displaced liquid is immobilized and remains as isolated clusters present in the porous matrix. This is called residual saturation and it is illustrated in figure 9.5. When a medium is initially fully saturated with liquid I, then only liquid I will present ahead of the front: $x \gg H$ with H [m] the position of the front. Within the front zone, $|x| \lesssim \xi$, networks of pores filled with liquids I and II coexist. Downstream of the front, the pore system is mainly filled with liquid II and immobile isolated clusters of pores filled with liquid I exist. To understand this process, first a better description of the flow is needed.

Here the simultaneous flow of a wetting (w) and a non-wetting (n) liquid will be discussed. So, the subscripts I and II will be dropped and replaced by the subscripts 'w' and 'n'. The local composition is defined by the saturations of the wetting, S_w, and non-wetting phase, S_n. The volume fluxes, $\vec{q}_w$ [m s^{-1}] and $\vec{q}_n$ [m s^{-1}], of both liquids can be coupled with the rate of change of the saturations:

$$\phi \frac{\partial S_w}{\partial t} = -\nabla \cdot \vec{q}_w \quad [\text{s}^{-1}] \tag{9.16}$$

and

$$\phi \frac{\partial S_n}{\partial t} = -\nabla \cdot \vec{q}_n. \quad [\text{s}^{-1}] \tag{9.17}$$

With the incompressibility condition $S_w + S_n = 1$ it can be shown that

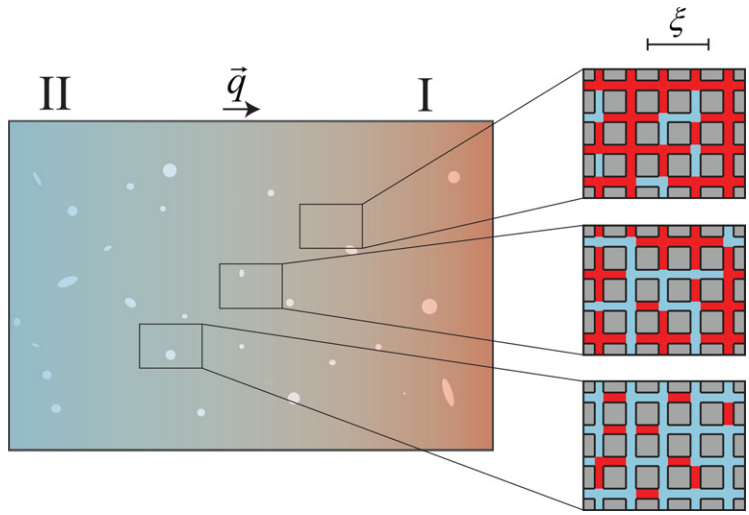


Figure 9.5. Fluid distributions at the pore scale level during imbibition. On the right, schematic pictures of the distribution of the fluids on the pore-scale level are shown. The sizes of the pores are not visualized in these pictures. Downstream of the front the saturation of the wetting fluid is low and only the small pores are invaded by the wetting fluid (top). Upstream of the front, the saturation of the wetting fluid is high: only a few big pores are still filled with non-wetting liquid.

$$\nabla \cdot \vec{q}_n + \nabla \cdot \vec{q}_w = 0. \tag{9.18}$$

With Darcy's law, equation (6.1), the volume fluxes $\vec{q}_{\mathrm{w}}$ and $\vec{q}_n$ of both liquids can be described:

$$\vec{q}_w = -\frac{kk_{rw}(S_w)}{\mu_w}\nabla p_w \quad [\mathrm{m\ s^{-1}}] \tag{9.19}$$

and

$$\vec{q}_n = -\frac{kk_{rn}(S_n)}{\mu_n}\nabla p_n. \quad [\mathrm{m\ s^{-1}}] \tag{9.20}$$

In these equations k [$\mathrm{m^2}$] is the single phase permeability of the porous medium. The quantities k_{rw} [−] and k_{nw} [−] are the relative permeabilities for the wetting and non-wetting liquids, respectively. These relative permeabilities are functions of the saturations. Note that kk_{rw} and kk_{nw} are the actual permeabilities for the wetting and non-wetting liquids, respectively. Further, p_w and p_n are the pressures in the wetting and non-wetting phase, respectively. We have ignored the influence of gravity as it is not relevant for the discussion in this section. The microscopic picture here is as sketched in figure 9.5: the wetting and non-wetting liquid flow through separate networks of pores.

In the case of local mechanical equilibrium, the pressures in both phases are coupled: $p_c = p_n - p_w$. As in the case of unsaturated flow, p_c is a function of $S_w = 1 - S_n$, given that the saturations define the state of the system and can be

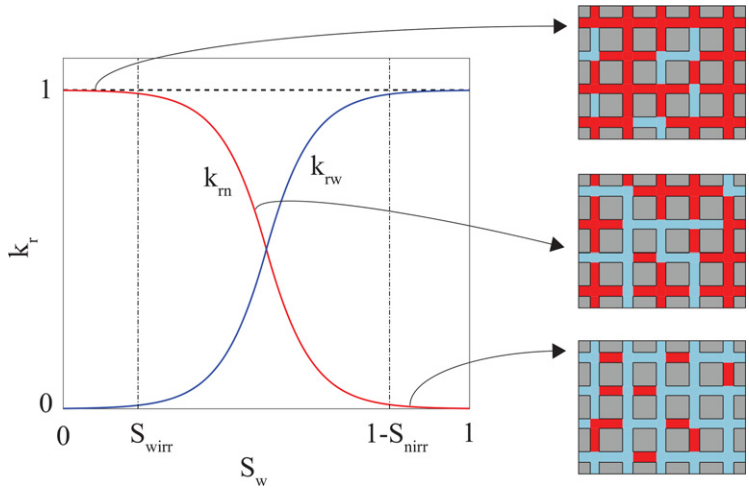


Figure 9.6. The relative permeabilities k_{rw} and k_{rn} as a function of S_w. The permeability for the wetting fluid k_{rw} vanishes for $S_w < S_{w,irr}$ as the wetting phase only consists of isolated clusters of pores filled with the wetting fluid. As a consequence, there is no long range network of the wetting liquid, facilitating its flow. The same holds for the permeability for the non-wetting phase k_{rn}, when $S_w > 1 - S_{n,irr}$. In that case the non-wetting phase consists of isolated clusters of pores filled with the non-wetting phase.

linked with the capillary pressure via the pore radii: equation (9.7) for the wetting phase and equation (9.8) for the non-wetting phase. Given that the concept of 'state' is valid, the relative permeabilities are unique functions of the saturations.

Experiments have shown that k_{rw} and k_{rn} increase and decrease with the saturation for the wetting phase S_w, respectively. This is easily understood. With increasing S_w the network of pores filled with wetting fluid densifies, whereas the network of pores filled with non-wetting fluid becomes more volatile. A typical example is given in figure 9.6. The second interesting aspect of the relative permeability functions are their behavior at high and low values of S_w: $k_w \approx 0$ for $S_w < S_{w,irr}$ and $k_n \approx 0$ for $S_w > 1 - S_{n,irr}$. This reflects the states of the pore networks of both liquids. When $S_w < S_{w,irr}$, there is no spanning network of pores filled with the wetting phase and flow of the wetting liquid is impossible. When $S_w > 1 - S_{n,irr}$, there is no spanning network of pores filled with the non-wetting phase and the non-wetting liquid cannot flow. The existence for such transitions has been proven within the framework of percolation theory, which will not be discussed in this book. The main finding of percolation theory is that the probability to find a system spanning a network of pores filled with a specific liquid equals zero, when the fraction of pores filled with this liquid drops below a certain threshold value (generally of the order 0.1–0.3). This explains why full displacement of a liquid by another liquid is extremely difficult.

Based on the definition of the Capillary number, equation (9.15), it can be estimated what has to be done to mobilize the trapped clusters. To push for example a non-wetting cluster through medium filled with a wetting phase, the viscous pressure drop over the cluster should be sufficiently large to outcompete the capillary

pressure difference. Since the length scale of interest is now the pore radius, $L = r$, it can be shown that

$$Ca_r = \frac{q\mu}{\phi\gamma\cos\theta}. \quad [-] \tag{9.21}$$

Interestingly the pore radius completely dropped out of the equation.

Equation (9.21) can also be used to assess the validity of the condition of local mechanical equilibrium and the idea that two immiscible liquids flow through separate networks of pores. When $Ca_r \gtrsim 1$, viscous forces also dominate over the capillary forces on the smallest length scale possible: the pore radius. In this case simultaneous flow of both phases within a single pore is enforced.

At the pore scale. The value Ca_r will be calculated for the same example as discussed in section 9.2. On the basis of this calculation we can evaluate whether or not pore-scale effects can be influenced by the viscous pressure gradient.

Here we limit the introduction and only list the parameters: $\phi = 0.2$ (porosity), $q = 2.0 \cdot 10^{-6}$ m s^{-1} (volume flux), $\gamma = 0.030$ N m^{-1} (interfacial tension), $\mu = 7.5 \cdot 10^{-3}$ Pa s (viscosity of the crude oil) and $\theta = 0°$ (contact angle). With equation (9.21) one finds

$$Ca_r = \frac{2.0 \cdot 10^{-6} \times 7.5 \cdot 10^{-3}}{0.2 \times 0.030} = 2.5 \cdot 10^{-6}.$$

Without any doubt it can be stated that in this particular example capillary forces rule on the pore scale level. Further, the calculation demonstrates that it is a major challenge to enforce simultaneous flow of oil and water through a single pore in order to mobilize trapped oil clusters (residual oil saturation). Handles for this purpose seem to be the contact angle (shift it towards 90°) and the interfacial tension makes it extremely small). This can only be done via interventions with chemicals, i.e. surfactants.

Finally, a warning regarding relative permeabilities is appropriate. Despite the fact that qualitatively relative permeabilities behave as discussed, many hysteresis effects have been observed in experiments. Depending on the way displacement processes are done, different relative permeability functions can be found. So the concept of 'state' and its connection to the local saturation should be treated with caution.

9.4 Viscous fingering

The opening section of this chapter was devoted to front motion, in the case where a liquid I is displaced by a liquid II and both liquids are immiscible. As the analysis was done in 1D, a stable flat front was enforced without justification. What happens when we allow the front to deform? Qualitatively the answer is already given in figure 9.3. In the case where $\mu_{II} > \mu_I$, the invading liquid has higher viscosity than the displaced liquid, the front remains stable. The friction is the highest in the zone

saturated with liquid II and the front slows down during the displacement process, see figure 9.2. This implies that perturbations will be damped as the local front velocity decreases with the local progression. When $\mu_{II} > \mu_I$, the invading liquid has lower viscosity than the displaced liquid, the front becomes unstable and fingers develop. The friction is now the highest in the zone saturated with liquid I and the front accelerates during the displacement process, see figure 9.2. Where the front has progressed the most, the friction is minimal and the front velocity the highest. As a consequence, perturbations will be amplified. This phenomenon is called the Saffman–Taylor instability. In later stages the perturbations develop into finger-like structures and the process is referred to as viscous fingering.

Viscous fingering plays an important role in oil recovery. In many oil recovery process water/brine is pumped via an injection well into an oil reservoir and forces the oil towards the production well. As the viscosity of water/brine is generally much lower than the viscosity of oil, the displacement front is unstable and finger formation occurs. As soon as a finger reaches a production well, a short-cut has developed and water/brine is directly pumped from the injection towards the production well. As a consequence the oil production stops.

The initial evolution of small perturbations can be predicted with linear stability analysis. In this book, only the main ingredients and the result of the analysis are summarized. The case considered is as described in section 9.1: a liquid II displaces a liquid I. The porous medium has porosity ϕ [–] and permeabilities in the zones saturated with liquids I and II are equal and constant: k [m^2]. We limit ourselves to a 2D system: the flow is parallel to x- and perpendicular to the y-axis and the typical size of the system is L [–]. The flow is driven by a pressure drop Δp [Pa]. The average volume flux equals q [m s^{-1}] and fluid velocity is $U = q/\phi$ [m s^{-1}]. A schematic picture is given in figure 9.7. A harmonically perturbed front can be described with

$$H(y, t) = h(t) + \delta \exp(\mathrm{i}\alpha y + \nu t), \quad [\mathrm{m}] \tag{9.22}$$

where δ [m], α [m^{-1}] and ν [s^{-1}] are initial amplitude, wavenumber and growth rate of the perturbation, respectively. The wavenumber α is related with the wavelength

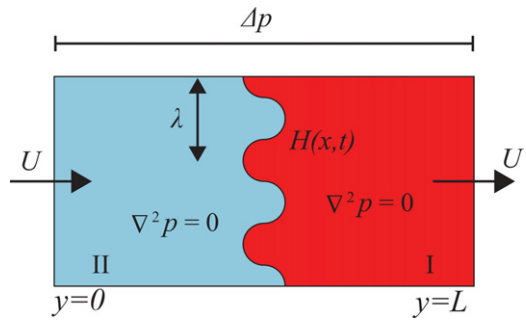


Figure 9.7. Front motion during the displacement of a liquid I by a liquid II. The stability of the front is analyzed by studying the growth of a sinusoidal perturbation. The local fluxes can be found by solving $\nabla^2 p = 0$ for both liquids. By coupling the local velocities with the fluid motion, the growth of the perturbation can be predicted.

λ [m]: $\lambda \equiv 2\pi/\alpha$. Further, h [m] is the average front position and $U = \partial_t h$. The perturbation grows when $\nu < 0$ and dies when $\nu < 0$. The local velocity of the front in the x-direction can be coupled with the pressure field by using Darcy's law (equation (6.1)):

$$-\frac{k}{\phi\mu_{\mathrm{I}}}\left(\frac{\partial p}{\partial x}\right)_{x=h^-} = -\frac{k}{\phi\mu_{\mathrm{II}}}\left(\frac{\partial p}{\partial x}\right)_{x=h^+} = U + \nu\delta\exp(\mathrm{i}\alpha y + \nu t). \quad [\mathrm{m\ s^{-1}}] \quad (9.23)$$

The pressure field can be found via the incompressibility condition: $\nabla^2 p = 0$. The resulting pressure fields for the domains saturated with the liquids I and II are equal to

$$p_{\mathrm{I}}(x, y, t) = -\frac{\phi\mu_{\mathrm{I}}}{k}\left[Ux - \left(\frac{\nu\delta}{\alpha}\right)\exp(+\mathrm{i}\alpha y - \alpha[x - H] + \nu t)\right] + C_{\mathrm{I}} \quad [\mathrm{Pa}] \ (9.24)$$

and

$$p_{\mathrm{II}}(x, y, t) = -\frac{\phi\mu_{\mathrm{II}}}{k}\left[Ux + \left(\frac{\nu\delta}{\alpha}\right)\exp(+\mathrm{i}\alpha y + \alpha[x - H] + \nu t)\right] + C_{\mathrm{II}} \quad [\mathrm{Pa}],(9.25)$$

respectively. C_{I} and C_{II} are constants. These solutions are applicable as long as $\lambda/L \ll 1$, which means that the perturbation in the pressure field at the system boundaries can be neglected.

The last step in the linear stability analysis consists of equating the pressures at the front, $p(H^+) = p(H^-) + \Delta p_f$. When liquid II is the wetting liquid, then $\Delta p_f = p_c$. When liquid I is the wetting liquid, then $\Delta p_f = -p_c$. By coupling the pressures, an expression for the growth rate of a perturbation can be found:

$$\nu = \alpha U\left(\frac{\mu_{\mathrm{I}} - \mu_{\mathrm{II}}}{\mu_{\mathrm{I}} + \mu_{\mathrm{II}}}\right). \quad [\mathrm{s^{-1}}] \quad (9.26)$$

The most interesting feature of this equation is its sign. Note that a front indeed becomes unstable when the viscosity of the displaced liquid exceeds the viscosity of the invading liquids: $\mu_{\mathrm{II}} < \mu_{\mathrm{I}}$. This is the case where in the 1D-analysis the front accelerates, see section 9.1.

Another aspect of equation (9.26) that deserves attention is the relation between ν and α. According to the equation, the highest rate is obtained when $\alpha \to \infty$ and $\lambda \to 0$. As the equation is based on Darcy's law, it is valid on length scales bigger than the size of a REV. As a consequence, this equation is not suitable for predictions of the typical wave length of the perturbation. Therefore, viscous fingering in porous media can only be described accurately with help of pore-scale models: models that describe the processes on the pore level.

Further reading

Sahimi M 2011 *Flow and Transport in Porous Media and Fractured Rock: From Classical Methods to Modern Approaches* 2nd edn (New York: Wiley) ch 14, 15: an extensive discussion of

multiphase flow. It covers continuum modeling, experimental methods and pore scale modeling.

Dake D P 1978 *Fundamentals of Reservoir Engineering* (New York: Elsevier): an introduction to one of the most important application areas of multiphase flow in porous media.

Meakin P 1998 *Fractals, scaling and growth far from equilibrium* (Cambridge: Cambridge University Press) ch 4, Appendix A]: a beautiful book describing beautiful phenomena. Its scope is much broader than porous media. The appendix is helpful for understanding the stability problem of moving interfaces in general and viscous fingering in particular.

IOP Concise Physics

Fluids in Porous Media
Transport and phase changes
Henk Huinink

Appendix A

Thermodynamic potentials

In this appendix it is shown how a thermodynamic potential Y [J] can be derived for system with fixed boundary conditions. It is proven that Y is the thermodynamic function that should be minimized to find equilibrium according to the second law of thermodynamics. A vector notation will be used for the work W,

$$\mathrm{d}W = \vec{f} \cdot \mathrm{d}\vec{X}, \quad [\mathrm{J}] \tag{A.1}$$

where $\vec{f}$ and $\vec{X}$ are the generalized forces (intensive variables) and generalized displacements (extensive variables). Note that $\vec{f} = (-p, \mu, \ldots)$ and $\vec{X} = (V, N, \ldots)$.

In our generic system T [K], $\vec{f}'$ and $\vec{X}''$ are fixed. The differential of the energy, equation (2.2), can now be written as

$$\mathrm{d}E = T\mathrm{d}S + \vec{f} \cdot \mathrm{d}\vec{X} = T\mathrm{d}S + \vec{f}' \cdot \mathrm{d}\vec{X}' + \vec{f}'' \cdot \mathrm{d}\vec{X}'', \quad [\mathrm{J}] \tag{A.2}$$

where the work terms have been split into two groups corresponding to either fixed intensive ($'$) or extensive variables ($''$). Given the boundary conditions the thermodynamic potential Y should obey the following differential

$$\mathrm{d}Y = \left(\frac{\partial Y}{\partial T}\right)_{\vec{f}',\vec{X}''} \mathrm{d}T + \left(\frac{\partial Y}{\partial \vec{f}'}\right)_{T,\vec{X}''} \mathrm{d}\vec{f}' + \left(\frac{\partial Y}{\partial \vec{X}''}\right)_{T,\vec{f}'} \mathrm{d}\vec{X}''. \quad [\mathrm{J}] \tag{A.3}$$

This can be achieved by defining Y as

$$Y \equiv E - TS - \vec{f}' \cdot \mathrm{d}\vec{X}'. \quad [\mathrm{J}] \tag{A.4}$$

This operation is the Legendre transformation. By combining this definition of Y with equation (A.2), the appropriate differential can obtained:

$$\mathrm{d}Y = -S\mathrm{d}T - \vec{X}' \cdot \mathrm{d}\vec{f}' + \vec{f}'' \cdot \mathrm{d}\vec{X}'', \quad [\mathrm{J}] \tag{A.5}$$

Note that Y varies with the appropriate boundary conditions, i.e. the variables that are under the control of the experimentalist.

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That Y is also the property determining equilibrium still has to proven. Therefore, we have to show that Y is minimal in equilibrium. To that end we embed the system in reservoir, see figure A.1. The reservoir is very large compared to the system. The total system and reservoir is isolated ($\mathrm{d}Q = 0$) and non-deformable ($\mathrm{d}\vec{X} = 0$). Via an internal constraint the system can be pushed out of equilibrium. The non-equilibrium state of the system is quantified via the variable Δ. The precise nature of Δ is not relevant for the exercise. According to the second law the total entropy S has to be maximal for the combination of system and reservoir

$$\mathrm{d}S = 0, \mathrm{d}^2S < 0. \tag{A.6}$$

As entropy is an additive property

$$S = S_R + S_S, \quad [\mathrm{J\ K^{-1}}] \tag{A.7}$$

where S_R and S_S [J K^{-1}] are the entropies of the reservoir and system, respectively.

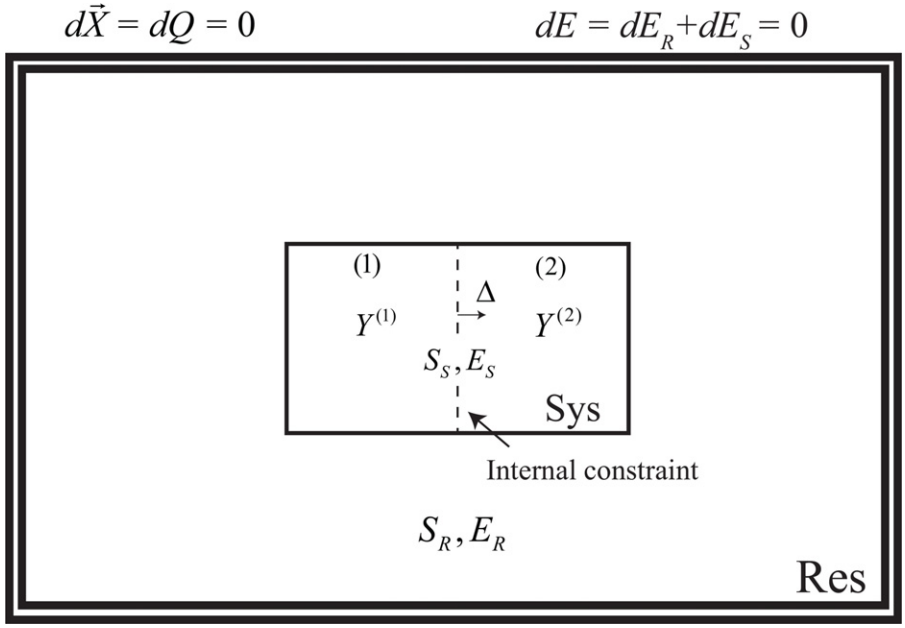


Figure A.1. A system (S) embedded in reservoir (R). The system can be brought out of equilibrium with the internal constraint Δ. The outer boundary of the reservoir does not allow heat transport and thereby isolates the whole of the system and reservoir. The equilibrium state of the system is found by minimizing its thermodynamic potential Y. This minimization condition follows from the condition that in equilibrium the entropy of the sum of the system and the reservoir should be maximal.

The total energy is also an additive property and since the combination of reservoir and system has been isolated

$$\mathrm{d}E = \mathrm{d}E_R + \mathrm{d}E_S = 0. \quad [\mathrm{J}] \tag{A.8}$$

As all extensive variables of the total (R+S) have to be constant, it should hold that

$$\mathrm{d}\vec{X}'_R + \mathrm{d}\vec{X}'_S = 0. \tag{A.9}$$

The extensive variables that are fixed for the system itself obey a more stringent condition:

$$\mathrm{d}\vec{X}''_R = 0,\ \mathrm{d}\vec{X}''_S = 0. \tag{A.10}$$

When the aforementioned internal constraint is used to push the system out of equilibrium, then the following should hold for the entropy:

$$\mathrm{d}S_S = \frac{\mathrm{d}E_S}{T_S} - \frac{\vec{f}'_S \cdot \mathrm{d}\vec{X}'_S}{T_S} \quad [\mathrm{J\ K^{-1}}] \tag{A.11}$$

and

$$\mathrm{d}S_R = \frac{\mathrm{d}E_R}{T_R} - \frac{\vec{f}'_R \cdot \mathrm{d}\vec{X}'_R}{T_R}. \quad [\mathrm{J\ K^{-1}}] \tag{A.12}$$

The absence of work terms of the form $\vec{f}''_{R,S} \cdot \mathrm{d}\vec{X}''_{R,S}$ is a consequence of equation (A.10).

According to equation (A.6), the entropy change equals

$$\mathrm{d}S = \mathrm{d}S_R + \mathrm{d}S_S = \left(\frac{1}{T_S} - \frac{1}{T_R}\right)\mathrm{d}E_S - \left(\frac{\vec{f}'_S}{T_S} - \frac{\vec{f}'_R}{T_R}\right) \cdot \mathrm{d}\vec{X}'_S = 0. \ [\mathrm{J\ K^{-1}}] \tag{A.13}$$

This equality can only be satisfied when

$$T_S = T_R = T, \vec{f}'_S = \vec{f}'_R = \vec{f}', \tag{A.14}$$

which is the case when the system and the reservoir are in equilibrium. With these equilibrium conditions the entropy change can be rewritten as

$$\mathrm{d}S = \mathrm{d}S_R + \mathrm{d}S_S = \frac{\mathrm{d}E_R}{T} - \frac{\vec{f}' \cdot \mathrm{d}\vec{X}'_R}{T} + \mathrm{d}S_S = 0. \quad [\mathrm{J\ K^{-1}}] \tag{A.15}$$

Using the fact that the total energy has to be constant, equation (A.8), one can rewrite equation (A.15):

$$-\frac{\mathrm{d}E_S}{T} + \frac{\vec{f}' \cdot \mathrm{d}\vec{X}'_S}{T} + \mathrm{d}S_S = 0. \quad [\mathrm{J\ K^{-1}}] \tag{A.16}$$

As the reservoir is very large compared to the system, the temperature T and $\vec{f}'$ will hardly vary. With this in mind we can use equation (A.16) to reformulate the original equilibrium condition $\mathrm{d}S = 0$ as

$$\mathrm{d}Y \equiv \mathrm{d}(E_S - TS_S - \vec{f}' \cdot \mathrm{d}\vec{X}'_S) = 0. \quad [\mathrm{J}] \tag{A.17}$$

Also, the second equilibrium condition $\mathrm{d}^2S < 0$ can now be found;

$$\mathrm{d}^2Y > 0. \quad [\mathrm{J}] \tag{A.18}$$

Equations (A.17) and (A.18) show that the potential Y is indeed the thermodynamical potential that should be minimized to find the equilibrium state of the system.

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Appendix B

Energy of a liquid film

The starting point for the derivation of equation (3.5) is equation (3.2):

$$E = \frac{1}{2}\int_V \int_V n(\vec{r}_1)n(\vec{r}_2)u(\vec{r})\mathrm{d}\vec{r}_1\mathrm{d}\vec{r}_2. \quad [\mathrm{J}] \tag{B.1}$$

Where n [m^{-3}] is the number density and $\vec{r} \equiv \vec{r}_2 - \vec{r}_1$ [m]. Further, $\vec{r}_1$ [m] and $\vec{r}_2$ [m] are the positions of the particles 1 and 2, respectively. We assume a homogeneous density distribution, that changes stepwise at the interface, as represented in figure 3.2. With this in mind, equation (B.1) can be rewritten in cylindrical coordinates:

$$\begin{aligned}
E &= \frac{1}{2}\int_V \int_V n(\vec{r}_1)n(\vec{r}_2)u(\vec{r})\mathrm{d}\vec{r}_1\mathrm{d}\vec{r}_2 \\
&= \frac{n^2}{2}\int_V \int_V u(\vec{r})\mathrm{d}\vec{r}_1\mathrm{d}\vec{r}_2 \\
&= \frac{n^2}{2}\int_0^L \int_{-\pi}^{\pi} \int_0^{\infty}\left[\int_0^L \int_{-\pi}^{\pi} \int_0^{\infty} u(R_1,\, \theta_1,\, z_1,\, R_2,\, \theta_2,\, z_2)R_2\mathrm{d}R_2\mathrm{d}\theta_2\mathrm{d}z_2\right]R_1\mathrm{d}R_1d\theta_1\mathrm{d}z_1 \\
&= \frac{n^2}{2}\int_0^L \int_0^{\infty}\left[\int_0^L \int_0^{\infty} u(R_1,\, z_1,\, R_2,\, z_2)2\pi R_2\mathrm{d}R_2\mathrm{d}z_2\right]2\pi R_1\mathrm{d}R_1\mathrm{d}z_1 \\
&= \frac{n^2}{2}\int_0^L \left[\int_0^L \int_0^{\infty} u(z_1,\, R_2,\, z_2)2\pi R_2\mathrm{d}R_2\mathrm{d}z_2\right]2\pi\left[\frac{R_1^2}{2}\right]_0^{\infty}\mathrm{d}z_1 \\
&= \frac{n^2 A}{2}\int_0^L \left[\int_{-z_1}^{L-z_1} \int_{r_{\min}(z_2)}^{\infty} u(R_2,\, z_2)2\pi R_2\mathrm{d}R_2\mathrm{d}z_2\right]\mathrm{d}z_1.
\end{aligned} \tag{B.2}$$

Where in the last line the coordinates z_2 and R_2 were transformed in reference to z_1 and R_1. The $r_{\min}(z_2)$ indicates that the minimal value of R_2 is dependant on z_2.

The potential is given by hard-sphere repulsion and a van der Waals attraction, equation (3.4):

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$$u(|\vec{r}|) = u_{HS}(|\vec{r}|) - \frac{C}{|\vec{r}|^6}. \quad [\mathrm{J}] \tag{B.3}$$

The first term in (B.3) only contributes to the lower limit of the inner integral: $r_{\min}(z_2)$. We focus therefore on the second term:

$$\begin{aligned} E &= \frac{n^2 A}{2} \int_0^L \left[\int_{-z_1}^{L-z_1} \int_{r_{\min}(z_2)}^{\infty} u(R_2, z_2) 2\pi R_2 \mathrm{d}R_2 \mathrm{d}z_2 \right] \mathrm{d}z_1 \\ &= n^2 A\pi \int_0^L \left[\int_{-z_1}^{L-z_1} \int_{r_{\min}(z_2)}^{\infty} -\frac{C}{(z_2^2 + R_2^2)^3} R_2 \mathrm{d}R_2 \mathrm{d}z_2 \right] \mathrm{d}z_1 \\ &= \frac{n^2 A\pi}{4} \int_0^L \left[\int_{-z_1}^{L-z_1} \left(\frac{C}{(z_2^2 + R_2^2)^2} \right)_{r_{\min}(z_2)}^{\infty} \mathrm{d}z_2 \right] \mathrm{d}z_1 \\ &= \frac{-n^2 A\pi}{4} \int_0^L \left[\int_{-z_1}^{L-z_1} \frac{C}{(z_2^2 + r_{\min}(z_2)^2)^2} \mathrm{d}z_2 \right] \mathrm{d}z_1. \end{aligned} \tag{B.4}$$

As can be seen in figure B.1, three regions of the film can be distinguished with each different values for $r_{\min}$. Below we treat these regions separately.

Region 1: $0 \leqslant z_1 \leqslant a$

In this region, the following holds:

$$r_{\min}(z_2) = \begin{cases} \sqrt{a^2 - z^2} & -z_1 \leqslant z_2 \leqslant a \\ 0 & a < z_2 \leqslant L - z_1. \end{cases} \tag{B.5}$$

The contribution of region 1 to the energy can thus be noted as:

$$\begin{aligned} \frac{E_{\mathrm{reg.1}}}{A} &= \frac{-n^2\pi}{4} \int_0^a \left[\int_{-z_1}^{a} \frac{C}{a^4} \mathrm{d}z_2 + \int_a^{L-z_1} \frac{C}{z_2^4} \mathrm{d}z_2 \right] \mathrm{d}z_1 \\ &= \frac{-n^2\pi C}{4} \int_0^a \left[\frac{a + z_1}{a^4} - \frac{1}{3(L - z_1)^3} + \frac{1}{3a^3} \right] \mathrm{d}z_1 \\ &= \frac{-n^2\pi C}{4} \left[\frac{3}{2a^2} - \frac{1}{6(L - a)^2} + \frac{1}{6L^2} + \frac{1}{3a^2} \right]. \end{aligned} \tag{B.6}$$

Region 2: $a \leqslant z_1 \leqslant L - a$

In this region, the following holds:

$$r_{\min}(z_2) = \begin{cases} 0 & -z_1 \leqslant z_2 < -a \\ \sqrt{a^2 - z^2} & -a \leqslant z_2 \leqslant a \\ 0 & a < z_2 \leqslant L - z_1 \end{cases} \tag{B.7}$$

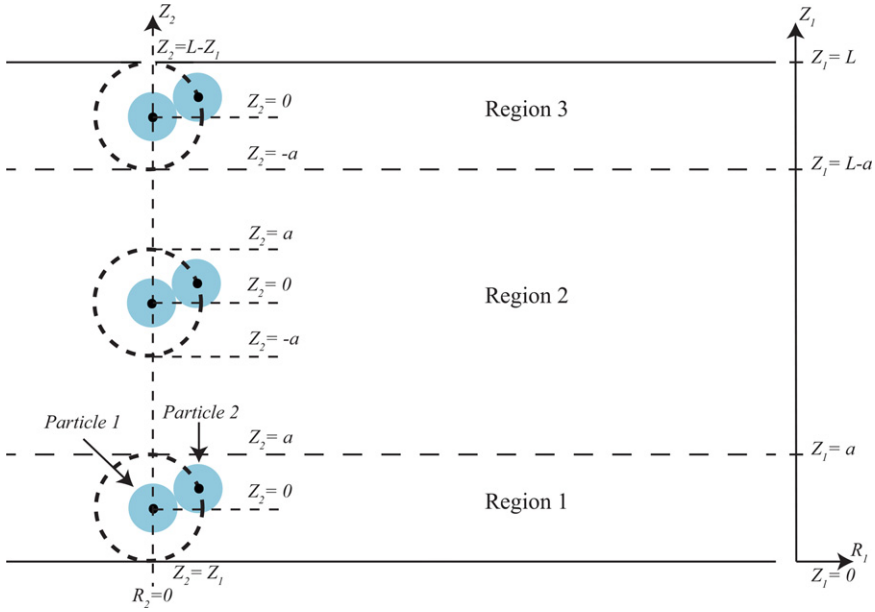


Figure B.1. The three integration regions of the film. The blue circles represent the two interacting particles.

The contribution of region 2 to the energy can thus be written as:

$$\begin{aligned}\frac{E_{\text{reg.2}}}{A} &= -n^2\pi\int_a^{L-a}\left[\int_{-z_1}^{-a}\frac{C}{z_2^4}\mathrm{d}z_2+\int_{-a}^{a}\frac{C}{a^4}\mathrm{d}z_2+\int_a^{L-z_1}\frac{C}{z_2^4}\mathrm{d}z_2\right]\mathrm{d}z_1\\ &= \frac{-n^2\pi C}{4}\int_a^{L-a}\left[\frac{1}{3a^3}-\frac{1}{3z_1^3}+\frac{2}{a^3}+\frac{1}{3a^3}-\frac{1}{3(L-z_1)^3}\right]\mathrm{d}z_1\\ &= \frac{-n^2\pi C}{4}\left[\frac{2(L-2a)}{3a^3}+\frac{1}{3(L-a)^2}-\frac{1}{3a^2}+\frac{2(L-2a)}{a^3}\right].\end{aligned} \tag{B.8}$$

Region 3: $L-a\leqslant z_1\leqslant L$

In this region, the following holds:

$$r_{\min}(z_2)=\begin{cases}0 & -z_1\leqslant z_2<-a\\ \sqrt{a^2-z^2} & -a\leqslant z_2\leqslant L-z_1\end{cases}. \tag{B.9}$$

The contribution of region 3 to the energy can be written as:

$$\begin{aligned}\frac{E_{\text{reg.3}}}{A} &= \frac{-n^2\pi C}{4}\int_{L-a}^{L}\left[\int_{-z_1}^{-a}\frac{C}{z_2^4}\mathrm{d}z_2+\int_{-a}^{L-z_1}\frac{C}{a^4}\mathrm{d}z_2\right]\mathrm{d}z_1\\ &= \frac{-n^2\pi C}{4}\int_{L-a}^{L}\left[\frac{1}{3a^3}-\frac{1}{3z_1^3}+\frac{L-z_1+a}{a^4}\right]\mathrm{d}z_1\\ &= \frac{-n^2\pi C}{4}\left[\frac{a}{3a^3}+\frac{1}{6L^2}-\frac{1}{6(L-a)^2}+\frac{(L+a)a}{a^4}-\frac{L^2-(L-a)^2}{2a^4}\right].\end{aligned} \tag{B.10}$$

The sum of all three regions gives the final result, equal to equation (3.5)

$$\begin{aligned}\frac{E}{A} &= E_{\text{reg.1}} + E_{\text{reg.2}} + E_{\text{reg.3}} \\ &= \frac{-n^2\pi C}{4}\left[\frac{3}{2a^2} - \frac{1}{6(L-a)^2} + \frac{1}{6L^2} + \frac{1}{3a^2}\right] \\ &\quad - \frac{n^2\pi C}{4}\left[\frac{L-2a}{3a^3} + \frac{1}{6(L-a)^2} - \frac{1}{6a^2} + \frac{2(L-2a)}{a^3} + \frac{L-2a}{3a^3} - \frac{1}{6a^2} + \frac{1}{6(L-a)^2}\right] \\ &\quad - \frac{n^2\pi C}{4}\left[\frac{a}{3a^3} + \frac{1}{6L^2} - \frac{1}{6(L-a)^2} + \frac{(L+a)(a)}{a^4} - \frac{L^2-(L-a)^2}{2a^4}\right] \\ &= \frac{\pi C n^2}{2a^2} - \frac{2\pi C n^2\, L}{3a^3} - \frac{\pi C n^2}{12L^2}. \quad [\text{J m}^{-2}]\end{aligned} \tag{B.11}$$

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Appendix C

Interfacial areas of a spherical cap

Our derivation of Young's law starts with equation (3.8). As we assume in our derivation the geometry of a spherical cap, some extra information on equation (3.9) is given. It is clear from figure 3.3 that the contact surface of the liquid and the solid is a circle with radius a [m], which can be written as $a = \sqrt{r^2 - (h - r)^2}$ using the Pythagoras theorem. Here, h [m] and r [m] are the height and radius of curvature of the droplet, respectively. The contact surface can thus be written as $A_{sl} = \pi a^2 = \pi(r^2 - (h - r)^2)$ [m^2]. The volume V [m^3] of the spherical cap is then the following integral:

$$\begin{aligned} V &= \int_0^h A_{sl}\mathrm{d}h' = \int_0^h \pi\left(r^2 - (h' - r)^2\right)\mathrm{d}h' = \frac{\pi h^2}{3}(3r - h) \\ &= \frac{\pi h}{3}\left(3rh - h^2\right) = \frac{\pi h}{3}\left(a^2 - rH\right) \\ &= \frac{\pi h}{6}\left(3a^2 + h^2\right). \quad [\mathrm{m}^3] \end{aligned} \tag{C.1}$$

Now, we rewrite this volume using $h = r(1 - \cos(\theta))$ and $a = r\sin(\theta)$, where the former is found using the angle $\pi - \theta$ as indicated in figure C.1. Note that θ is the contact angle of the droplet. This thus gives:

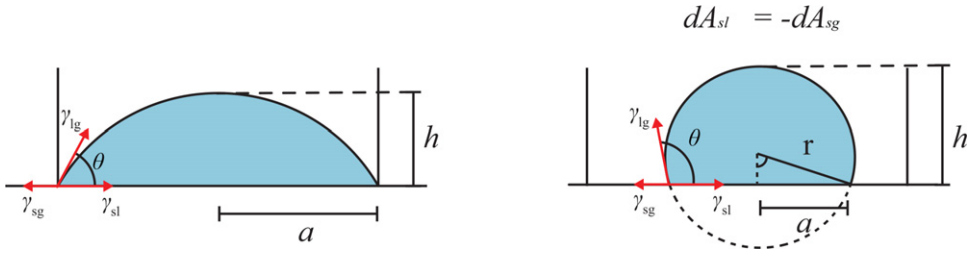


Figure C.1. The spherical cap of a wetting (left) and a non-wetting liquid (right).

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$$
\begin{aligned}
V &= \frac{\pi h}{6}\left(3a^2 + h^2\right) \\
&= \frac{\pi r}{6}\left(3r^2 \sin(\theta)^2 + r^2(1 - 2\cos(\theta) + \cos(\theta)^2)\right)(1 - \cos(\theta)) \\
&= \frac{\pi r^3}{3}\left(\sin(\theta)^2 - \cos(\theta) + 1\right)(1 - \cos(\theta)) \\
&= \frac{\pi r^3}{3}(2 + \cos(\theta))(1 - \cos(\theta))^2 \\
&= \frac{\pi r^3}{3}(2 + z)(1 - z)^2. \quad [\mathrm{m}^3]
\end{aligned} \tag{C.2}
$$

In the last line the parameter $z \equiv \cos(\theta)$ is introduced to simplify the equations. This form could have been more easily derived by using $\frac{\pi h}{3}(3rh - h^2)$ from (C.1) and the different forms for h and a, but seeing as (C.1) is the usually given form for the volume of a spherical cap, the extended version has been given.

As the volume of the spherical cap is constant, we are able to find a form for $\mathrm{d}z$:

$$
\begin{aligned}
\mathrm{d}V &= \frac{\partial V}{\partial r}\mathrm{d}r + \frac{\partial V}{\partial z}\mathrm{d}z = 0 \\
&= \pi r^2(2 + z)(1 - z)^2\mathrm{d}r + \frac{\pi r^3}{3}\left[(1 - z)^2 - 2(1 - z)(2 + z)\right]\mathrm{d}z \\
\Rightarrow \mathrm{d}z &= \frac{(1 - z)(2 + z)}{r(1 + z)}\mathrm{d}r.
\end{aligned} \tag{C.3}
$$

The following step is writing the surfaces A_{lg} and A_{sl} in terms of z:

$$
A_{lg} = 2\pi hr = 2\pi r^2\left(1 - \cos(\theta)\right) = \pi\left(a^2 + h^2\right) = 2\pi r^2\left(1 - z\right) \tag{C.4}
$$

$$
A_{sl} = \pi a^2 = \pi r^2 \sin(\theta)^2 = \pi r^2\left(1 - \cos(\theta)^2\right) = \pi r^2\left(1 - z^2\right). \tag{C.5}
$$

And as for the volume, the second step is taking the derivative:

$$
\mathrm{d}A_{lg} = \frac{\partial \mathrm{d}A_{lg}}{\partial r}\mathrm{d}r + \frac{\partial \mathrm{d}A_{lg}}{\partial z}\mathrm{d}z = 2\pi r(2(1 - z)\mathrm{d}r - \mathrm{d}z) \tag{C.6}
$$

$$
\mathrm{d}A_{sl} = \frac{\partial \mathrm{d}A_{sl}}{\partial r}\mathrm{d}r + \frac{\partial \mathrm{d}A_{sl}}{\partial z}\mathrm{d}z = 2\pi r(1 - z^2)\mathrm{d}r - 2\pi r^2\mathrm{d}z. \tag{C.7}
$$

Rewriting these using equation (C.3):

$$
\begin{aligned}
\mathrm{d}A_{lg} &= \frac{\partial \mathrm{d}A_{lg}}{\partial r}\mathrm{d}r + \frac{\partial \mathrm{d}A_{lg}}{\partial z}\mathrm{d}z = 2\pi r(2(1 - z)\mathrm{d}r - \mathrm{d}z) \\
&= \frac{\partial \mathrm{d}A_{lg}}{\partial r}\mathrm{d}r + \frac{\partial \mathrm{d}A_{lg}}{\partial z}\mathrm{d}z = 2\pi r\left(2(1 - z)\mathrm{d}r - \left(\frac{(1 - z)(2 + z)}{r(1 + z)}\mathrm{d}r\right)\right) \\
&= \frac{2\pi rz(1 - z)}{(1 + z)}\mathrm{d}r
\end{aligned} \tag{C.8}
$$

$$\begin{aligned} \mathrm{d}A_{sl} &= \frac{\partial \mathrm{d}A_{sl}}{\partial r}\mathrm{d}r + \frac{\partial \mathrm{d}A_{sl}}{\partial z}\mathrm{d}z = 2\pi r\left(1 - z^2\right)\mathrm{d}r - 2\pi r^2 \mathrm{d}z \\ &= \frac{\partial \mathrm{d}A_{sl}}{\partial r}\mathrm{d}r + \frac{\partial \mathrm{d}A_{sl}}{\partial z}\mathrm{d}z = 2\pi r\left(1 - z^2\right)\mathrm{d}r - 2\pi r^2\left(\frac{(1 - z)(2 + z)}{r(1 + z)}\mathrm{d}r\right) \\ &= \frac{2\pi r(1 - z)}{(1 + z)}\mathrm{d}r. \end{aligned} \tag{C.9}$$

Now finding the extremum of the free energy, using the fact that $\mathrm{d}A_{sl} = -\mathrm{d}A_{sg}$, and using the equations (C.8), (C.9) and $z = \cos(\theta)$:

$$\begin{aligned} \mathrm{d}F &= \gamma_{sl}\mathrm{d}A_{sl} + \gamma_{sg}\mathrm{d}A_{sg} + \gamma_{lg}\mathrm{d}A_{lg} = 0 \\ &= (\gamma_{sl} - \gamma_{sg})\mathrm{d}A_{sl} + \gamma_{lg}\mathrm{d}A_{lg} = 0 \\ &= (\gamma_{sl} - \gamma_{sg})\frac{2\pi r(1 - z)}{(1 + z)}\mathrm{d}r + \gamma_{lg}\frac{2\pi r z(1 - z)}{(1 + z)}\mathrm{d}r = 0 \\ &\Rightarrow \left(\gamma_{sl} - \gamma_{sg}\right) + \gamma_{lg}\cos(\theta) = 0. \end{aligned} \tag{C.10}$$

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Appendix D

Bruggeman equation

In this appendix it is shown how the effective dielectric constant of heterogeneous medium can be estimated with the so-called effective medium approximation (EMA). Actually, the Bruggeman equation for the dielectric constant of multiphase material is derived.

We consider a heterogeneous material with grains having different dielectric constants, as shown in the left picture of figure D.1. The permittivity of each region is indicated by ϵ_i [F m^{-1}]. In the EMA a real grain with a permittivity ϵ_{in} [F m^{-1}] is embedded in a homogeneous environment with an average permittivity ϵ_m [F m^{-1}]. To derive the Bruggeman equation this inclusion is assumed to be spherical and has a radius a [m], see figure D.1.

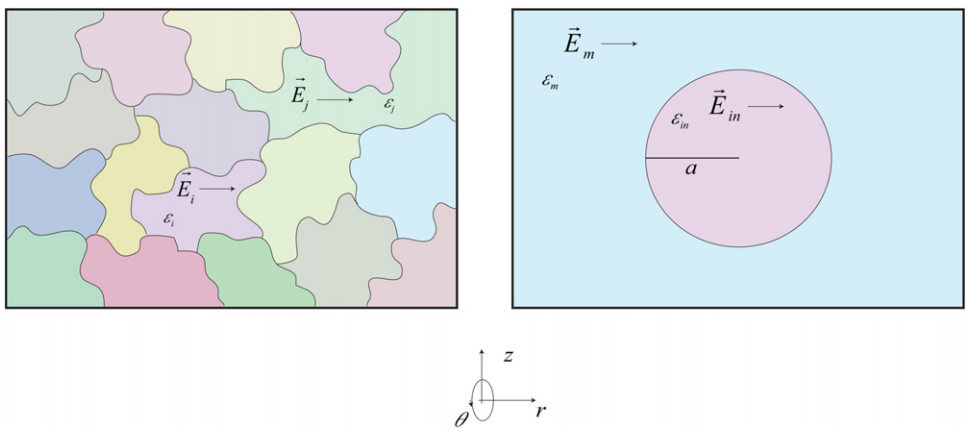


Figure D.1. A heterogeneous dielectric medium: the actual material (left) and the representation as used in the EMA (right). In the EMA framework the electric field is calculated in a sphere embedded in an averaged medium.

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The second step in the derivation consists of solving the electric potentials in both regions, i.e. ψ_{in} [V] and ψ_m [V]. This can be done by applying Poisson's equation, (7.12), to both regions.

$$\nabla^2\psi = 0. \quad [\mathrm{C\ m^{-3}}] \tag{D.1}$$

The most general solution is given by using the infinite sum of Legendre polynomials $P_l(x)$, with A_l,B_l and C_l generic constants:

$$\psi_{in} = \sum_{l=0}^{\infty} A_l r^l P_l(\cos(\theta)) \quad [\mathrm{V}] \quad (r < a) \tag{D.2}$$

$$\psi_m = \sum_{l=0}^{\infty} \Big(B_l r^l + C_l r(-l-1) \Big) P_l(\cos(\theta)) \quad [\mathrm{V}] \quad (r > a) \tag{D.3}$$

where r [m] and θ are the radial coordinate and angle, respectively. Note that cylindrical symmetry is assumed. At a big distance from the spherical inclusion, one thus has:

$$\psi_{m,\infty} = -E_0 r\cos(\theta), \quad [\mathrm{V}] \quad (r \gg a) \tag{D.4}$$

where E_0 [V m^{-1}] is the strength of the average electric field. The following equations, respectively, represent the continuous boundary conditions at $r = a$:

$$\psi_{in}\big|_{r=a} = \psi_m\big|_{r=a} \quad [\mathrm{V}] \tag{D.5}$$

$$-\epsilon_{in}\frac{\partial\psi_{in}}{\partial r}\bigg|_{r=a} = -\epsilon_m\frac{\partial\psi_m}{\partial r}\bigg|_{r=a}. \quad [\mathrm{Cm^{-2}}] \tag{D.6}$$

These can be used to obtain the constants A_l,B_l and C_l. The result is the following:

$$\psi_{in} = -\frac{3\epsilon_m}{\epsilon_{in} + 2\epsilon_m} E_0 r\cos(\theta) \quad [\mathrm{V}] \quad (r < a) \tag{D.7}$$

$$\psi_m = -E_0 r\cos(\theta) + \frac{\epsilon_{in} - \epsilon_m}{\epsilon_{in} + 2\epsilon_m}\frac{a^3}{r^2} E_0\cos(\theta) \quad [\mathrm{V}] \quad (r > a). \tag{D.8}$$

This equation shows that the effective electric field strength E_i in a grain with dielectric constant ϵ_i equals

$$E_i = \frac{3\epsilon_m}{\epsilon_{in} + 2\epsilon_m} E_0. \tag{D.9}$$

The last step in the derivation of the Bruggeman equation is as follows. As the average field strength in the material should be E_0, it should hold for the local deviations that

$$\frac{1}{V}\int_V \big(E(\vec{x}) - E_0\big)\,\mathrm{d}\vec{x} = 0, \quad \Big[\mathrm{V\ m^{-1}}\Big] \tag{D.10}$$

where $E(\vec{x})$ is the local field strength. In the case of N different types of spherical inclusions, equation (D.10) can be rewritten as

$$\sum_{i=1}^{N} \varphi_i \left(E_0 - E_i \right) = \sum_{i=1}^{N} \varphi_i \left(\frac{\epsilon_i - \epsilon_m}{\epsilon_i + 2\epsilon_m} \right) E_0 = 0. \quad \left[\mathrm{V\ m^{-1}} \right] \tag{D.11}$$

From this equation, the Bruggeman is obtained:

$$\sum_{i=1}^{N} \varphi_i \left(\frac{\epsilon_i - \epsilon_m}{\epsilon_i + 2\epsilon_m} \right) = 0. \tag{D.12}$$